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INTERACTION OF HYDROGEN WITH SOLID SURFACES

K. CHRISTMANN

Institut für Physikalische Chemie der Freien Universität Berlin, D-1000 Berlin 33, Germany

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The present article focusses on the chemisorptive and physisorptive behavior of hydrogen interacting with solid surfaces. Depending on the electronic structure of the solid hydrogen will either physically interact and form a weak van der Waals type of bond, or the H_2 molecule will dissociate and the H atoms will form a chemical bond with the surface atoms. In some cases the interaction is not just restricted to the surface but may lead to absorption or even hydride formation. Also, reversible or irreversible changes of the structure of the solid may occur as the hydrogen is interacting. This leads to relaxation, reconstruction and facetting phenomena.

After some introductory remarks recent results on H_2 physisorption and H chemisorption phenomena will be presented and compiled whereby emphasis is put on metal single crystal work. Particular attention will be paid to the energetics and kinetics of the adsorption process, and to the structural properties of hydrogen adsorbed layers. In this context, order–disorder transitions within chemisorbed phases will be considered as well as structural phase transformations involving the underlying solid surfaces (H-induced surface reconstruction). Thereafter a section will be devoted to theoretical model calculations describing the mechanisms of the H chemisorption and the bonding.

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1. Introduction

"By virtue of its single valence electron hydrogen should be the simplest chemically reacting adsorbate... However, the vast literature on H adsorption had demonstrated considerable complexity... To date little more than speculation has been attempted with regard to the nature of the binding states of hydrogen on metals." This statement has been taken from an early work by Tamm and Schmidt in 1969 [1]. An article by Lagos in 1982 [2] states in the beginning: "The bonding of hydrogen to the surface of transition metals has received extensive attention in the last years. Nevertheless no definite answer on the nature of the bonding seems to be available at present..."

Phrases like these can frequently be found in the introductions of articles which are concerned with hydrogen adsorption on surfaces and shed some light on the still existing difficulties in the fundamental understanding of the interaction of hydrogen with surfaces up to now. Whilst the static properties are fairly well known – at least for metal surfaces – and even some knowledge exists about various aspects of the potential energy hyper-surface there is much less information about the interaction *dynamics*, i.e. the hydrogen motion on the potential energy hyper-surface. It is sincerely hoped that progress will be made soon, owing to the more and more sophisticated experimental and theoretical investigations. Particularly molecular beam experiments are suited to elucidate the dynamical character of adsorption.

A review written these days can help to survey and compile the enormous data body that has been gathered during the past decades. It should then be possible to point to some more general lines and characteristic features of the interaction of hydrogen with solid surfaces. The limitation of space, however, requires a careful selection of data to be reviewed and greatly determines the concept of this work. Accordingly, primarily *experimental* studies will be dealt with, whereby emphasis is put on the most important interaction of hydrogen with *transition metal* surfaces. Nevertheless, also some recent promising theoretical concepts to rationalize the H adsorption process will be briefly reviewed.

The motivation to investigate hydrogen–surface interaction phenomena arises from various sources. In the first place, we have to mention the field of heterogeneous catalysis in which a great part of the reactions involve hydrogen as a reactant or product molecule. The synthesis of ammonia, of hydrocarbons or alcohols by means of the Fischer–Tropsch reaction, or solidification of unsaturated fatty acids are only some examples. Another important aspect is in the field of materials science and metallurgy where particularly hydrogen has long been known as a major source of embrittlement and fracture phenomena [3]. Here, the elementary steps are even less understood than in the case of heterogeneous catalysis. Finally, there is an increasing interest to use certain alloys or intermetallic compounds to dissolve and store gaseous

hydrogen [4]. This, along with the development of the fuel cell technology requires a detailed knowledge of the underlying reaction mechanism, in particular of the adsorption of hydrogen which is always the preceding interaction step.

The outlines of the present article are as follows: First, the general principles of the interaction of a single hydrogen molecule or atom with a solid surface will be described with the aid of appropriate examples. In particular, energetic and kinetic properties will be reviewed and compared. Thereafter, a chapter presents a collection of data and features that demonstrate the multiparticle interaction pattern. Finally, some recent theoretical treatments of the H–surface interaction problem including the dissociation step will follow, before a general conclusion will end the article.

There have been a number of review articles on hydrogen adsorption in the past, mostly as part of textbooks on chemisorption and catalysis [5–10]. Actually, only little change has arisen in the problems of interest since the early fifties: The aim was and still is to gain more insight into the thermodynamics and kinetics of the adsorption process, and one has concentrated on measuring heats of adsorption [11], lateral interaction energies [12], activation energies for desorption and diffusion, accommodation and sticking probabilities, entropies of adsorption and, of course, heats of solution of hydrogen in bulk materials [13]. The problem of hydrogen interacting with bulk materials will, however, be ignored in this article because there exists a great many exhaustive review papers on this subject [4]. Also, a basic understanding of the chemical and physical bonding mechanism of adsorbed H is attempted since many years, which seems to be even more difficult.

Whereas in the fourties and fifties mainly polycrystalline samples were investigated (evaporated films, powders, foils and wires) mostly with respect to thermodynamic (binding energy) properties, there was a significant change in the interest coinciding with the technical development of surface sensitive electron spectroscopies. Here the diffraction of low energy electrons (LEED) has to be mentioned in the first place which, for the first time, allowed a determination of *surface* structure although it took almost twenty years to develop this method to a stage where the analysis of the structure of a clean metal surface could be performed routinely [15]. Suddenly, there was a call for single crystal studies in which a well-defined surface geometry allowed the correlation of various experimentally determined quantities with the surface structure (dipole moments of the adsorbate, binding energies of chemisorbed gases [16] or the catalytic activity of surface reactions [17]). Yet, the surface cleanliness was a major obstacle that could not be overcome and could be regarded as the main source for the irreproducibility of the data until the Auger electron spectroscopy was developed and introduced by Harris and by Palmberg in 1968 [18]. Since then it was realized that impurity atoms like carbon, sulfur or oxygen usually alter the adsorption properties of surfaces to

a great extent, and in particular hydrogen reacts extremely sensitively upon even small concentrations of foreign atoms on surfaces in that the binding energy of the H atoms is dramatically lowered or characteristic adsorbate structures are inhibited. It took another ten years until reproducible data could be obtained in independent laboratories (for which the determination of the c(2 × 2)-2H structure on Ni(111) surface [19] and of the c(2 × 6)-H phases on Ni(110) [20,21] are good examples). Yet, recent studies of the dynamical process of the H adsorption have revealed that there is one obstacle left for getting reproducible kinetic data, particularly for the zero coverage limit, namely, the *characterization of kind and number of crystallographical defects* of the solid surface which the H_2 is exposed to. Indeed, as electron micrographs of even carefully prepared single crystal surfaces show these surfaces exhibit by no means large defect-free areas but rather consist of only fairly small (~ 100 Å^2) well-ordered patches interrupted by steps, dislocations, facets or grain boundaries. At a first glance, these surfaces produce excellent LEED patterns and show no anomalies at medium or high adsorbate concentrations. However, at small coverages especially the kinetics but also the energetics of adsorption may be significantly altered. It appears as if the small H atom would be particularly susceptible to these defects (see section 2.3).

Despite the major experimental problem to unequivocally characterize the microscopical crystallographical state of a surface in all details, the present state of the experimental knowledge is nevertheless a favorable starting point to write a review article on hydrogen adsorption, to compare the chemisorption properties of various systems and to draw from this some more general conclusions as to the nature of the hydrogen–surface interaction.

2. The single-particle interaction

2.1. The hydrogen (H_2) molecule

In order to understand the interaction of hydrogen with a solid it is inevitable to know as much about the free molecule as possible, i.e., its geometrical shape, its electronic structure, its dissociation (or binding) energy, its vibrational and rotational structure, etc. It is trivial that the energetic position of both the highest occupied (ionization potential) and the lowest unoccupied (electron affinity) molecular orbital with respect to the electron states of the solid to which the molecule couples is of primary importance for the formation of a chemical bond.

Of course, one must always keep in mind that one is dealing with a diatomic *molecule* which is able to dissociate when getting in contact with the surface. It is much easier to consider the interaction of a *single* H atom with a surface than that of a H_2 molecule which, owing to its strong covalent

Table 1
Some selected physical and thermodynamical properties of molecular hydrogen [a)]

Property	Unit	H_2 ($_1H^1$)	D_2 ($_1H^2$)	HD
Molar mass	kg kmol^{-1}	2.1059	4.0282	6.03222
Absolute mass	kg	3.497×10^{-27}	6.6847×10^{-27}	
Natural	%	99.985	0.015	
abundance	kJ mol^{-1}	431.37	439.50	435.4
			$_1H^3$: 442.89	
Binding energy	eV mol^{-1}	4.4763	4.553	4.511
(heat of			$_1H^3$: 4.5881	
dissociation)	kcal mol^{-1}	103.05	104.91	104.00
			$_1H^3$: 105.80	
Internuclear				
distance	m	0.7416×10^{-10}	0.7417×10^{-10}	0.4713×10^{-10}
Ionization				
potential	eV	15.427	15.46	
Moment of inertia	kg m^2	0.460×10^{-47}	0.919×10^{-47}	0.611×10^{-47}
Rotational				
constant B_0	cm^{-1}	60.809	30.4	
Polarization α_0	m^3	$\sim 0.830\times10^{-30}$		
Ground state				
configuration		$^1\Sigma_{g^+}$		
Ground state				
vibration				
frequency $\tilde{\nu}$	cm^{-1}	4395.2	3118.4	3817
Triple point	K	13.95		
Melting point	K	14.025	18.65	
Boiling point	K	20.39	23.6	
Melting entropy	J mol^{-1} K^{-1}	8.41		
Vaporization				
entropy	J mol^{-1} K^{-1}	44.75		
Heat of fusion at				
13.9 K	J mol^{-1}	117.2		
Heat of vaporiza-				
tion at 18 K	J mol^{-1}	912.5	1264.2	
Molar volume	m^3			
at 13.9 K				
(liquid)		26.15×10^{-6}	23.14×10^{-6}	
(crystal)		23.31×10^{-6}	20.48×10^{-6}	
Crystal structure				
of solid H_2		hcp/fcc		
Space group		D_{6h}^4		
Lattice constants				
(hcp)	10^{-10} m			
a		3.75		
c		6.12		
c/a		1.632		
Characteristic				
temperature Θ	K	91		
Maximum vibra-				
tion frequency	s^{-1}	1.91×10^{12}		

[a)] Spectroscopic data taken from: G. Herzberg, "Molecular Spectra and Molecular Structure", Vol. 1, "Spectra of Diatomic Molecules", 2nd ed. (Van Nostrand–Reinhold, New York, 1950). Thermodynamic data taken from: E.A. Moelwyn-Hughes, "Physical Chemistry", 2nd ed. (Pergamon, Oxford, 1961).

innermolecular bond, is chemically rather inert and will exert more or less weak van der Waals type interactions only with a solid surface. On various surfaces, however, particularly transition metal surfaces, the hydrogen molecule will spontaneously dissociate into the atoms which then form strong metal–hydrogen bonds. Energetic arguments would suggest that this dissociation process should always occur as soon as the net energy of this bond formation exceeds the heat of dissociation of the molecule. There is, however, also an important influence of the *kinetics* of interaction in that for various reasons quite a pronounced activation energy barrier may exist for the dissociation step to take place. One factor here is the dissipation of the kinetic energy of the incoming molecule which is governed by several physical mechanisms (cf. section 2.3). Only recently there have been theoretical [22] and experimental [23] attempts to tackle the dissociation problem which is believed to be crucial for the entire hydrogen adsorption process.

It is certainly beyond the scope of this article to present or discuss the physical and chemical properties of the H_2 molecule which has been dealt with in several excellent review articles [24]. When dealing with hydrogen physisorption and dissociation phenomena it may nevertheless be worthwhile to have a brief look at some of the more important physical parameters such as the geometric dimensions, the bond length, the ground state energy level, the ionization potential or the electron affinity, etc. Also, a knowledge of the crystal structure of solid H_2, melting and boiling points, latent heats of phase transitions, et. may be useful in the context of hydrogen adsorption. Some of these data are therefore compiled in table 1.

There are various properties of the ground state molecule that deserve particular attention, namely

(i) its very small size (H–H distance only 0.74 Å);

(ii) its almost spherical shape;

(iii) the existence of two stable isotopes which differ in mass by a factor of two!

(iv) the strong influence of the nuclear spin leading to the two H_2 species "ortho"- and "para"-hydrogen with the total nuclear spin being symmetric and antisymmetric, respectively;

(v) its very small electron density;

(vi) its ability to dissociate into the atoms.

Only mentioned are the very interesting features of the H_2 molecule in the solid state which distinguish it from other molecular solids: Apart from the already described ortho- and para-species it is remarkable that the free rotor states that govern the rotational motions of an isolated molecule are almost unperturbed by the interactions with neighbors. Therefore, in the H_2 solid, all the molecules are translationally localized at their lattice sites but rotate nearly freely even at $T = 0$ K. This is a consequence of the large molecular rotational constant and the weak anisotropic forces arising from the almost spherical

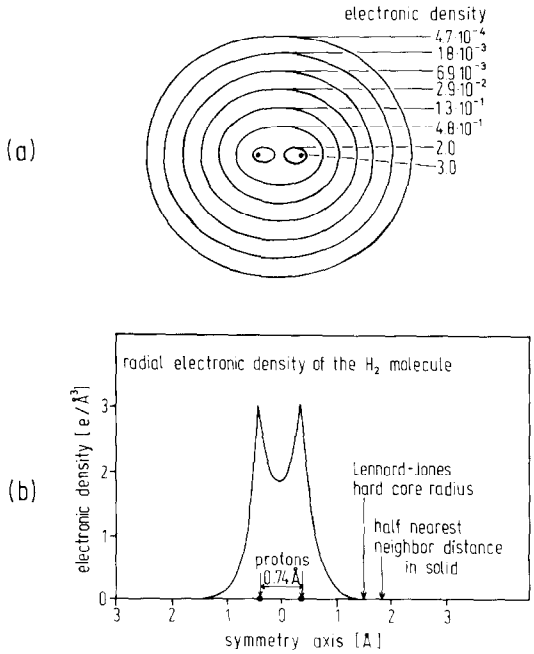


Fig. 1. Map of the electronic charge density of a hydrogen molecule. Upper part: Contour lines of equal charge density. Lower part: Charge density along the molecular axis. After Silvera [24].

charge distribution along with the fairly large intermolecular distance (3.79 Å) in the zero-pressure solid. Fig. 1 which is taken from the article by Silvera [24] shows the electronic charge distribution of the H_2 molecule which is nearly spherical.

2.2. The hydrogen atom

Upon dissociation, H atoms are formed which then interact independently with the solid surface. This process is referred to as hydrogen *chemisorption* and it occurs on surfaces of selected elements only. Fig. 2 shows the periodic table of the elements where all elements have been labelled which chemisorb hydrogen with appreciable sticking probability.

Again, for the sake of understanding the details of the H chemisorption, it is helpful to have a short list of the most prominent physical and chemical properties of the H atom. This is given in table 2.

As in the H_2 molecule case some particular properties of the H atom and its interaction with solids deserve attention. Again, the H atom exhibits the most pronounced isotope effect of all elements, it shows (owing to its light mass) quite frequently its wave character, e.g., in tunneling or diffraction

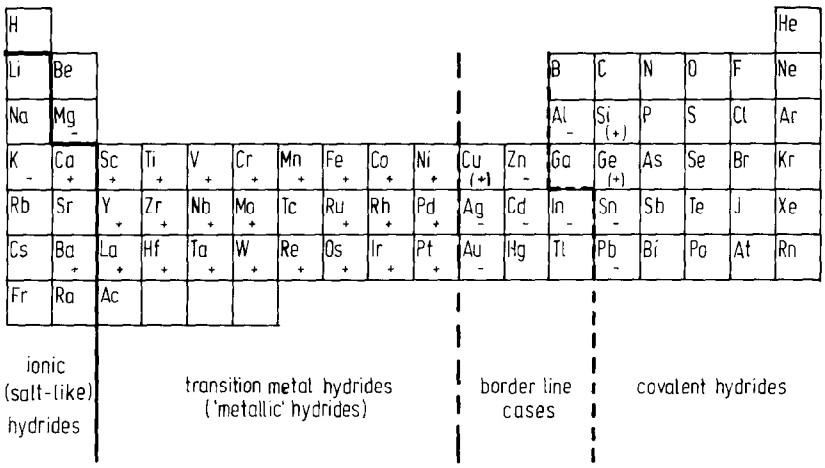


Fig. 2. Periodic table of the elements indicating those metals which chemisorb hydrogen with high sticking probability (+) and those which do not (−). Also marked are the various types of hydrides formed by the elements.

Table 2
Some selected physical properties of the H atom

Property	Unit	H	D	T
Molar mass	kg kmol^{-1}	1.00797	2.01410	3.0161
Absolute mass	kg	1.67261×10^{-27}	3.3423×10^{-27}	5.0049×10^{-27}
Atomic radius (from internuclear distance)	m	0.3707×10^{-10}		
Bohr radius	m	0.528×10^{-10}		
Velocity of electron in the innermost orbit	m s^{-1}	2.187×10^{6}		
Ground state configuration		$^2S_{1/2}$	$^2S_{1/2}$	$^2S_{1/2}$
First ionization potential I	eV	13.59		
Electron affinity A ($1s^2$)	eV	0.80		
Pauling's electronegativity		2.20		
Nuclear magnetic moment μ (in nuclear magnetons)		+2.79278	+0.85742	

processes. One very important consequence of this is its ability for the so-called "hydrogen-bonding" which may be of some significance also in the interaction of hydrogen-containing molecules with the surface. Moreover, hydrogen diffusion on surfaces may be governed to some extent by tunneling processes which lead to abnormal kinetic features.

The ability of H atoms to diffuse into the interior of various metals such as palladium or titanium is another unique property of this element. Under certain conditions hydrides may be formed readily. These often exhibit likewise remarkable properties, for example, a very high packing density of H atoms in the crystalline state. In the titanium dihydride TiH_2 the hydrogen nuclei are more densely packed than in solid hydrogen itself! This fact makes this kind of material (and some others) very advantageous for the storage of hydrogen.

Certainly the small physical size of a H atom is the major source for its enhanced solubility in some solids, yet there are several other factors which play a role as can be found in more detail in the original literature [25].

Quite remarkable is also the tendency of the hydrogen atom to form stable compounds with most of the metals, the so-called "hydrides", which may have ionic, covalent or metallic character, depending upon the position of the metal in the periodic table. This is, more or less, a consequence of the H atom to either donate an electron (to become a proton, H^+) or to accept an electron (to become a hydride ion, H^-). The electronegativity of the H atom is given by Pauling [26] to be 2.2, whereas most of the elements on the left-hand side of the periodic table have electronegativities smaller than or equal to 2 which simply means that the H atom will accept electron(s) from these elements to form essentially ionic hydrides. A very detailed review on the hydride subject was written by Gibb, and for further details the reader is referred to this article [27]. In fig. 2 we have also marked the various types of hydrides formed by the elements.

Due to its well-known electronic structure the H atom is sometimes regarded as a very good adsorbate to treat theoretically – surprisingly, this is rather not the case, and for reasons pointed out in this article the interaction of a H atom with a metal surface is much less understood than that of, e.g., carbon monoxide. One reason here is that the small H atom can get very close to a surface, it can be screened almost completely, and convergence problems may occur in numerical calculations. Theoretical reports on the H adsorption subject are numerous – there are classical papers by Newns [28], van der Avoird [29], Toya [13], Grimley [31] and Louie [32]. Quite recently, successful treatments of the H chemisorption were reported by Feibelman et al. [33] using self-consistent layer calculations and by Nørskov and Lang [34] on the basis of an effective medium theory, as well as by Daw and Baskes using the related concept of the embedded atom method (EAM) [35]. More details can be found in section 4.

2.3. Trapping and sticking

Before a thermal H_2 molecule can really enter any interaction potential well provided by a surface and stay therein for longer than just one vibrational period it has to dissipate at least part of its initial kinetic energy. In order to understand the various possible physical phenomena during the impact process we accompany a hydrogen molecule on its way to the surface. At 300 K, a H_2 molecule has a mean thermal velocity of ~ 1775 m s^{-1}; if its motion is directed perpendicular to the surface, it approaches the surface at a rate of approximately 18 Å per picosecond before it "feels" the van der Waals interaction potential. There are now various "reaction channels":

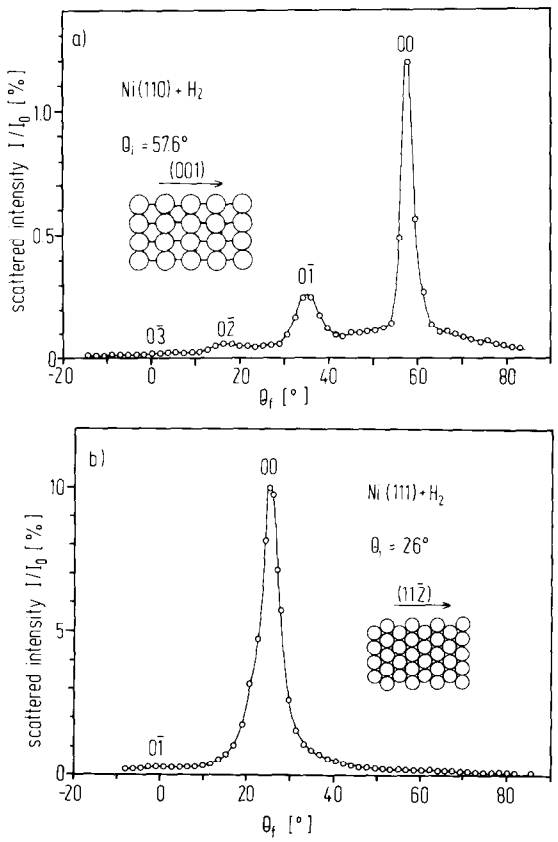


Fig. 3. (a) Angular distribution of H_2 molecules scattered elastically and direct-inelastically from a Ni(110) surface along the (001) direction. Angle of incident beam $\Theta_i = 57.6°$, surface temperature $T_s = 400$ K. (b) Same as (a) except that a Ni(111) surface at $T_s = 420$ K, $\Theta_i = 26°$ and (11$\bar{2}$) azimuth was used. Note the absence of diffraction features! After Robota et al. [39]

(i) The molecule may be reflected directly, that is, specularly, without any exchange of energy. In this so-called "elastic" scattering path the molecule obeys the rules for conservation of energy and momentum. However, the possibility for internal transformation of translational into rotational energy states has to be considered, particularly with non-symmetric molecules like HD as was, e.g., demonstrated by Cowin et al. for the Pt(111) [36] and by Wöll [37] for the Au(111) system. Translation–vibration excitations are unlikely because they require too much energy under the normal experimental conditions. Experimentally, the angular distribution of scattered molecules is a direct probe to discriminate elastically scattered from inelastically scattered molecules. [38]. A sharp specularly reflected beam is always evidence for predominantly elastic scattering. Figs. 3a and 3b display the angular distributions obtained by Robota et al. [39] for H_2 molecules scattered from Ni(110) and Ni(111) surfaces in the zero-coverage limit, for a given azimutal orientation. For Ni(110) (fig. 3a) another physical phenomenon, namely particle *diffraction*, can be observed: The angular distribution exhibits pronounced diffraction peaks, in addition to the intense specular beam. The position of the diffraction maxima agrees with the prediction from the known lattice constant and the initial velocity of the particle. With Ni(111), practically no diffraction features can be observed (fig. 3b) which fact simply reflects the much smaller corrugation of this densely packed metal surface. Both surfaces show only very low background scattering which means that those molecules which are reflected back undergo true elastic scattering. Using deuterium (D_2) instead of hydrogen for scattering changes the angular distributions quite markedly: as figs. 4a and 4b demonstrate, there is now a much larger fraction of background intensity observed which tends to a maximum in the specular direction. This background intensity is due to the so-called "*direct-inelastic*" scattering which represents one other possible interaction channel:

(ii) This type of scattering arises from energy exchange processes between the impact molecule and the surface during a *single* collision event. This means that the molecule has suffered some exchange of energy and momentum with the surface prior to reflection. The process is described by the equations:

$$\boldsymbol{k}_i^2 = \boldsymbol{k}_f^2 + (2m/\hbar^2)\left[E(J_f) - E(J_i)\right], \tag{1a}$$

$$\boldsymbol{k}_i^{\parallel} = \boldsymbol{k}_f^{\parallel} + \boldsymbol{g}, \tag{1b}$$

with $\boldsymbol{k}_i$ and $\boldsymbol{k}_f$ being the initial and final momenta (with components $\boldsymbol{k}_i^{\parallel}$ and $\boldsymbol{k}_f^{\parallel}$, respectively, parallel to the surface) and $\boldsymbol{g}$ being a reciprocal lattice vector parallel to the surface. $J_{i,f}$ denote the respective rotational states, and eq. (1a) principally contains the possibility of translation $\rightarrow$ rotation transformation during collision with the surface. The energy exchange can occur through various channels: (a) electron–hole (e–h) pairs can be excited, that is, electrons from just below the Fermi level are excited to states above E_F, a process

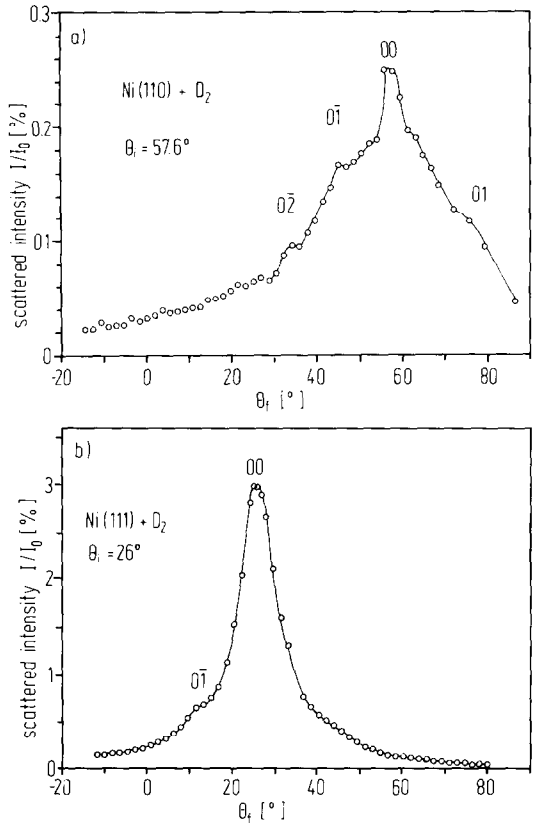


Fig. 4. (a) Angular distribution of D_2 molecules scattered elastically and direct-inelastically from a Ni(110) surface. (b) Same as (a) except that Ni(111) was used. Otherwise the conditions are exactly the same as in fig. 3. After Robota et al. [39].

also referred to as "electronic friction" [40]. (b) Excitation of phonons which couple to the resonance states of the incoming particle [41]. (c) Other elementary excitations in the solid (plasmons, magnons). Depending on the time scale of the energy dissipation, one has to distinguish between non-adiabatic and adiabatic processes (for which the Born–Oppenheimer approximation holds): The electronic excitations certainly belong to the class of non-adiabatic processes, whereas the phonon-coupling mechanism can be treated within the Born–Oppenheimer approximation. More detailed considerations on the quantum mechanics of the energy accommodation mechanism can be found in various theoretical publications [42–46]. Experimentally, one may distinguish between the accommodation mechanisms in the following way: Evidently, any mechanism based on electronic excitations should depend preferentially on the

electronic structure of the substrate, particularly on the density of states near the Fermi level, and not so much on the *mass* of the incoming particle. On the other hand, the excitations of phonons in the solid should be dependent on the relative mass of the impact molecule, i.e., the mass ratio m/M with m denoting the mass of the molecule and M the mass of the substrate atom, according to the relative energy transfer expression [41]:

$$\frac{\Delta E}{E_i} = 4\frac{m}{M}\left(1 + \frac{m}{M}\right)^2. \tag{2}$$

For the H_2 molecule interacting with typical transition metal surfaces (Ni, Pd, Pt) this mass ratio is rather small, and one would expect the phonon excitation mechanism not to be very effective. The electronic friction, however, should occur very effectively on those metals which have a high density of states near E_F. There have been a number of experiments to resolve this question. One simple way is to compare the scattering behavior of hydrogen H_2 and deuterium D_2 which have the same electronic structure but differ in mass by a factor two. If the sticking process is dominated by electronic excitations in the solid, H_2 and D_2 should exhibit a very similar sticking probability (or scattering behavior), whereas substantial differences would occur if the phonon mechanism provides the energy dissipation. The experimental answers on this are somewhat controversial. Ertl and coworkers [39] in their molecular beam study of H_2/Ni come to the conclusion that D_2 actually exhibits much more pronounced direct-inelastic scattering than H_2 which could not be rationalized if the e–h pair mechanism would dominate the energy exchange with the solid. El-Batanouny et al. [47] in their investigations on the H_2 uptake of Nb surfaces covered with Pd films speculated on evidence of e–h pair excitation: Only a certain Pd overlayer on a niobium substrate led to a substantial uptake of hydrogen, whereby the role of the Pd was assumed to essentially provide a high density of states near the Fermi level in order for the electron–hole pair excitation mechanism to occur. Further systematic experiments, particularly molecular beam measurements, are certainly required to ascertain the mechanism of the energy accommodation of the H_2 molecule. If the e–h pair mechanism is at all valid the adsorption of the *light* hydrogen molecule would be one of the few examples where this process could play a role.

(iii) Neither the elastic nor the direct inelastic channel can lead to a real adsorption situation in which the molecules remain on the surface for an appreciable period of time (preferentially in the dissociated state as atoms). We shall now consider these "dissociative" reaction paths. Actually, *two* different mechanisms leading to dissociation seem to exist, namely, the so-called "direct" process, in which the impact molecule has enough kinetic (translational) energy to overcome a possible activation barrier located in the entrance channel (cf. section 2.5.2), or the opposite "indirect" process in

which the molecule is kept as an entity for a certain amount of time in a molecular interaction potential from which it dissociates eventually. In the direct path, the only decisive parameter is the translational energy of the incoming particle: the surface temperature, T_s, has no effect on the dissociation and sticking probability. A good example here is represented by the H/Ni(111) system as pointed out by Robota et al. [39]. The other possibility which involves a so-called "molecular precursor state", is characterized by an activation barrier in the exit channel which basically consists of a stretching of the innermolecular H–H bond, for which vibrational excitation is necessary. Contrary to the direct path, the surface temperature does play a role in this precursor-mediated dissociation channel in that the phonon-assisted vibrational excitation of the trapped molecule is enhanced at higher T_s. Examples here are probably the adsorption of hydrogen on Pt(111) and Cu(111). In the precursor state the trapped molecule may have an appreciable residence time; it is able to move nearly freely across the surface until it finds an appropriate dissociation site.

(iv) The subsequent atomic adsorption ("chemisorption") then may be regarded as the final stage of energy accommodation. Particles leaving the surface again first recombine to H_2 molecules before they enter the gas phase. Actually there is no single experiment known which gave evidence for an *atomic* desorption. According to the detailed balancing concept the two dissociation reaction paths must show up also in the desorption path: The reversal of the indirect process (in which the molecules desorb from the precursor) will lead to a cosine distribution whilst *directly* desorbing molecules come off translationally "hot", i.e., in a sharper than cosine distribution. This was, among others, confirmed for the H/Ni(111) system, where the angular distribution of the desorbed molecules obeyed a $\cos^5$ power law [39,48,49].

Once trapped in the atomic chemisorption potential with its substantial depth of ~ 1 eV, the H atoms will only occasionally and only at elevated temperatures be able to leave this potential, to recombine and to desorb as a molecule. This just points to another view of the dynamics of adsorption, namely, the *residence time* of a given particle on the surface, τ.

All stages (i)–(iv) can also be expressed in terms of this residence time τ. Stage (i), the elastic scattering, occurs certainly within an extremely short time (an upper limit would be one vibrational period, $\tau = 10^{-13}$ s). The direct inelastic scattering event is probably as fast as this and an upper limit would perhaps be 10^{-12} s for this stage of "transient trapping". A molecule trapped in the mobile precursor state may stay on the surface for as long as 10^{-6} s before it either desorbs again or enters the chemisorption potential. In this latter case, the residence time ranges, depending on the depths of the potential energy well, from milliseconds up to hours (10^{-3}–10^{3} s), and commonly only these particles are regarded to have stuck on the surface. Of course, also those particles which have been trapped and dissociated via the direct channel must

be reckoned among this group. A somewhat better defined criterion for sticking has been given by Tully [52]: Thereafter all those particles have stuck which have dissipated more than their thermal energy kT to the surface.

While staying in the chemisorption potential, the H atoms can, of course, be vibrationally excited, and particularly with hydrogen this leads to a remarkable consequence, the so-called "quantum motion" [50,51]: For a proper description of the ground state and the vibrational excitation spectrum of H adsorbed on a periodic surface, the three-dimensional Schrödinger

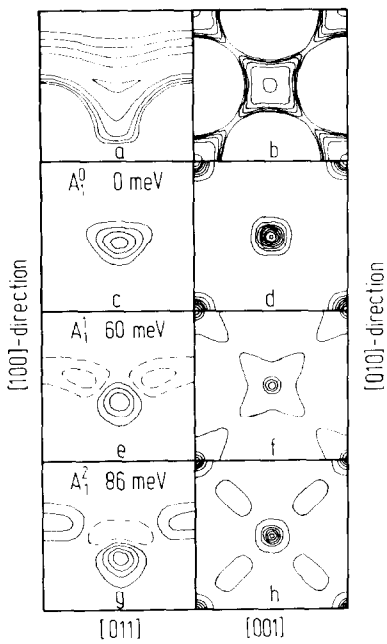


Fig. 5. Calculated potential (a, b), wave functions with A_1 symmetry (c, e, g) and density contours (d, f, h) for a hydrogen atom chemisorbed on a Ni(100) surface. The left panel shows the potential and wave functions in a *vertical* cut through the fourfold center position along the [110] direction (ΓM) where the potential has its minimum. The lengths of the cuts are the Ni nearest neighbor distances ($4.7a_0$) both in the parallel and the perpendicular directions. The right panel shows the potential and the charge densities, respectively, in cuts *parallel* to the surface. In detail, (b) is a cut through the absolute minimum of the potential. The ground state is shown in panels (c) and (d), the first excited state in (e) and (f) and the second excited state in (g) and (h). Note that – while the ground state is well localized – the first and second excited states are rather delocalized thus enabling parallel motion to the surface in a band. In the right panel the cuts are one Ni lattice constant ($6.65a_0$) in each direction. The lowest energy contour is -2.70 eV and the spacing between the adjacent contours is 0.15 and 0.22 eV for (a) and (b), respectively. The wave function and the density contours each have a constant spacing, dashed lines indicate negative contours. All wave functions are evaluated at the Γ point of the surface Brillouin zone. After Puska et al. [51].

equation must be solved. This was performed for H on Ni(100) by Puska et al. [51], and the results are reproduced in fig. 5. Actually, the H atoms can be regarded to be highly delocalized in energy bands which are separated from one another by ~ 50 meV and even show some dispersion. Particularly the excited states have a considerable width and can account for the high mobility of the adsorbed hydrogen on that surface.

There are various experimental methods to determine the absolute sticking probability, s_0, which is defined as the ratio between the sticking and actually impinging H_2 molecules (whereby the sticking portion can, e.g., be measured by a desorption experiment), for the zero coverage limit.

It is quite remarkable that in many cases s_0 for hydrogen is substantially smaller than 1, often in the range between 0.01 and 0.1 (this observation holds also for other diatomic homonuclear molecules such as N_2 or O_2). For the (predominantly associative) adsorption of carbon monoxide, on the other hand, sticking coefficients close to 1 are the rule. One possible reason for this behavior can be sought in the existence of (small) activation energy barriers for adsorption as in the case of Ni(111)/H [39]. There are several recent thorough re-investigations of this adsorption system using angular dependent thermal desorption [48], isotope exchange [53] and molecular beam measurements [54,55]. It turns out from these studies that the two-state character of the H adsorption (leading to a β_2 and a β_1 state) is confirmed but that the sticking into these states is thermally activated, whereby the activation energy of the low energy β_1 state is strongly influenced by surface *defects* (which provide sites with vanishing activation energy for dissociation [55]). Recent work on the same system by Russell et al. [53] reaches basically quite similar results but concludes that not a site specific but rather an interaction and coverage dependent activation barrier for desorption from these states exists. Since these latter investigations consider high-coverage (multi-particle) effects we shall not expand on them here.

Hints for a small activation barrier for adsorption were also obtained for the other low-index Ni plane for which a small sticking probability was reported [56], namely, the Ni(100) surface. Hamza and Madix [57] performed a molecular beam study using H_2 beams of different kinetic energy and found on the basis of isotope effects in the sticking probability (H_2 adsorbs faster than D_2 because the lighter H_2 molecule can tunnel more efficiently through the barrier), an activation energy of adsorption of about 5 kJ/mol. A theoretical work by Jackson and Metiu [58] concludes that the H_2 molecule dissociates over on-top sites, whereby a translational activation is required. The barrier is calculated to have a height of ~ 7 kJ/mol and a width of ~ 0.6 Å, in good agreement with Hamza and Madix's results [57].

Even more exciting phenomena have been reported during the past decade for the other frequently studied adsorption system H on Pt(111). They point to a pronounced influence of surface defects (predominantly steps) on the

probability of dissociation. We want to mention here the relevant investigations by Gale, Salmeron and Somorjai [59]. They used a deliberately stepped Pt(111) surface in a molecular beam study in which a beam of D_2 was directed on a surface covered with preadsorbed hydrogen under different azimuthal angles. It was found that the rate of the H_2–D_2 exchange reaction (cf. eq. (9)) depended sensitively on this angle: If the beam was directed towards the steps, a high rate was obtained, but it dropped strongly, if the beam was adjusted parallel to the steps. Very much in line with this are more recent results obtained in the group of Comsa [60,61]. Utilizing thermal atom scattering techniques (TAES) they concentrated on the long ongoing question about the origin of the very small sticking coefficient reported for H_2 on Pt(111) which ranges between 0.01 and 0.1. They found that this value of s_0 depended strongly on the kind and number of crystallographical defects of the Pt(111) surface in that extremely carefully prepared Pt(111) surfaces revealed s_0 values as small as 10^{-4}. Also, the coverage dependence of the sticking probability was affected by the defect concentration in that ultimately defect-free Pt(111) samples revealed a constant function s (coverage) up to 60% of the saturation coverage at 25 K [61]. These data can be quantitatively understood in terms of a model in which the H_2 dissociation occurs exclusively at step sites. These "dissociation sites" are reached after trapping of the H_2 molecule in a mobile molecular precursor state on the flat terraces and subsequent migration to the steps or by direct collision with step sites from the gas phase. Molecular beam measurements by Karner et al. [62] on a Ni(997) surface showed likewise non-activated adsorption on the step sites, but activated adsorption on the flat terraces which again can be traced back to the different dissociation probability during the sticking process.

In view of these recent experiments doubt must be raised on many s_0 and $s(\Theta)$ measurements which were performed with surfaces whose crystallographical state was not carefully determined, i.e., which exhibited a residual concentration of steps, kinks, dislocations, etc. Unfortunately, these defects are very difficult to quantify. High-resolution LEED [63] or LEED microscopy [64] as well as tunneling spectroscopy [65] or high-resolution electron microscopy [66] could be employed to tackle this problem. Particularly on *smooth* crystal planes the above listed defects are crucial for the H_2 dissociation process. Among these are the (111) faces of the fcc metals (to a lesser extent also the (100) faces), the (110) faces of the bcc metals and the (0001) faces of the hcp metals. On all of these surfaces there is a general trend that small but not extremely small sticking values have been reported in the past, and the existence of (small) activation barriers for dissociation is likely.

The situation is entirely different for crystallographically more "open" surfaces, namely, the (110) faces of the fcc the $(10\bar{1}0)$ faces of the hcp and the (211) faces of the bcc system. These and all higher-index surfaces can be regarded as consisting of a sequence of steps of higher or lower density.

Taking all the experimental sticking data for H_2 together it appears that on most, if not all of these surfaces sticking coefficients *close to unity are the rule*. Examples are Ni(110) [21,67], Pd(110) [68], Co(10$\bar{1}$0) [69], Rh(110) [70], Ru(10$\bar{1}$0) [71], or W(111) [72]. Also, there are no anomalies with regard to the angular distribution of desorbing H_2 molecules as the angle-resolved thermal desorption study by Steinrück et al. [48] and by Robota et al. [39] demonstrates – in all these cases a simple cosine distribution of the emitted H_2 molecules is observed. This means that the desorbing molecules are in thermal equilibrium with the surface, unlike molecules desorbing, e.g., from Ni(111) whose rate components are peaked normal to the surface up to $\cos^5\Theta$ and which are therefore referred to as translationally "hot". As pointed out before, the concept of "detailed balancing" (for more details see, e.g., refs. [73,74]) would just predict this behavior. Only those particles come off translationally "hot", i.e., with an excess kinetic energy, which have passed an activation barrier for adsorption in the entrance channel, on their way from the gas phase into the adsorbed state, and vice versa. Again, there exist several interesting investigations by Comsa et al. [75,76] who studied the velocity distribution of D_2 molecules desorbing from a Cu(100) or a Pd(100) surface by means of a time-of-flight technique and found also that part of the desorbing molecules showed fluxes that were strongly peaked normal to the surface. However, for the particular case of Pd there is no activation energy for adsorption of hydrogen as a detailed study of the H/Pd(100) system revealed [77] and can therefore not explain the observed effects. Instead, the fast molecules are absorbed in the Pd bulk, travel to the surface upon heating and recombine there directly, without equilibrating in the chemisorption well. Moreover, sulfur impurities played an important role in the penetration and recombination mechanism. This is just an example to demonstrate that care has to be taken when interpreting angular distributions of desorbing hydrogen.

Turning back to more "normal" sticking phenomena, we have, from the data body presently available, compiled a table (table 3) which gives a variety of initial sticking probabilities for H_2 adsorption on different surfaces of metals. A collection of such data can be found in the article by Morris et al. [78].

2.4. Physisorption

We now consider a hydrogen molecule in its ground state, infinitely far away from the surface which is assumed to be ideal, that is, exhibits a sinusoidal lateral interaction potential in the x–y direction. The H_2 molecule now approaches the surface along the reaction coordinate, z, perpendicular to the surface plane. The situation can very conveniently be described in terms of the well-known one-dimensional Lennard-Jones potential energy diagram $U(z)$ which is shown in fig. 6a [79]. As was pointed out before, there will be only

Table 3
Some selected values of initial sticking probabilities, s_0, for H_2 on metal surfaces for dissociative adsorption (single crystal data only)

Crystal face (*hkl*)	Temperature (K)	s_0	Remarks	References
Fe(100)	300	0.03		J. Benziger & R.J. Madix, Surface Sci. 96 (1980) 114
(110)	150	0.16		[159]
Co(0001)	300	0.045		M.E. Bridge, C.M. Comrie & R.M. Lambert, J. Catalysis 58 (1979) 28
$(10\bar{1}0)$	120	1.0	TDS	[69]
Ni(100)	120	0.06–0.1	TDS	[56]
			Activated adsorption	[57]
(111)	120	0.1	TDS	[50]
	120	~1	Activated adsorption	[39]
	140	0.05		[67]
(110)	120	0.96	TDS	[21,67]
			MB	
	300	0.4	TDS	A. Winkler, personal communication
Cu(100)			Activated adsorption	[96]
(111)		< 0.1	MB	[97]
(110)				
Mo(100)	300	0.7		R. Chappell & D.O. Hayward, J. Vacuum Sci. Technol. 9 (1972) 1092
(110)	300	0.35		
(111)	300	0.7		
Ru(0001)	100	0.4		L.R. Danielsson et al., Surface Sci. 71 (1978) 599
	100	0.25		[135]
		0.3		[113]
$(10\bar{1}0)$	100	1.0	TDS	[71]
Rh(110)	85	1.0	TDS	[70]
(111)	100	0.65	TDS	[243]
Pd(100)	120	0.5	TDS	[77]
(110)	120	0.7	TDS	[68]
(111)	80	0.5	LEED, TDS	[266]

Ta(100)	300	0.04		L. Johnson, M.J. Dresser & E.E. Donaldson, J. Vacuum Sci. Technol. 9 (1972) 852
W(100)	300	0.65		R.M. Prince & G.R. Floyd, Surface Sci. 74 (1978) 342
	300	0.51	Calibrated capillary	[204,251]
	300	0.6		D.H. Winicur et al., Surface Sci. 109 (1981) 263
	300	0.56		B.D. Barford & R.R. Rye, J. Chem. Phys. 60 (1974) 1046
(110)	300	0.22		B.D. Barford & R.R. Rye, J. Chem. Phys. 60 (1974) 1046
(111)	300	1		
(211)	300	0.57		
Ir(111)	100	7×10^{-3}		[244]
(110)	130	1	β_2 state	[245]
	300	1		[245]
	130	0.007	β_1 state	[245]
Pt(100)-hex	78	0.15		[272]
	300	0.17		[267]
	300	0.07		[270]
(110)-1×2	78	0.2		[272]
	300	0.31		R.W. McCabe & L.D. Schmidt, Surface Sci. 60 (1976) 85
	300	0.33		R. Chappell & D.O. Hayward, J. Vacuum Sci. Technol. 9 (1972) 1052
(111)	150	0.1	Activated adsorption	[158]
	300	0.07	MB	[59]
	78	0.1		[272]
	300	0.016		J.A. Schwarz, R.S. Polizzotti & J.J. Burton, Surface Sci. 67 (1977) 10
	300	0.2		[272]
(210)	78	0.4		R.W. McCabe & L.D. Schmidt, in: Proc. 7th Intern. Vacuum Congr. Vol. 2 (1977), p. 120
(997)	120	0.34	TDS	[131]

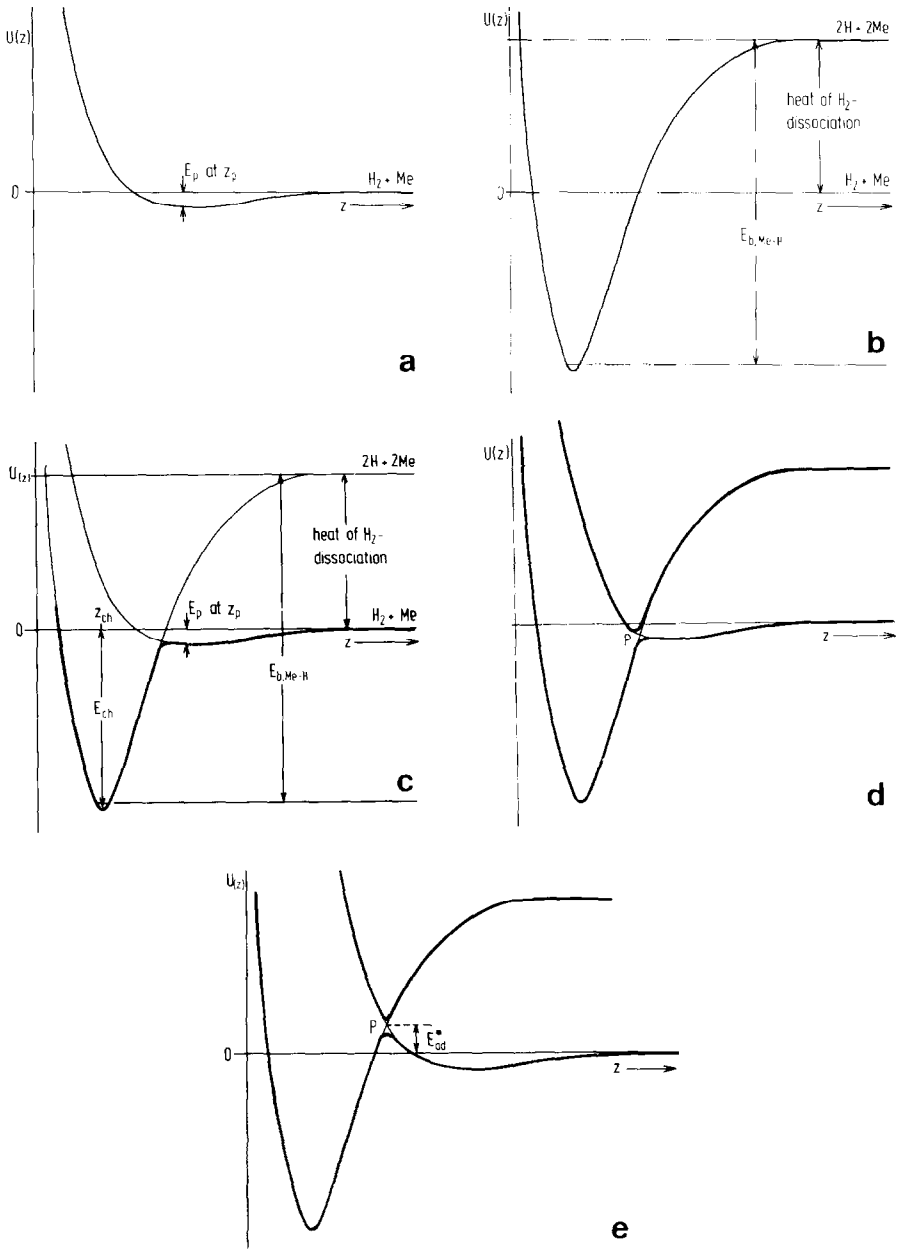


Fig. 6. Types of interaction potential $U(z)$ of H_2 and H with a solid surface in perpendicular direction according to Lennard-Jones. (a) Physisorptive (molecular) interaction. (b) Interaction of pre-dissociated hydrogen with the surface. (c) Superposition of curves (a) and (b) to illustrate the mechanism of spontaneous dissociation. Resulting potential energy curves for: (d) non-activated adsorption (P below $E = 0$), (e) activated adsorption (P above $E = 0$).

small attractive interaction forces of van der Waals origin as the H_2 molecule is sufficiently close to the surface giving rise to a shallow interaction potential (depth E_{phys}) at a distance z_{phys} from the surface. At present, there are not too many experimental verifications of this co-called "physisorptive" interaction, but a variety of theoretical predictions of this process [80–82]. The respective binding energies of the H_2 molecule to the surface (here, *any* solid surface can be visualized) are very small; they range between 3.5 and 15 kJ/mol, depending on the system. For the experiment this means that the respective investigations have to be carried out at liquid helium temperatures. Reports on the physisorption of H_2 on graphite [83,84] or on copper [85,86] and silver [87] are mentioned in this context; also, the interaction potential depths for HD (D_2) on Au(111) have been determined recently by molecular beam scattering experiments [37]. This method is most commonly employed for the determination of H_2–alkali halide interaction potentials, e.g., for the system $NaCl/H_2$ [88].

Quite frequently the question comes up as to the possibility of *molecular* hydrogen (H_2) adsorption at temperatures well above 20 K (which is regarded as the highest possible temperature at which true physisorption of hydrogen can occur). Reports in the past in which it was claimed that molecular hydrogen had been detected between 100 and 300 K mostly turned out to suffer from H_2O impurities, thermal gradients and spurious desorption in temperature programmed thermal desorption spectroscopy.

However, there was a report quite recently by Mårtensson et al. [89] in which molecular H_2 adsorption was found on edge sites of a stepped Ni(100) ($\triangleq$ Ni(510)) surface by means of vibrational loss (EELS) spectroscopy, at a temperature around 100 K. A broad weak vibrational band (ν_{H-H}) appeared after saturation of the atomic H states, between 350 and 450 meV. It is stated that this molecular adsorption can only occur after the chemisorption channel (see section 2.5) is blocked by a saturated adsorbed layer. An electronic model is proposed to reconcile both the molecular "chemisorption" and the location of the adsorption site at the edges of the terraces.

2.5. Chemisorption

A completely different interaction pattern is observed, if the H_2 molecule is dissociated into the atoms prior to the interaction. Since the dissociation energy has to be spent on the system first, the potential energy curve in fig. 6b starts 435 kJ *above* the energy zero level on the right-hand side of the diagram. Now, the 1s wave functions of the H atoms tend to form *chemical* bonds with the surface material which in many cases will lead to the situation depicted in fig. 6b: Strong attractive interaction forces lead to a pronounced lowering of the potential energy of the system as the H atoms are brought close to the surface. The depths of the potential wells range between 500 and 600 kJ/mol

and simply represent the energy content of a single H–surface atom bond. Note that this quantity is gained *twice* since one H_2 molecule dissociates into two atoms. Again, experiments are scarce in which this interaction energy is determined *directly* – the problem is that a beam of H *atoms* has to be prepared which either requires a very hot tungsten filament [90] or a high-frequency discharge [91] in order to supply the H_2 gas atmosphere with sufficient energy. Recent experiments in which silicon surfaces were subjected to interaction with a H atom beam showed a strong and rapid chemical attack of the H atoms to form silicon hydride (silane) compounds [92,93]. Although there are no experiments known at present it is believed that the interaction of H atoms with, e.g., alkali or alkaline earth metals will immediately lead to the formation of the respective hydrides, whereby these processes are highly exothermic. On Ni(110) single crystal surfaces the interaction with a H atom beam has been reported to reveal the existence of excited interaction complexes, the so-called "hot-precursor states" [94] which have also been treated theoretically.

2.5.1. The spontaneous (non-activated) adsorption

Quite frequently the interaction of a non-excited H_2 molecule with solid surfaces will lead to spontaneous dissociation and to the release of energy, the so-called "heat of adsorption". Fig. 6c illustrates this: The superposition of the physisorptive and the atomic interaction potential energy curves yields a crossover point P which may lie above or below the energy zero level. Accordingly, we obtain two principal potential energy curves demonstrated in figs. 6d and 6e, respectively. In the case where the point of intersection is below $E = 0$, spontaneous dissociation will occur at P without any need for additional energy; the H atoms then readily enter the so-called "chemisorption" potential energy well and the heat of chemisorption, E_{ad}, is released and can be measured in an experiment quite easily. This is the "non-activated" adsorption of hydrogen, which is frequently observed in the interaction of H_2 with high-index planes of transition metal surfaces such as Ni, Pd or Pt. The well-known activation of hydrogen over Raney-nickel or platinum black are examples of spontaneous dissociation of H_2 molecules. The energy balance of this exothermic process is as follows: One gains the heat of adsorption, E_{ad}, and twice the energy of a H–metal bond, E_{Me-H}, whereas the heat of dissociation, E_{diss}, must be spent on the system. Accordingly, the experimental determination of the heat of adsorption allows the Me–H bond energy to be calculated:

$$E_{Me-H} = \tfrac{1}{2}(E_{diss} + E_{ad}). \tag{3}$$

There are numerous examples where the metal–H surface bond energies have been determined – they can be compared with the dissociation energies of the bulk metal hydrides in order to get an idea about the strength and nature of

the H chemisorptive bond. Some selected examples for E_{ad} and E_{Me-H} are given in table 4. This table also contains the ground state dissociation energies, $D_{0,Me-H}$, of the respective diatomic metal-hydride molecules as far as they are available from spectroscopic measurements. It is immediately realized that the metal–H binding energies as they arise from the chemisorption are of similar magnitude as $D_{0,Me-H}$. The reader is also referred to a theoretical article by Nordlander et al. [99] who performed extensive calculations of the ground state properties of H chemisorbed on transitional metal surfaces. Their results for the chemisorption energies on all the 3d, 4d and 5d metals are in remarkable agreement with the experiment.

Interestingly, the differences between various systems are not too pronounced – the values range around 2.6 eV for the H–Me binding energy, although quite large differences exist in the heats of adsorption. This points to a fairly similar binding mechanism of hydrogen to many metals as has also been confirmed theoretically [34]. We note here that in all these calculations the binding energy, heat of adsorption, etc. refers to the *zero coverage limit* for adsorption of the very first H atom or H_2 molecule on an *ideal* surface plane. Spurious effects of defects or higher adsorbate concentrations are *not* considered in most of the theories although they may be very important practically. In section 3.2 which is concerned with the energy of adsorption we shall expand more on these effects.

2.5.2. The activated adsorption

The other case of the interaction (fig. 6e) describes the so-called "activated" adsorption with the intersection of the two potential energy curves occurring *above* $E = 0$. This means that, in order for the dissociation to take place, an extra portion of energy has to be supplied to the system, the so-called "activation energy", E^*. In the recent past, chemisorption systems which were formerly believed to belong to the class of non-activated systems seem to require at least a small activation energy as indicated by unusual small sticking probabilities for molecular hydrogen; a typical example is the H/Ni(111) interaction system [39,50]. Known for a fairly long time is the system Cu/H where the unusual temperature dependence of the sticking probability pointed to the existence of an activation barrier of a few kcal/mol [96]. This was confirmed later by molecular beam experiments and trajectory calculations [97].

Most of the early work on hydrogen interaction with Cu was performed with powders or polycrystalline evaporated films [96a,b]. Interestingly enough, there is a wealth of data material as far as the interaction of O_2 or CO with Cu single crystals is concerned, but actually very few reports are available for H_2 interaction. One of these rare investigations was done by Pritchard and his group [96c] in 1975 in which the (100), (111), (110) and (211) Cu surfaces were exposed to *atomic* hydrogen generated by a hot W filament in front of the

Table 4
Initial heats of adsorption of hydrogen, $E_{ad,0}$, hydrogen–metal binding energies E_{Me-H} as derived from $E_{ad,0}$, bond dissociation energies of the diatomic hydride molecules, $D_{0,Me-H}$, and bond lengths, r_0, of diatomic hydride molecules [a)]

Surface	$E_{ad,0}$ (kJ mol^{-1})	E_{Me-H} (kJ mol^{-1})	$D_{0,Me-H}$ (kJ mol^{-1})	r_0 (Å)	References
Li			241.3	2.5963	
Be			212.4	1.321	
Na			212.4	3.208	
Mg			240.4	1.664	
Al			295.4	1.76	
Si				1.519	
K			179.5	3.61	
Ca			164.1	1.89	
Mn			231.7	1.64	
Fe(110)	109	271			[159]
(100)	100	265			[159]
(111)	88	260			[159]
Co(0001)	67	250		1.593	M.E. Bridge, C.M. Comrie & R.M. Lambert, J. Catalysis 58 (1979) 28
$(10\bar{1}0)$	70	251			[69]
Ni(100)	95	264	299.2	1.665	[155]
(110)	90	261			[155]
(111)	95	264			[155]
Cu(311)	39	236	279.0	1.482	J. Pritchard, T. Catterick & A.K. Gupta, Surface Sci. 53 (1975) 1
Zn			81.1	2.24	
Rb			183.4	3.708	
Sr			162.2	2.054	
Nb(100)	111	272			D.I. Hagen & F.F. Donaldson, Surface Sci. 45 (1974) 61
Mo(110)	142	287			S.M. Ko & L.D. Schmidt, Surface Sci. 42 (1974) 508

Ru(0001)	80	256			[135]
	120	276			[113]
Rh(111)	78	255			[243]
(110)	77	255			[70]
Pd(100)	99	266			[77]
(111)	88	260			[106]
(110)	102	267			[106]
Ag(111)	~15	241	241.3	1.641	[87]
Cd			68.0	1.665	
In			239.4	1.761	
Sn			308.9	1.853	
Cs			183.4	3.86	
Ba			175.7	2.18	
W(100)	134	283			[2,255,268]
(110)	138	285			[2,255,268]
(111)	155	294			[2,255,268]
Ir(110)	96.2	264			[245]
(111)	53	243			[244]
Pt(100-hex	62.8	247.5			[267]
(110)-hex	54.8	243			R.W. McCabe & L.D. Schmidt, in: Proc. 7th Intern. Vacuum Congr. Vol. 2 (1977) p. 120
(111)	42	237	347		[158]
	77	255			[23]
	75	254			K. Wangemann, J. Rüstig & K. Christmann, to be published
Au			299.2	1.709	
Hg			36.3	1.89	
Tl			210.4	2.319	
Pb			96.5	2.60	
Bi			260.6	2.08	

[a] After ref. [107].

surface. Thereafter, the change of the work function due to the H uptake was measured. With the (100) and (211) surfaces an increase of $\Delta\varphi$ was found, whereas the (100) and (110) surfaces showed a final decrease of the work function, after an intermediate maximum. Exposure of the low-index faces (100), (110) and (111) to *molecular* hydrogen at 300 K did *not* produce any measurable $\Delta\varphi$ even through H_2 pressures up to several Torrs were applied. However, the higher-index faces (211), (311) and (755) did give rise to work function changes under similar conditions thus confirming the occurrence of dissociative H chemisorption. Among others, H_2 adsorption isotherms were measured for the Cu(311) surface, and heats of adsorption between 39 kJ/mol at low and 41.5 kJ/mol at medium H coverages were derived. Clearly, the apparent inertness of the low-index Cu faces points to a noticeable activation barrier for H adsorption which was confirmed and quantified in the already

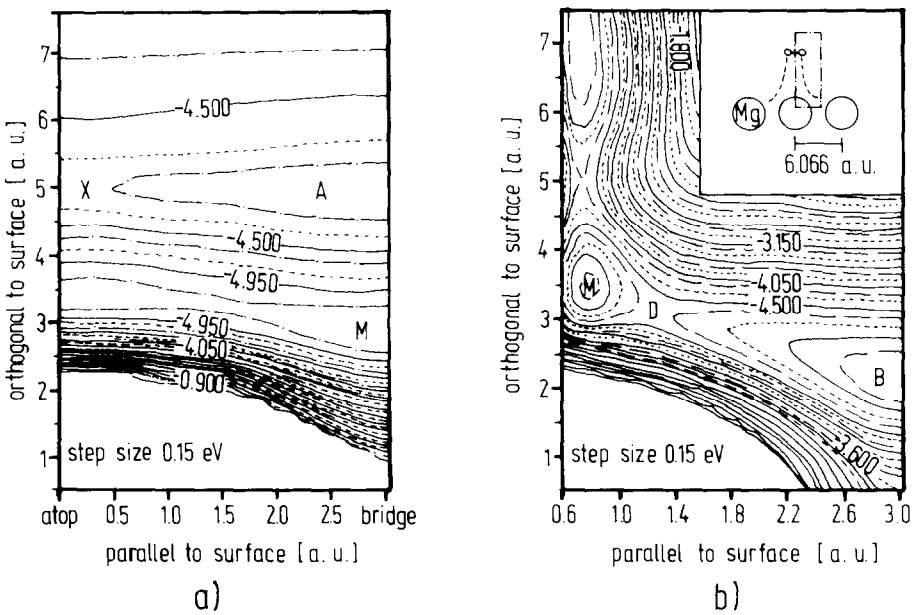


Fig. 7. Calculation of the dissociation and bonding geometry of a H_2 molecule interacting with a Mg(0001) surface. Panel (a) shows the equipotential energy contours for a H_2 molecule *parallel* to the surface and a bond length of 1.5 a.u. Panel (b) shows the same situation for a H_2 molecule dissociating over an atop site into the bridge sites. The energies are given in eV relative to those of the free atoms. The molecular adsorption energy in M is only 0.4 eV relative to the free molecule. The distance from the surface is measured from the first atomic layer and the distance parallel to the surface (which in panel (b) is equal to one-half of the H–H separation) is measured from the atop site towards the bridge site. The coordinate of panel (a) gives the midpoint between two protons. After Nørskov et al. [98].

mentioned molecular beam experiments by Stickney, Cardillo and coworkers. In their study [97a] using a mixed supersonic H_2 and D_2 beam with variable kinetic energy from 6.7 to 45 kJ/mol the probability for dissociative adsorption (by monitoring the species HD) was measured as a function of the translational energy of the incoming molecules for Cu(110), Cu(310) and Cu(100) surfaces. In all three cases the HD formation started at 10 to 20 kJ/mol beam energy which corresponds to an activation barrier for adsorption of about that height.

The depth of the molecular interaction potential well for H_2/Cu surfaces was later investigated by Lapujoulade and his group by means of molecular beam scattering experiments [94].

To explain the origin of the activation barrier is a fairly complex problem and has developed to a subject of extensive study in the dynamics of adsorption. It may be reduced to the question *how* the H_2 molecule has to be activated in order to dissociate.

Following the treatment by Nørskov et al. [98], the trajectory for the dissociation of a H_2 molecule on a magnesium surface can be divided into four steps (fig. 7):

(i) The incoming molecule meets a first barrier, the height of which depends on the spatial orientation of the molecule and, more importantly, on the local geometry of the surface where the molecule is supposed to arrive. According to Nørskov, the H_2 molecule is oriented *parallel* to the surface and finds the lowest activation barrier over the atop position.

(ii) Having passed this barrier the molecule is trapped in a weak potential, with a bond length only slightly larger than in the free molecule. It is almost free to move along the surface (the activation energy for surface migration is negligible). This state of the H_2 molecule is typically known as the "precursor" state for chemisorption which has been deduced only indirectly from measurements of the coverage dependence of the sticking probability, e.g., with the system Pd(100)/H [77]. Detailed considerations of the adsorption kinetics suggest a differentiation between the so-called "intrinsic" and an "extrinsic" precursor. The intrinsic state exists at empty surface sites only whilst the extrinsic precursor is present over sites filled with chemisorbed species.

(iii) There is a second activation barrier which hinders the trapped molecule to spontaneously dissociate into the atoms: No longer is the atop site most favorable but the bridge position; the corresponding activation barrier is, however, one fifth of that of the first barrier.

(iv) On that bridge site the molecule easily dissociates into the atoms which are then chemically bound to adjacent center sites.

However, the actual location of the second activation barrier is determined by the mechanism of the dissociation as mentioned already in section 2.3: Either a "direct" or an indirect (precursor mediated) reaction path can lead to dissociation. From Nørskov's calculation this latter mechanism seems to hold

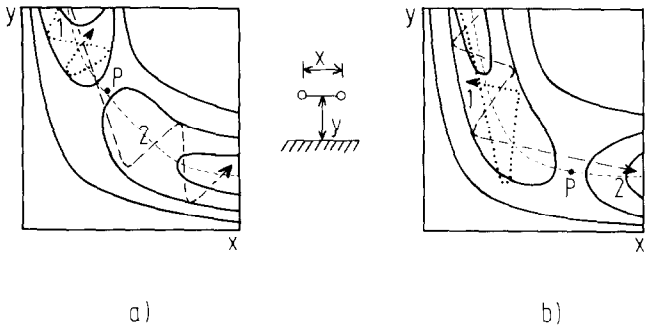


Fig. 8. Potential energy surface in a two-dimensional diagram illustrating the interaction of the H_2 molecule with a solid surface. (a) Activation barrier for dissociation (saddle point P) in the entrance channel, (b) in the exit channel. Several possible trajectories are drawn: (1) indicating reflection without chemisorption, (2) indicating dissociative adsorption.

for the Mg(0001)/H system. It may be very instructive in this context to transfer a classification to the dissociation process which was originally adapted by Polanyi and Schreiber [100] for homogeneous gas phase reactions. The situation is illustrated in figs. 8a and 8b: If the barrier is located in the entrance (reactant) channel (the molecule is still fairly far away when the activation occurs) (the coordinate x describes the internuclear separation of the H_2 molecule, the coordinate y the distance of the molecule from the surface) a translational excitation of the H_2 molecule would be favorable for the dissociation to occur *directly*, without a noticeable residence time in a molecular (precursor) state. In other words, the impinging molecule would automatically transfer its kinetic energy normal to the surface into a H–H vibration parallel to the surface. This is illustrated in fig. 8a: As the coordinate y becomes only slightly smaller, the internuclear separation x increases markedly, which means that the molecule has undergone dissociation. On the exit side, it exists already as atoms. An example of this kind seems to be provided by the H/Ni(111) system where only translationally excited molecules dissociate effectively [39]. On the other hand, "*indirect*" dissociation may take place, whereby the incoming particle is first molecularly adsorbed in the precursor state as described before; it cannot easily transfer its kinetic energy into an internal vibration. As fig. 8b illustrates the activation barrier is now located in the exit (product) channel, that is, closer to the surface: Up to fairly short distances y the molecule remains essentially intact (almost constant internuclear distance x) until it finally surmounts P. In this case, vibrational excitation is crucial and is mainly provided by accumulation of vibrational energy in the innermolecular bond. This is certainly facilitated by coupling of the molecular vibrations to the surface phonons. Examples for this indirect process in which a fairly long-lived molecular precursor is essential, have been

reported for the systems N_2/Fe(111) and O_2/Pt(111) [101]. A comprehensive consideration to the dissociation problem was also given in a theoretical work by Harris and Andersson [102]. These authors proposed a quantum-chemical explanation of the activation energy for dissociation and could, within their model, also rationalize the difference in dissociation activity between d-electron and sp-electron metals. As a H_2 molecule is brought to a metal surface the wave functions of the metal and those of the H_2 molecule interact with the consequence that the metal orbitals must orthogonalize with respect to the occupied H_2 σ_g wave function, a phenomenon referred to as Pauli repulsion. This shifts the energies of the electrons of the metal bands upwards thus causing an appreciable energy barrier for dissociation. This is particularly so for the s wave functions with their considerable extension in space. On the other hand, the typical transition metal possesses d-electron holes, and the unfilled d-band offers an alternative route: No longer is the energy-consuming orthogonalization necessary, but a partial filling of the d-holes by H_2 bonding electrons bypasses the Pauli repulsion. Accordingly, no or only small activation barriers for dissociation are built up on transition metal surfaces.

2.6. *Isotope and quantum-statistical effects*

We now suppose that the H atom has found its equilibrium position on the solid surface and that the temperature is low enough so that it is adsorbed in its ground state, viz., localized in the minimum of the potential well. In other words, the H atom has *occupied an adsorption site* on the respective surface. Before we consider the actual local geometry of that site (which, of course, reflects the net electronic interaction of the H atom with the surrounding atoms of the solid) a few remarks are worthwhile about the possible isotope and quantum-statistical effects in the (equilibrium) interaction energy which is equivalent to the heat of adsorption, E_{ad}. If they are at all measurable, isotope effects should be most prominent with hydrogen (deuterium, tritium) as an adsorbate.

As a matter of fact, the dissociation energy of deuterium, D_2, is somewhat larger than that of the H_2 molecule. We have [103]

$$\Delta E_{diss} = \Delta E^{H}_{diss} - \Delta E^{D}_{diss} \tag{4}$$

$= 431.37 - 439.5$ kJ/mol $= -8.1$ kJ/mol. This means that in the first instance the whole interaction curve shown in fig. 6b is shifted by ΔE_{diss} to positive interaction energies, if one considers D_2 instead of H_2. This is illustrated in fig. 9. On the other hand, there is a lower zero point energy associated with the surface bond of the D atom, owing to its greater mass, when compared to the H atom surface bond. If we denote the respective zero point energy difference as

$$\Delta E_0 = E_0^{H} - E_0^{D} \tag{5}$$

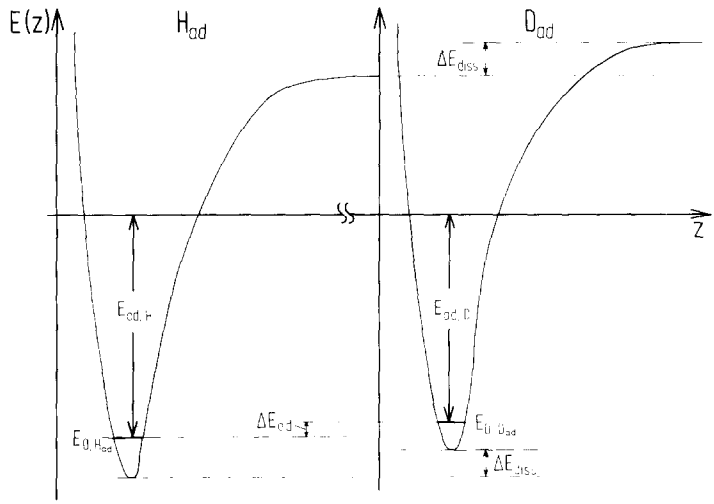


Fig. 9. One-dimensional potential energy diagram for dissociative H_2 (D_2) adsorption. Left-hand side: $E(z)$ (H_{ad}), right-hand side $E(z)$ (D_{ad}).

we obtain for the difference in the heat of adsorption between H_2 and D_2, $\Delta E_{ad} = E_{ad}^{H} - E_{ad}^{D}$, according to fig. 9 the following relation:

$$\Delta E_{ad} = -\Delta E_0 - \Delta E_{diss}. \tag{6}$$

This means that the isotope effects in the heat of adsorption are usually of the order of a few kJ/mol as hydrogen is more strongly bound than deuterium, since ΔE_0 ranges between 1 and 3 kJ/mol, based on measurements of H–surface vibrational frequencies for H (D) adsorbed on transition metals. Experimental examples were reported, e.g., for H on Ni(100) [104], Pd(100) [105], Pd(110) [106] and on Rh(110) [70,103]. Note that the above considerations are only correct if the excitation of a single bound H (D) atom from the bottom of the chemisorption potential well to the zero energy level is considered. In cases where *molecular* states are dealt with in which a dissociation is not required, different relations exist. However, in all cases where H_2 or D_2 *molecules* or molecularly bound states are involved quantum-statistical effects may become relevant.

A homonuclear diatomic molecule can exist either in the ortho- or in the para-modification, depending on the orientation of the nuclear spins of the individual atoms. With ortho-hydrogen, we have parallel proton spins and the nuclear spin degeneracy 3 with odd rotational quantum numbers J, whereas para-hydrogen has antiparallel proton spins, spin multiplicity 1 and therefore even rotational quantum numbers. (This leads to the well-known anomaly in the specific heat of these H_2 modifications [107].) In thermal equilibrium, i.e., at temperatures > 100 K, the ratio o : p H_2 is 3, whereas at lower temperatures

para-H_2 represents the energetically more favorable species. With the deuterium D_2 molecule, on the other hand, o-D_2 has even and p-D_2 odd J numbers. At high temperatures we have o : p $D_2 = 2$ and, opposite to H_2, at low temperatures o-D_2 is more stable. It is well known that a certain distribution of o : p H_2 is relatively stable against nuclear spin scrambling, because the nuclei are screened against spin-flip processes by the molecular electronic shell and by selection rules, but that the presence of, e.g., charcoal as an adsorptive leads immediately to the equilibrium distribution between the two modifications [108]. It has been argued that inhomogeneous magnetic fields arising from paramagnetic impurity atoms on the surface of the charcoal remove certain selection rules so that forbidden rotational transitions can occur [109] or that the H_2 wave functions are distorted owing to a physisorption process which will result in the same effect. For more details about these interesting phenomena the reader is referred to the original literature [108,109].

The *HD molecule* has different nuclei, and there is no distinction between the o- and p-modifications possible. Since this species is frequently formed immediately after co-adsorption of deuterium and hydrogen on active transition metal surfaces which spontaneously dissociate the molecules according to the scheme (* denotes an empty site):

$$2\,H_{2(g)} + 4* = 4\,H_{ad}, \tag{7}$$

$$2\,D_{2(g)} + 4* = 4\,D_{ad}, \tag{8}$$

$$4\,H_{ad} + 4\,D_{ad} = H_{2(g)} + 2\,HD_g + D_{2(g)} + 8* \tag{9}$$

(H_2 to HD to D_2 ratio = 1 : 2 : 1 for statistical mixture) we not only have a hint to a complete isotope exchange reaction and thus to *atomic* adsorption, but we also visualize that any quantum-statistical effects will disappear during this isotope scrambling. Again, we state that those phenomena will only play a role for weakly interacting molecular hydrogen (deuterium) species. As one of the very few examples that have been reported so far we mention a high-resolution electron energy loss study for H_2 physisorbed on Ag(111) by Avouris et al. [87] in which rotational and rotational and vibrational excitations of molecularly adsorbed and condensed H_2 were studied. The authors report on a quite rapid conversion from predominantly ortho-hydrogen to para-hydrogen upon adsorption at 10 K. A (short-range) surface effect was concluded from the fact that the rate of o–p conversion decreased considerably if thicker condensed H_2 layers were prepared. A somewhat contradictory result was obtained by Andersson and Harris [86] with H_2, D_2 and HD adsorbed at 15 K on a Cu(100) surface. Both o- and p-H_2 physisorb on the surface, but there was only negligible conversion found within the time scale of the experiment. Very recently, Palmer and Willis [110] investigated (by means of EELS) the rotational and vibrational excitation of H_2 physisorbed on graphite. They found that the molecules could freely rotate in three dimensions and concluded from

this that the molecular axes were not aligned in the adsorbed H_2 layers. Also, only the rotational ground state (p-H_2) could be detected.

Clearly, more experimental data is required from similar physisorption systems to learn, e.g., more about the occurrence of short-range magnetic fields on different metal and graphite surfaces and their role in o–p H_2 conversion. A more recent theoretical paper by Ishii and Sugano [112] focussed in somewhat more detail on the role of magnetic surface in the conversion reaction.

2.7. *The structure of the H adsorption complex*

So far we have focussed on the properties of the hydrogen adsorbate and have not considered the other part in the interaction, namely, the solid surface which provides the aforementioned adsorption *sites*.

For the sake of simplicity we assume an ideal (that is, defect-free) crystal surface with a well-defined two-dimensional lattice, for example, a (100) surface of a fcc metal or an alkali halide. If we neglect for a moment the certainly most important point of how any interaction and bonding between the H_2 molecule or H atom and the solid surface can come about (here the electronic theories described in more detail in section 4 provide some answer) and only regard the *structural* aspect, the first questions that come to one's mind are: (1) *Where*, i.e., on *which site* does the dissociation of the molecule take place? (2) Where are the H atoms actually adsorbed, that is, where on the surface is the location of the chemisorption potential well? (3) How many surface atoms participate in or are required for the adsorptive bonding of *one* H atom? (4) What then is the absolute configuration of the H—surface complex, i.e., what are the bond lengths to the neighbors, what are the bond angles, etc.?

Since it is also possible that the structure of the solid changes upon the adsorption of hydrogen either locally or on a long-range scale (reconstruction, relaxation, facetting phenomena) one has to extend this catalogue of questions by the problem of the structure of the solid substrate before and after the adsorption.

There are, in principle, many structure-sensitive surface spectroscopies which are commonly and routinely used to determine both the structure of the solid surface and the adsorption complex. Before we enter the presentation of the results it is necessary to premise some remarks about the applicability of the important structure-sensitive techniques to adsorbed hydrogen layers, in particular, about LEED. The interested reader can find much more details about surface-sensitive spectroscopic techniques, for example, in the monograph by Woodruff and Delchar [111].

As we have already stressed the first and most important experimental prerequisite for obtaining reliable and reproducible results, regardless of which

spectroscopy is applied, is the utmost cleanliness of sample and vacuum conditions. Perhaps more than with any other adsorbate both the interaction mechanism and the state of the adsorbed hydrogen can be affected by the chemical cleanliness and by the crystallography of the substrate. The reason is, as repeatedly mentioned before, that the small H atom is an excellent probe for all kinds of irregularities on a surface – foreign atoms with their effect on the local electronic charge distribution as well as crystallographical defects which provide an altered local coordination. As we have learnt, this is likewise sensitively felt by an adsorbing H atom. Also, coadsorption of, e.g., carbon monoxide or water vapor usually changes the adsorbed state of hydrogen markedly, and the same effect is observed with carbon or sulfur surface impurities. Not only is the *kinetics* of adsorption strongly affected (typically, the sticking probability decreases significantly) but also the H *binding states* are shifted or even suppressed. As an example, the β_2 H desorption state of a Ni(110) surface is shifted to lower temperatures by coadsorbed CO. Moreover, long-range order effects are quite often inhibited by impurities, not to mention surface reconstruction phenomena which are also greatly modified even by traces of coadsorbed CO. Again, the H–Ni(110) system provides a good example here since the well-known 1×2 reconstruction induced by 1.5 monolayers of H is inhibited by coadsorbed CO.

An extremely sensitive monitor of the surface cleanliness is the H-induced work function change which is immediately reduced as soon as traces of contaminants are present on the surface. A good example here is the H/Ru(0001) system, for which Feulner and Menzel [113] reported a strong effect of C impurities on the H-induced $\Delta\varphi$. In order to avoid sample contamination the ultimate pressure of the vacuum system should be kept as low as possible to maintain the partial pressure of active gases below, say, 10^{-12} mbar. "Active" means here that they stick with high probability and compete with hydrogen for the adsorption sites, such as CO.

Quite generally many of the common UHV compatible surface spectroscopies are applicable with good success also for adsorbed H layers. An absolute exception, however, is the Auger electron spectroscopy which cannot monitor H for physical reasons. Furthermore, all methods that probe electron charge densities are expected to yield rather poor results if applied to adsorbed H atoms because of their weak Coulomb field which makes the cross section for electron scattering small. Actually, it is the effective potential that is felt by the low-energy electrons, and just with adsorbed H atoms also the protonic charge determines the scattering cross section substantially. Despite these difficulties, low energy electron diffraction has proven to be a very helpful tool to determine the structure of adsorbed H layers. Besides LEED, also He scattering appears to very sensitively map the location of adsorbed H as could be shown in the pioneering work by Rieder and Engel [20,114]. Furthermore high-resolution EELS has frequently been used in the past to obtain informa-

tion about the state and configuration of adsorbed H, and it seems to work particularly well with H on semiconductor surfaces. UV photoelectron spectroscopy is another valuable technique which monitors H induced alterations of the surface electronic structure quite sensitively. Besides these spectroscopies, standard techniques such as thermal desorption spectroscopy (TDS) or work function change measurements ($\Delta\varphi$) can equally well be employed for monitoring H adsorption phenomena. More sophisticated methods, for example molecular beam measurements, have also provided invaluable information about H adsorption processes, particularly with respect to the dynamics of the interaction. However, LEED may still be regarded as the most important tool that yields accurate *structural* information, also in the case of H layers (which must, of course, exhibit long-range order).

The small diffractive power of H atoms has concerned the scientists since the beginning of LEED. The observation of "intense" H-induced "extra" spots led immediately to the conclusion that not a H phase with long-range order but distorted substrate atoms were causing these diffraction features. Particularly early LEED investigations with the H/Ni(110) system brought up some controversy with regard to this subject [115–118]. Later calculations by Ying et al. [119] showed that the major diffraction contribution of adsorbed H atoms arises from forward scattering events (which gain importance with increasing electron energy). This means that the electrons scattered off a H atom on the surface are primarily directed towards the bulk of the crystal to then undergo further diffraction at the Coulombic potentials of the metal atoms, and only after a variety of such scattering events the electrons leave the crystal and appear as "reflected" or "backscattered" electrons. A more recent calculations as to quantify the cross section for elastic electron scattering off an adsorbed H layer was performed in the context of LEED investigations of the H/Rh(110) systems [120]. In detail, the ratio of the integer and fractional order beam intensity was calculated over a certain energy range. The main result was that there should be approximately a factor of 100 difference between the scattering cross section of a typical heavy metal atom such as Rh and the light H atom. For the particular system H/Rh(110), however, considerably larger intensities of the fractional-order beams were observed, and thus an at least small additional displacement of Rh atoms had to be taken into account [121]. Generally, the small scattering power of a H atom has led some investigators to completely neglect the diffraction contribution of H atoms, e.g., in dynamical LEED calculations of adsorbed hydrocarbon molecules, with good success for predicting the location of the carbon skeleton.

One may argue, however, that upon adsorption of a H atom an exchange of electronic charge is taking place with a net flow to the H atom which should increase its electron density to a certain extent. The generally observed H-induced work function *increase* supports this conclusion. The net effect, however, is not very pronounced as the small dipole moments (cf. table 6)

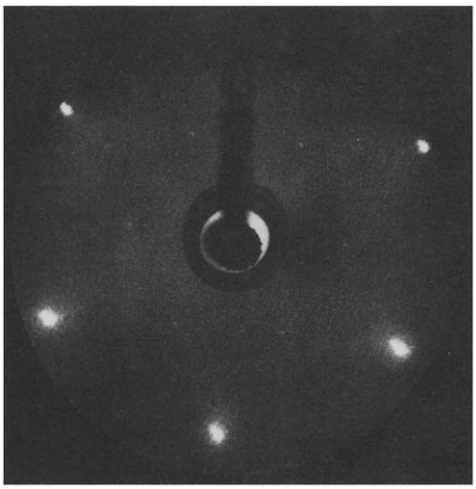


Fig. 10. Photograph of the $(\sqrt{3} \times \sqrt{3})R30°$ LEED pattern produced by H_2 adsorption on graphite(0001) at $T = 10$ K and $p_H = 1.2 \times 10^{-8}$ Torr, electron energy = 142 V. The intense spots are due to the graphite, the weak spots belong to the H_2 overlayer. After Seguin and Suzanne [83]. Reproduced with permission.

around 0.05 D demonstrate. Even for a small Me–H dipole length the net charge flow is usually not more than about 1/10 of the elementary charge unit, e_0. This amount certainly cannot account for the experimentally observed LEED intensity of ordered chemisorbed H phases, except perhaps the H/W(100) system where μ_0 is substantially larger. Here, however, reconstruction effects are clearly involved and could account for the large μ_0. Thus, the forward scattering effect appears to be quite important. The theoretically predicted low intensity of H-induced LEED beams is nicely demonstrated by the $\sqrt{3} \times \sqrt{3}$ LEED pattern observed with H_2 physisorbed on graphite by Seguin and Suzanne [83] which is presented in fig. 10. Here, the H_2 molecule is actually physisorbed, and owing to the very weak interaction, the charge distribution around the protons is not noticeably altered.Hence the scattering properties of this system should more or less reflect that of the unperturbed H_2 molecule.

The high mobility of an adsorbed H atom is another serious problem which may obscure the formation of ordered phases. In this situation, cooling of the sample to below ~ 200 K or even lower temperatures may help to fix the H atoms in periodic adsorption sites and to increase their long-range order. This is documented by the fact that actually all order–disorder transition temperatures of H phases are well below 300 K (cf. table 9) which causes the invisibility of these phases at room temperature. This behavior is, of course, a consequence of the fairly weak lateral interaction energies (cf. section 3.6.2).

Unlike any other adsorbate H may also form a 1×1 phase near monolayer coverage because of the smallness of the H atoms. Then the experimental detection may likewise be difficult, since the H-induced changes of the integer-order LEED beams are usually only in the percent range, and careful intensity–voltage measurements must be performed. Examples are H on Ni(100), Pd(100), Ru(0001), Pt(111), W(100) as will be made clear below. We shall now return to a variety of experimental structure determinations of adsorbed H on single crystal surfaces.

2.7.1. The geometrical structure

Whilst there is no absolute structure determination of adsorbed *molecular* hydrogen yet (which makes it difficult to argue about the geometrical requirements during the dissociation process) a fairly complete picture of the geometry of the atomic H–surface bond has been obtained during the last decade, primarily based on dynamical LEED structure analyses [122] or high-resolution electron energy loss spectroscopy (EELS) measurements [123]. As an example, we present recent data of the LEED structure determination of H on Fe(110) [124] where intensity–voltage ($I(V)$) curves were measured among others for the (0,1), $(-\frac{1}{2},1)$, $(\frac{1}{2},0)$ beams of the hydrogen-induced 2×1-H structure and compared with various theoretical curves calculated for different binding geometries. As can be seen from fig. 11, these curves are sensitively affected by the coordination of the adsorption site but also by the distance between the H overlayer plane with respect to the first Fe atomic layer.

The EELS method predicts, according to a simple central force model, different positions of the Me–H vibrational frequencies for H bound in atop, twofold (bridge), threefold or fourfold hollow sites, and the validity of this correlation (which also exists for CO and other molecules) has been proven many times. Thus, for H on Ni(100) [104] and on Pd(100) [105] the fourfold hollow site was determined by EELS, in excellent agreement with theoretical predictions by Upton and Goddard [125] and Nordlander et al. [99].

It turns out from all of these studies that there is a general tendency of the H atom to occupy surface sites with a *high local coordination*, with actually very few exceptions. Some recent results of structural analyses of adsorbed H layers are given in table 5; the data are based on LEED and vibrational loss (EELS) measurements. Independent of the surface geometry the site with the highest coordination is preferred: On the (111) surfaces the threefold coordinated site, on the (100) surfaces the fourfold and on the (110) surfaces the threefold site. So far, the W(100)/H system seems the only example where this rule is not obeyed: Here, the EELS data clearly suggest the occupation of a twofold (bridge) site [126–128]. The other relevant information that we can extract from the table is the magnitude of the H–metal bond length. The available data ranges from 1.72 Å for H on Ni(110) [129] up to 1.91 Å for H on Ru(0001) [130]. After subtraction of the ionic radii of the respective metals

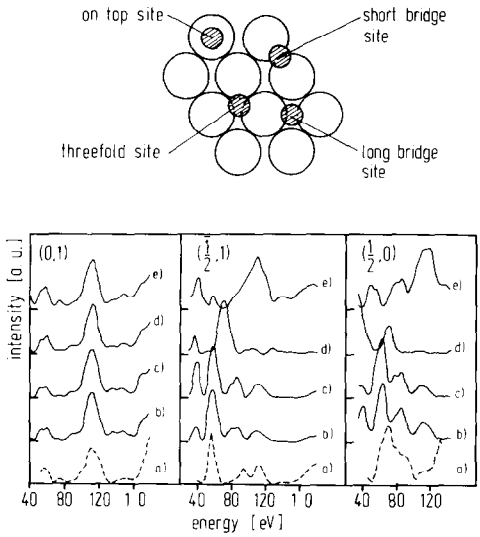


Fig. 11. Top: H adsorption sites on a Fe(110) surface. The small shaded circles indicate a H atom. Bottom: Intensity–voltage curves at normal electron incidence for the 2×1-H structure: (a) and dashed: experimental curves; (b)–(e) calculated curves at their respective best fit. Fe–H layer spacing $d_{Fe-Fe} = 2.0$ Å: (b) threefold site, $d_{Fe-H} = 1.0$ Å; (c) long bridge site, $d_{Fe-H} = 1.0$ Å; (d) short bridge site, $d_{Fe-H} = 1.2$ Å; (e) atop site, $d_{Fe-H} = 1.7$ Å. After Moritz et al. [125].

we arrive at a "hard core" radius of the chemisorbed H atom of ~ 0.4 Å in the Ni and ~ 0.60 Å in the Ru case, with an uncertainty of ±0.1 Å. This means that (at least for the examples presented) the radius of the chemisorbed H is very close to the Bohr radius of the free atom, $a_0 = 0.53$ Å. This may imply that the actual charge transfer from or to the adsorbed H atom is probably fairly small – which agrees with the results of work function measurements [10] indicating quite small dipole moments of chemisorbed H atoms. In other words, we expect neither a clear protonic ($Me-H^+$) nor a hydridic ($Me-H^-$) configuration of the adsorption complex (whereby this statement is, of course, only valid for transition metals). This in line with the general result of many quantum-mechanical calculations whereafter the H–Me bond has predominantly *covalent* character. As has become evident already in the presentation of the trapping and sticking data the possible presence of defect sites on a surface can drastically alter its adsorptive properties, for example the energy accommodation process. But also the formation of the adsorbate complex is frequently affected. Owing to the tendency of the H atom to occupy surface sites with a high coordination number step and kink sites (which provide a higher local coordination) are preferentially populated. This tendency can be quantified in that the chemisorption potential well is by ΔE_{ad} deeper on highly coordinated than on less coordinated surface sites.

Table 5
Results of some structure analyses of adsorbed H layers based on LEED, EELS and other techniques

Surface (*hkl*)	H phase	Local coordination of H atom	Me–H layer distance (Å)	Me–H bond length (Å)	Method	Reference
Fe(110)	$p(2\times1)$	3-fold	0.9 ± 0.1	1.75 ± 0.05	LEED	[124]
	3×1-2H	3-fold	0.9 ± 0.1	1.75 ± 0.05	LEED	[124]
Co($10\bar{1}0$)	2×1 p2mg	3-fold			LEED	[69]
Ni(111)	$c(2\times2)$	3-fold	1.15 ± 0.05	1.84 ± 0.06	LEED	[50]
Ni(100)		4-fold			EELS	[104]
	1×1	4-fold			He scattering	K.H. Rieder & H. Wilsch, Surface Sci. 131 (1983) 245
Ni(110)	2×1 p2mg	3-fold	0.41 ± 0.1	1.72 ± 0.1	LEED	[129]
	2×1 p2mg	3-fold			He scattering	[20]
Ru(0001)	1×1	3-fold		1.91	VLEED	[130]
		3-fold			EELS	[179]
Rh(110)	1×1-2H	3-fold	0.8 ± 0.1		LEED	W. Nichtl et al., to be published
Pd(111)	$c(\sqrt{3}\times\sqrt{3})R30°$	3-fold			Phase diagram calc.	[327]
Pd(100)	$c(2\times2)$	4-fold			EELS	[105]
Pd(110)	2×1 p2mg	3-fold	0.6 ± 0.05	2.0 ± 0.1	LEED	M. Skottke, R.J. Behm, G. Ertl, V. Penka & W. Moritz, J. Chem. Phys., in press
W(100)	$\sqrt{2}\times\sqrt{2}$	2-fold			EELS	[126–128]
W(110)		2-fold			ARUPS	
					EELS	[189]
Pt(111)	1×1	3-fold	0.71	1.76	EELS	[170]
	1×1	3-fold	1.0	1.9	He scattering	I.P. Batra, J.A. Barker & D.J. Auerbach, J. Vacuum Sci. Technol. 2 (1984) 943

Indeed there are several experimental observations which support this statement. As an example, measurements of the isosteric heat of hydrogen adsorption as a function of coverage on stepped Pd(111) [106] and Pt(111) surfaces [131] clearly revealed a somewhat enhanced binding energy for small coverages. This is illustrated in fig. 12 which is taken from the work by Conrad et al. [106] and compares the isosteric heat of adsorption of H_2 on a stepped and on a well-oriented Pd(111) surface. Quite similar results were obtained also by thermal desorption spectroscopy with a Pt(997) surface [131], where a high energy binding state was observed which could be attributed to the exclusive population of step sites. From the available data the binding energy difference between "regular" and step or kink sites, ΔE_{ad}, is around 5–10 kJ/mol. This relatively small value along with the easily excitable surface diffusion of H atoms (cf. section 2.9) lead to an exclusive population of step sites taking place at sufficiently low temperatures; otherwise this population is governed by Boltzmann statistics according to

$$N_{steps}/N_{total} = \exp(-\Delta E/kT) \tag{10}$$

(if N_i denotes the concentration of the respective sites) which may wipe out the differences to a certain extent at higher temperatures. Whilst the studies of the role of defects in H chemisorption certainly support the view about the favored high coordination numbers in the H surface chemical bond they can – as well as the binding energy measurements – only indirectly contribute to the question of *how many atoms* of the solid are involved in the bonding of *one* H_2 molecule or *one* H atom. They do suggest that this atom experiences only a sufficiently large binding energy if *more than one* atom of the solid surface participates in the chemical bonding. In other words, there is a certain *ensemble* of adjacent surface atoms required to form a stable H adsorption complex. This famous "ensemble effect" (which term was introduced by Sachtler [132]) is well-known in heterogeneous catalysis and plays a dominant role in the so-called "structure sensitive" surface reactions – which run only

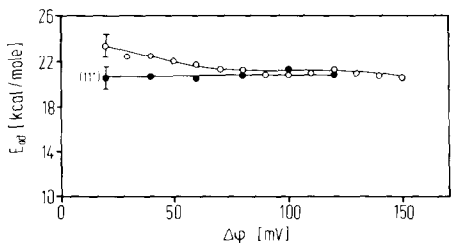


Fig. 12. Isosteric heat of adsorption of H_2 on a Pd(111) surface as a function of coverage (for which the H-induced work function change, $\Delta\varphi$, has been taken as a monitor). Open circles: stepped Pd(111) surface. Full circles: flat Pd(111) surface. After Conrad et al. [106].

over sites with a certain size and local geometry [133,134]. As was pointed out by Knor [8] there is apparently some need to add more terms to the energy balance of eq. (3) that account for the participation of neighboring surface atoms in the bonding of one H atom. Also, the weakening of metal–metal bonds in the vicinity of an adsorbed H atom must principally be considered. We then obtain for the energy balance:

$$E_{ad} = 2\sum_i n_i E^i_{Me-H} - E_{diss} - 2\sum_j p_j E^j_{Me-Me}, \tag{11}$$

in which $\sum n_i E^i_{Me-H}$ denotes the sum over all n_i Me–H bonds which contribute to the binding of a given H atom ($\sum n_i E^i_{Me-H} = E_{Me-H}$ only, if just one metal–H bond is formed by each H atom). $\sum p_j E^j_{Me-Me}$ represents the total energy drain according to the weakening of metallic bonds in the vicinity of the adsorbed H atom. Particularly for surface reconstruction processes this term may become important.

More quantitative information about the ensemble size required for hydrogen chemisorption can be obtained from measurements of the uptake of strongly chemisorbed hydrogen on *chemically modified* single crystal surfaces, for example, "bimetallic" surfaces, which consist of one inert and one active component for H adsorption (systems like Cu–Ni, Cu–Ru, Au–Ru fulfill these requirements – with the noble metal being the inert constituent). Particularly suited are immiscible systems like Cu–Ru because any alloying, segregation or depletion effects do not occur. The H uptake of Ru(0001) surfaces covered with various (submonolayer) amounts of Cu [135] decreased the amount of strongly adsorbed H considerably because few Cu atoms statistically distributed across the active Ru surface could very efficiently destroy the (apparently fairly large) Ru ensembles required for strong hydrogen adsorption. From these studies an average ensemble size of $n \geq 4$ adjacent Ru atoms was concluded, in agreement with H chemisorption data reported by Yu et al. [136] who also found $n = 4$ for CuNi surfaces. This clear picture is somewhat spoiled by results obtained with CuRu(0001) surfaces in the group of Goodman [137], because these authors could not find a H chemisorption site blocking ratio of the Cu greater than 1 : 1. This may be caused by some differences in the experimental set-up and procedure in the preparation of the bimetallic surfaces, particularly with respect to the lateral dispersion of the Cu atoms or clusters on the Ru(0001) surface which is absolutely crucial. Very recent measurements in our own laboratory using Au–Ru surfaces confirm the approximate ensemble size of $n \approx 4$ [138].

The other general problem in interpreting the uptake data is the difficulty to decide whether it is the *dissociation* of the H_2 molecule that needs the ensemble or (as tacitly assumed) the formation of the actual *chemisorptive bond*. In other words, it may be a *kinetic* or an *energetical* effect. There are some arguments in favor of this latter interpretation (e.g. the absence of

spill-over processes) and the reader is referred to the original work for further details [135,137].

2.7.2. The electronic structure

The just discussed geometrical properties are to a great extent a consequence of the electronic interaction between a H_2 molecule and/or a H atom with the local chemical environment of the adsorption site on the surface. How does this interaction come about?

We can be sure that both the approaching H_2 molecule and the surface are electrically neutral, and no attractive or repulsive coulombic forces are operating. The H_2 molecule has no permanent dipole moment, whereas, e.g. an electron-rich metal surface has a more or less pronounced tail of "spilling out" negative charge which may somewhat polarize the adsorbing molecule and direct it to certain sites with the highest local field inhomogeneity. Moreover, as in the intermolecular bonding, there are certainly van der Waals forces operative which are, e.g., responsible for the condensation of H_2. The sum of these forces lead to the shallow potential of fig. 6a. In this stage of the interaction there is certainly no charge transfer between the surface and the approaching molecule which is also still relatively far away from the surface. However, as it gets closer there is the possibility of electron tunneling between the occupied hydrogen (1s or 1σ) orbital and those wave functions of the surface which have the right symmetry for electron exchange. At this point the dissociation may occur on metal surfaces. The adatom states are shifted and broadened and both the ionization and the electron affinity levels degenerate to narrow bands, and they may also change their energetic position. As photoemission studies show which are explained further below this is particularly so for an adsorbed H atom because it can get *quite close* to the surface and its 1s orbital can strongly overlap with metallic wave functions. In a more delocalized view, the tails of the filled 1s and the empty electron affinity bands reach into the Fermi level of the surface with the consequence that both bands are now *partially* filled and contribute to the chemisorptive bonding. Nørskov and Stoltze have summarized the dissociation and chemisorption process for H_2 interaction with a Mg surface. The aforementioned shift and broadening of the involved electron states are illustrated in fig. 13 taken from their work [139]. It is pointed out therein that antibonding molecular orbitals (the $2\sigma_u^*$ in the particular case of H_2) are more and more populated by metal electrons as the molecule approaches the surface. Not only leads this to a weakening of the innermolecular bond but the respective MO is broadened and pulled down to below the Fermi level of the metal. This provides an attraction which bends the (originally repulsive) potential energy curve over and eventually gives rise to the minimum which, according to Nørskov and Stoltze, represents a chemisorbed molecular state. In spite of the pronounced filling of the antibonding H_2 level the covalent H–H bond is not completely destroyed yet, and

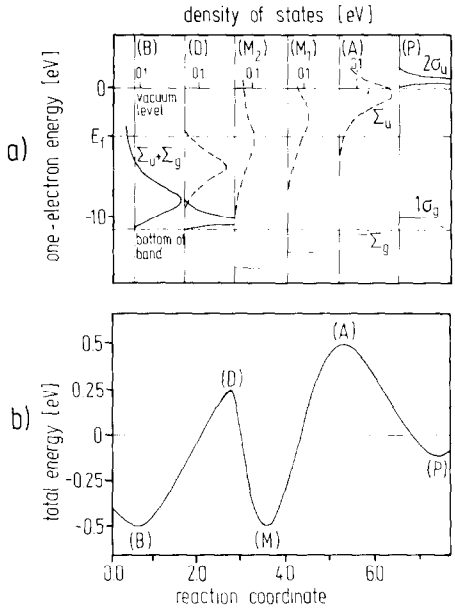


Fig. 13. (a) One-electron density states and (b) total binding energy for H_2 approaching a Mg(0001) surface with its molecular axis parallel to the surface and oriented along the [100] direction above a Mg atom, as a function of position along the reaction coordinate connecting the local minima and saddle points of the adiabatic potential energy surface. (P) = physisorbed state; (A) = activation barrier for molecular adsorption; (M_1, M_2) = molecularly adsorbed state with density of states in the outer (M_1) and inner (M_2) part of the well; (D) = activation barrier for dissociation; and (B) = two separate H atoms chemisorbed in the twofold bridge site. After Nørskov and Stoltze [139].

some additional energy is required to make the molecule dissociate completely. This causes the second activation barrier at closer H_2–metal distances. The dissociated state then exhibits a single adsorbate-induced resonance level between 6 and 10 eV below the Fermi energy which is observable with UV photoemission. Whilst this picture was developed for a typical sp-metal (Mg) it may well be transferred also to d-electron metals, whereby the d-electrons generally decrease the stability of the molecular H_2–metal complex and increase the rate with which the dissociation occurs, since the activation barriers are substantially lowered.

The situation is much simpler in alkali adsorption, where a more or less *complete* donation of one electron from the adatom to the surface occurs resulting in a largely positive adion, because the level of the highest occupied adatom state is located much higher than the Fermi level. Things are more complicated with the adsorption of a H atom whose (unperturbed) ionization level I is 13.6 eV (referenced to the vacuum level) whereas the Fermi levels of

most of the transition metals are of the order of 4 to 6 eV below E_{vac}. Here the adatom is expected to be stable in the neutral state. But since the levels broaden and shift, any kind of smaller or larger electron exchange from the H atom to the metal and vice versa can occur. So the electrons making the chemisorptive bond can be thought of as moving in orbitals contributed from both the H atom and the surface atom(s) underneath which is the typical criterion for a *covalent* bond.

One important experimental tool to map the electronic interactions in the chemisorption process is UV photoelectron spectroscopy (UPS), whereby one can distinguish angle-integrated and angle-resolved (ARUPS) techniques. An overview has been given, e.g., by Plummer [140] or Feuerbacher and Fitton [141]. UPS has been employed for studying the H chemisorption on Pd [142–145], Ni [142–147], Pt [142,143], Co [148], Ru [149] and W surfaces [150,151] to give only a few examples. Quite generally, the electronic interaction shows up as changes in the density of states near the Fermi level, E_F (where d-states become often suppressed) and as new broad emission features at ~ 5 to 10 eV below E_F which are due to the H 1s orbital interacting with metallic states. These features can be followed through k-space in order to determine the symmetry of the bands involved in the H chemisorption and the location of the adsorption site [147–149]. The fairly extended width of the H-induced extra emission seems to be characteristic; really sharp features have never been observed. An example from our own work is presented in fig. 14 which shows a H-induced UPS peak of a Ni(110) surface and its energy–momentum dependence [147]. Another instructive example is given in the work by Hofmann and Menzel [149] who studied, with ARUPS, the H adsorption on a Ru(0001) surface. Again, pronounced changes in the photoemission spectra were observed indicative of a 1×1 overlayer of H adsorbed in threefold sites. Also, the location of the H 1s derived band, split off from the metallic d-band was followed through k-space, varying from -8.1 eV below E_F at the Γ point up to -5.9 eV at the K point of the surface Brillouin zone. Also worthwhile to mention is the interesting phenomenon (which was observed previously by Eberhardt et al. [144] in their UPS study of H adsorption on Ni(111), Pd(111) and Pt(111)) that room temperature adsorption of hydrogen did *not* produce any such photoemission features which was tentatively explained by a reversible overlayer–underlayer phase transformation of the adsorbed hydrogen. For more details see section 3.6.4.

Experimental UPS data can be analyzed and compared with quantum-chemical calculations in order to learn more about the orbitals involved in the bonding (e.g., via electron binding energies, polarization dependences and cross sections). As an example, first principles Hartree–Fock calculations have been reported by Upton and Goddard [125] for binding of H atoms at high-symmetry sites of low-index faces of Ni surfaces with a 20 atom Ni cluster used as a model. They find that the 3d orbitals determine the shape of

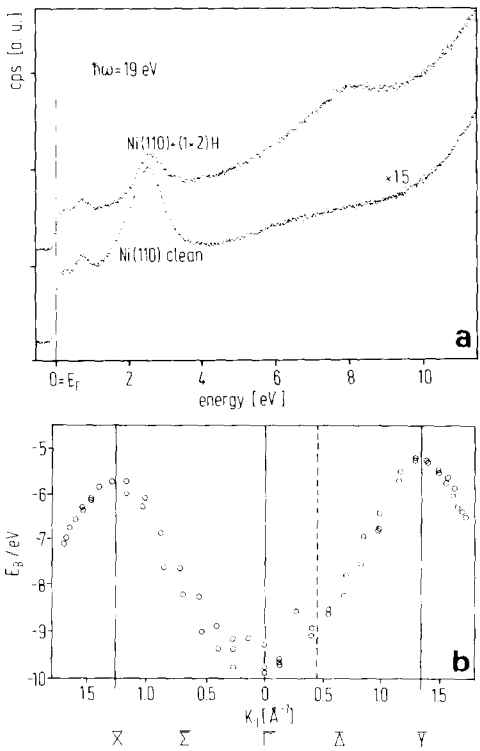


Fig. 14. (a) UV-photoelectron spectra ($\hbar\omega = 19$ eV) of a clean (lower curve) and a 1×2-H covered Ni(110) surface (upper curve) showing the broad H-induced resonance level at $E \approx 7.5$–8 eV. (b) Part of the $E(k_{\parallel})$ dependence showing the band dispersion of the H-induced electronic levels in the surface Brillouin zone ($\bar{\Gamma}\bar{Y}$ direction = [100] and $\bar{\Gamma}\bar{X}$ direction = [1$\bar{1}$0]). The dashed line indicates the boundary of the first Brillouin zone of the 1×2 reconstructed surface, the full lines the periodicity of the H-band dispersion.

the conduction band whereas the bonding is dominated by delocalized 4s-like orbitals. On each possible adsorption site the H binding energy was calculated, and the result (1.6 eV for onefold, 2.8 eV for twofold, 3.2 eV for threefold and 3.0 eV for fourfold site symmetry) confirms the statement made before about the tendency of H to highly coordinated sites. Another example deals with the system H on Pd(111), where a fully self-consistent calculation was performed by Louie [32]. The full effects of the underlying band structure and all interactions (H–H, H–Pd and Pd–Pd) were taken into account with very few variable input parameters. Again, several possible sites for adsorption were checked, and once more the threefold coordinated site (either tetrahedral or octahedral) came out to be most favorable. During the last years attempts were made to detect H-induced unoccupied states above E_F by means of inverse photoemission, for example in the group of Küppers [152] for H

adsorption on Ni(111) and Ni(100). According to Upton and Goddard [125] there should be a split-off hydrogen state ~ 1–2 eV above the Fermi level for all binding sites. Reimer et al. [152] found, with Ni(100), a weak extra feature ~ 1.3 eV above E_F which could perhaps reflect this predicted state. A more recent theoretical work by Fournier and Salahub [153] reveals that upon H adsorption metallic d-states in the vicinity of the adatom are pushed above E_F through antibonding interaction with H s-states by ~ 1 eV. This would also be in line with Reimer et al.'s result [152].

There is another less sophisticated and more commonly used technique for monitoring charge transfer processes on and in a surface, namely, work function change ($\Delta\varphi$) measurements. Almost any adsorption process leads to a minor or major change of the work function of the solid surface, φ_0. With metals, φ_0 can be measured fairly easily, and so its change during adsorption, $\Delta\varphi$. This change can range from a few mV up to several volts. The Kelvin method [154] is particularly suited to follow the H-induced work function change as the adsorption proceeds. The general problem though of work function measurements is, as will be pointed out further below, that φ and $\Delta\varphi$ can be composed of several contributions (surface and bulk effects) which cannot always be separated from each other. Nevertheless, work function effects are commonly considered experimentally.

For small coverages where each adsorbing H atom can be regarded as not being affected by neighbors one may apply the Helmholtz equation

$$\Delta\varphi = \mu_0 4\pi\sigma_0 f^* \tag{12}$$

(σ_0 is the number of adsorbed H atoms/cm^2, $f^* = 1/(4\pi\epsilon_0)$ is the conversion factor) in order to deduce the initial dipole moment μ_0 of the adsorbed complex. The $\Delta\varphi$ measurement yields two pieces of primary information: First, the *sign* of the change of the work function, i.e., whether the adspecies is positively polarized (in this case $\Delta\varphi$ is negative, that is, φ_0 decreases – as in the alkali adsorption), or $\Delta\varphi > 0$, for negatively polarized dipoles, that is, φ_0 increases (as for CO adsorption on Pd surfaces). This statement is, however, only valid, if the surface in question does not change its structure upon adsorption and if the H–Me dipole is not totally screened. In this context it still remains an open question why on Pt(111), W(110) and Fe(110) the chemisorbed H forms a nominally positively polarized complex. The other information can be obtained from the expression $\partial\Delta\varphi/\partial\Theta$, i.e., the slope of the work function–coverage function for vanishing coverages, i.e., limit($\Theta \to 0$) ($\partial\Delta\varphi/\partial\Theta$). According to eq. (12), a large slope usually indicates a high initial dipole moment and vice versa. For hydrogen adsorption on clean metal surfaces, quite often an *increase* of the work function is found (for example, with the systems H on Ni(100) [56], Ni(111) [50,155], Ni(110) [21,156,157], Pd(100) [177], Pd(111) [106], Pd(110) [68,106], Rh(110) [70], Ru(10$\bar{1}$0) [71], Co(10$\bar{1}$0) [69]), but in some cases also a decrease of φ, e.g. with the systems

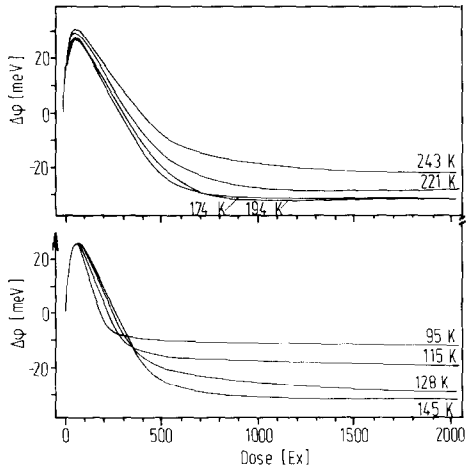


Fig. 15. H-induced work function change of a Ru(0001) surface for different temperatures as a function of H_2 exposure [Ex]. 1 [Ex] equals the impingment of 10^{14} molecules/cm^2. Note the initial $\Delta\varphi$ maximum and its temperature dependence! After Feulner and Menzel [113].

Pt(111) [158] and Fe(110) [159]. For more information see table 8 in section 3.3. $\Delta\varphi$ may even change its sign at very small coverages as observed with the system H on Ru(0001) which is illustrated in fig. 15 taken from the work by Feulner and Menzel [113]. In this case there was no hint to an a-priori heterogeneity of the Ru(0001) surface, the peculiar $\Delta\varphi$ behavior seems to be an intrinsic property of the H/Ru(0001) system.

Similar $\Delta\varphi$ effects were found with H adsorption on a stepped Pt(111) surface (fig. 16) which, however, could be traced back to preferential adsorption on step sites on which – opposite to the case of the flat terrace sites – a negatively polarized hydrogen surface species was stable. A quantitative correlation between the number of such step sites and the $\Delta\varphi$ contributions even allowed the distinction of two different step sites [131] which are marked in fig. 16 as shaded and full circles, respectively.

Several other examples are known in which $\Delta\varphi(\Theta)$ shows alterations of its slope at very early stages of adsorption, i.e., for small coverages which may indicate an a-priori or a H-induced surface heterogeneity. More information about $\Delta\varphi(\Theta)$ relations can be found in section 3.3. Here we concentrate on zero coverage effects only and on the discussion how the bonding of a H atom (or the respective electronic interaction) shows up in the work function variation. The predominantly found $\Delta\varphi$ *increase* suggests a net flow of charge from the metal to the adatom which, however, is quite small. Table 6 summarizes for some systems dipole moments that have been determined from $\Delta\varphi$ measurements according to eq. (12). It is evident that the dipole moments μ_0 scatter around 0.05 debyes (1 D = 3.3×10^{-30} A s m) which is very small as

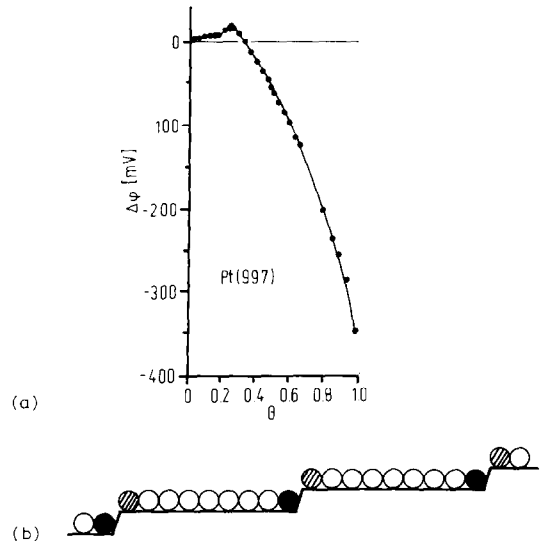


Fig. 16. (a) H-induced change of the work function of a stepped Pt(111) surface, as a function of H coverage, Θ. (b) Schematic drawing (cut perpendicular through the surface and normal to the steps) of the Pt(997) surface with three different adsorption sites of H indicated. Full and shaded circles: H atoms adsorbed at step sites, open circles: H atoms adsorbed on terrace sites. After ref. [131].

Table 6
Initial dipole moments, μ_0, for some H–metal adsorption systems

Metal surface (*hkl*)	$\Delta\varphi$ at $\Theta_H = 0.1$ (mV)	Initial dipole moment μ_0		Reference
		(A s m)	(D)	
Fe(110)	−20	-1.04×10^{-31}	-3.2×10^{-2}	[159]
Co(10$\bar{1}$0)	+5	$+4.52 \times 10^{-32}$	$+1.37 \times 10^{-2}$	[69]
Ni(111)	+20	$+9.49 \times 10^{-32}$	$+2.9 \times 10^{-2}$	[50]
Ni(110)	+23	$+1.782 \times 10^{-31}$	$+5.4 \times 10^{-2}$	[21]
Ru(10$\bar{1}$0)	+19	$+1.939 \times 10^{-31}$	$+5.9 \times 10^{-2}$	[71]
Ru(0001)	Pos. state	$+3.3 \times 10^{-32}$	$+1.0 \times 10^{-2}$	[113]
	Neg. state	-2.6×10^{-32}	-0.79×10^{-2}	[113]
Rh(110)	+13	$+1.174 \times 10^{-31}$	$+3.56 \times 10^{-2}$	[70]
Pd(100)	+11	$+7.33 \times 10^{-32}$	$+2.22 \times 10^{-2}$	[77]
Pd(110)	+25	$+2.356 \times 10^{-31}$	$+7.14 \times 10^{-2}$	[68]
W(100)		$+7.09 \times 10^{-31}$	+0.21	[253]
Ir(110)-1×2		$+4.62 \times 10^{-31}$	+0.14	[245]
Pt(111)	−20	-1.176×10^{-31}	-3.56×10^{-2}	[158]
Pt(110)-1×2	+50	$+3.96 \times 10^{-31}$	+0.12	[244]

compared to μ_0 values for CO ($0.1 < \mu_0 < 1$ D) or adsorbed alkali atoms ($1 < \mu_0 < 10$ D).

More difficult is an estimation of how much charge is actually transferred to the H atom because this requires the exact knowledge of the local adsorption geometry (see section 2.7.1). For the system H/Ni(111) a charge transfer of less than $0.1e_0$ was concluded on the basis of a LEED structure determination. The fact that in some cases H adsorption even leads to a *decrease* of the work function (H on Pt(111)!) underlines that the direction of this small charge transfer is quite difficult to predict. Again, the relative positions of the energy levels I and A of the H atom and φ_0 of the metal surface are crucial as pointed out before. Any systematic study of the dependence of $(\partial\Delta\varphi/\partial\Theta_H)$ of φ_0 for various metals has not been performed so far. Moreover (and this makes any predictions almost impossible) the H adsorption may locally induce alterations of the surface crystallography. According to eq. (11) metal–metal bond energies may become affected resulting in new equilibrium positions of the substrate atoms (relaxation and reconstruction). Since the definition of the work function for $T > 0$ K reads [60]:

$$\varphi_0 = E_{vac} - \mu, \tag{13}$$

with E_{vac} the vacuum level and μ the electrochemical potential of the solid, i.e., position of the Fermi level as measured from the bottom of the conduction band, φ_0 clearly contains always a contribution of the bulk properties of the respective solid, and any changes in the surface charge density (as, e.g., caused by relaxation or reconstruction phenomena) will also affect φ_0 and hence $\Delta\varphi$.

An excellent probe for the electronic interactions occurring between an adsorbate atom and the surface are low energy electrons, particularly elastically scattered electrons as they appear in the specularly reflected LEED or EELS beams. The typical energy range is from 1 to 10 eV. There is a variety of very recent studies on this subject.

Conrad et al. investigated the electron reflectivity of ruthenium [161] and palladium surfaces [162] with and without chemisorbed hydrogen as well as the primary energy dependence of the hydrogen vibrational cross sections. This dependence indicates a breakdown of the dipole selection rules and emphasizes the role of trapping of electrons in surface states and resonances in the electron energy loss process. An important factor here is the *inclusion* of the H atom in the electron density of the metal surface, owing to the proximity of the hydrogen to the surface. There are model calculations by Sunjić and coworkers [163] which predict this behavior.

A recent work by Lindroos et al. [130] reports on LEED $I(V)$ measurements at very low electron energies ($7 < E_p < 40$ eV) ("VLEED"). The system under investigation was Ru(0001) without and with adsorbed hydrogen. The reflectivity of the surface does so sensitively depend on the adsorption of hydrogen and its location on the surface that conclusions are possible about

the local adsorption geometry. This extreme sensitivity is explained by a quenching of an unoccupied electron surface state 11.1 eV above E_F by adsorbed H. In this particular example Lindroos et al. obtain a H–Ru distance for H in the threefold fcc sites within the monolayer coverage of $r_{H-Ru} = 1.91 \pm 0.10$ Å from which a "hard sphere" radius of ~ 0.6 Å for the H atom is deduced, in agreement with the statements made in section 2.7.1. Other measurements of very low energy electron reflectivities have recently been reported by Herlt and Bauer for the H/W(100) and H/W(110) systems [164].

2.8. Surface vibrations

Besides the diffraction techniques vibrational spectroscopies have become powerful tools to determine the binding geometry (coordination, local structure, site symmetry) of an adsorption complex. There exist numerous review articles on this subject among which we only want to mention the work by Hollins and Pritchard [165] which focusses on infrared (IR) studies of chemisorbed layers on single crystals, the article by Ueba [166] which considers the vibrational lineshapes of adsorbates on solid surfaces and the comprehensive volume "Vibrational Spectroscopy of Adsorbates" of the Springer Series in Chemical Physics [167]. Besides the common optical IR technique it is the high-resolution electron energy loss spectroscopy (HREELS) which has gained much interest in the past and nowadays represents one of the most frequently and successfully used vibrational spectroscopies. It is well applicable also to adsorbed hydrogen, and most of the present information gathered about H–surface vibrations, lineshapes, ground state frequencies, force constants, etc. arises from this technique which is, e.g., described in detail in the monograph by Ibach and Mills [168]. At present, the instrumental limit of IR spectroscopy is around 800 cm^{-1} which means that it cannot detect vibrations with lower frequency. This makes EELS which does not have this instrumental cut-off (despite its limited spectral resolution) an especially valuable technique to investigate H–metal surface vibrations which often lie in this frequency range.

In this section we shall review some of the more important vibrational data obtained so far for adsorbed H systems as well as the conclusions drawn from this data with regard to the symmetry of the H adsorption site. The reliability of any conclusion about the structure and the chemical bonding of the adsorbed hydrogen (H_2 or H) rests on the correct assignment of the various measured vibrational modes. Whilst this may be a rather difficult task in the case of big molecules with their large number of normal vibrational modes it should be relatively easy if one deals with diatomic molecules or even atoms. This is, for example, documented by the large number of successful vibrational analyses reported for carbon monoxide adsorbed on transition metal surfaces. Although one would expect an equally simple situation with adsorbed H_2 or H

there are nevertheless several complications with this adsorbate, the most awkward being the comparatively small cross section for dipole scattering which is responsible for only weak loss signals and may sometimes even obscure a particular vibrational mode. One reason for this behavior may be that the dynamic dipole moment associated with a H–Me vibration is small because of the largely covalent character of the bonding (note that the Pauling electronegativities of H and the common transition metals do not differ very much). Another fact is that the H–Me bond lengths are usually short, that is to say, the H atom is located quite close to the surface. In many cases a site with a high coordination is occupied with the consequence that the H atom is practically immersed in the surface and thus electrically almost completely screened. Nevertheless all the well-known dipole selection rules apply for the specularly reflected beam [169] and principally allow the determination of a particular adsorption site. Another (quite useful) property of H adsorbate systems is the comparatively large isotope shift when using deuterium instead of hydrogen which facilitates the assignment of a certain observed vibrational band. There are, of course, also some systems such as H/W(100) which exhibit quite pronounced electron energy loss features, and numerous studies have concentrated on that adsorption system. Among others, cross sections for the excitation of certain vibrational modes have been determined from angle-dependent EELS measurements, and we refer for further details to the exhaustive description given by Willis [123]. Another quite interesting complication arises from the possibility of electronic excitations of surface states of the metallic substrate which in turn can be sensitively modified by adsorbed hydrogen. Hints to a strong energy dependence of the electron reflectivity were reported by Baró et al. [170] in their investigation of the H/Pt(111) system. Later, Conrad et al. studied this effect more thoroughly in the course of their vibrational analysis of the H/Ru(0001) [161], H/Pd(111) [162] and H/Rh(110) systems [171]. In these latter studies they showed convincingly that the electron reflectivity curve of the clean surfaces reveals maxima and minima at particular electron energies due to surface resonances. Depending on the primary beam energy the cross section for excitation of H–surface vibrations shows maxima which are correlated with the appearance of minima in the reflectivity curve. Consequently, an analysis of these resonances and an understanding of the details of the coupling mechanism is crucial since it can significantly modify the intensity of a certain H vibrational excitation and sometimes can even make a particular mode visible or invisible.

Before we actually present some examples of H–surface vibrations we shall briefly describe the correlation between the local symmetry or coordination of an adsorption site and the related vibrational frequency. Some hints as to where the particular modes are located are provided by IR spectroscopy data from transition metal hydrides. H atoms bound to a single metal atom absorb in the 2200–1600 wavenumber region whereas edge-bridging H atoms produce

fairly weak and broad absorption bands between 1400 and 800 cm^{-1} [172]. From this one may deduce that triply coordinated hydrogen should lead to absorption features in a region with even smaller wavenumbers.

Besides the *energetic position* of a vibrational loss the *number* of detectable vibrational modes is quite an important and revealing guide in assigning the adsorption site symmetry correctly. A particularly nice example for this is provided by the recent EELS analysis of the Ni(110)/H system by Ibach et al. [173] who could – on the basis of a careful analysis of the number and energetic position of H-induced losses – contribute to a solution of the long-standing question of the kind and number of H adsorption sites before and after a 1×2 surface reconstruction driven by one monolayer of hydrogen occurring with this system.

Following Froitzheim [174] we may state that with a homonuclear diatomic molecule such as H_2 adsorbed parallel to the surface in a symmetric bridge site the loss spectrum in the specular direction (here we only consider *dipole scattering*!) should consist of two loss peaks, whereby the one at the lower frequency is due to the surface whereas the high frequency mode reflects the stretching vibration of the molecular entity provided that it creates a perpendicular component of the dipole moment. This in turn depends on the force constant of the bond between the two adatoms and the tilt angle of the force constant of the bond between each of the adatoms and the substrate atom(s).

If there is "head-on" terminal bonding, again two losses should occur, corresponding to the vibration of the whole molecule against the surface (lower frequency mode) and to the stretching vibration of the molecular bond (higher frequency mode).

Bonding of an atom in a symmetric bridge position leads only to a single dipole-active mode whereas an asymmetric bridge bonding configuration causes a second additional mode with a perpendicular component that can be dipole excited.

Totally symmetric bonding of an atom generally leads to a single dipole-active vibrational loss regardless of whether the atom is located in a terminal, bridge, threefold or fourfold coordination, the so-called symmetric stretch mode. However, as soon as the site symmetry is reduced more vibrations will become dipole-active owing to the various perpendicular components of the force constants of the bonds between the H atom and the involved surface atoms. Again, the crucial ingredient is the total symmetry of the adsorption complex; the existence of, e.g., a mirror plane perpendicular to the surface reduces immediately the number of observable dipole-active modes. In this situation, off-specular measurements (which exhibit also the dipole-forbidden parallel modes) may be very helpful in that they allow one to find all possible normal vibrations of an adsorption complex.

Another complication should be mentioned, namely, the possible excitation

of overtones (second and higher harmonics of a vibration) and combination frequencies. Although for a harmonic oscillator the first overtone has no dynamic dipole moment anharmonicities of the potential can easily produce a dynamic dipole moment which in turn causes (fairly weak though) loss features.

We shall now compile some recently obtained vibrational loss data for H adsorbed on single crystal surfaces of metals and semiconductors. For each H bonding geometry we shall select a typical representative example.

2.8.1. The H adsorption site with fourfold symmetry

As typical examples of this bonding geometry we can regard the adsorption of hydrogen on many surfaces with square symmetry (but not all of them!) on which the H atom–surface complex belongs to the space group C_{4v}. We mention the systems H on Ni(100) and H on Pd(100) which have been investigated by Andersson [104] and Nyberg and Tengstål [105,175] respectively, and focus first on Andersson's early work. H_2 and D_2, respectively, were adsorbed onto a Ni(100) surface at 20 K up to a coverage of 0.9 Θ_{max} and then subjected to an EELS measurement. The spectra which are reproduced in fig. 17 exhibit a single loss feature at 74 ± 1 meV ($= 596$ cm^{-1}) for adsorbed hydrogen and at 52 ± 1 meV ($= 419$ cm^{-1}) for adsorbed deuterium. (Note that the loss intensities are fairly small!) The ratio of these energies is – as expected from the mass ratio of the isotopes – close to the square root of 2. The absence of any other high frequency mode indicates *dissociative* chemisorption as well as the exclusive population of *one* kind of adsorption site. From a comparison of the experimentally observed and theoretically predicted loss frequencies the author could unequivocally conclude on the symmetrical fourfold hollow adsorption site.

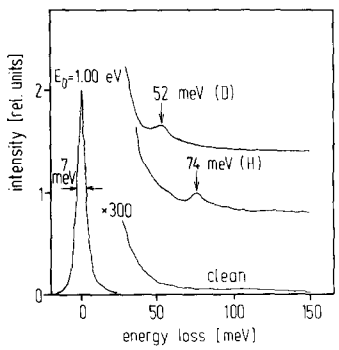


Fig. 17. HREEL spectra of H_2 and D_2 adsorbed to a coverage of $0.9\Theta_{max}$ on a Ni(100) surface at $T = 200$ K, under specular conditions. The primary electron energy E_p is 1.0 eV, the angle of incidence 47.7°. The H-(D-)induced losses are indicated. After Andersson [104].

The second example is, as mentioned above, the Pd(100)/H system. Whereas there is no well-ordered H phase formed on Ni(100) under Andersson's experimental conditions (in fact, his observation of a quasi-ordered H LEED phase has never been reproduced) it is well established that H forms a $c(2 \times 2)$ overlayer at $\Theta_H = 0.5$ on Pd(100) and a 1×1 layer at $\Theta_H = 1.0$ whereby the H atoms are likewise occupying the symmetrical fourfold hollow site. The existence of mutual H–H interactions which increase with increasing surface concentration can and will have an influence on the frequency of the H–Pd vibrations. This was convincingly shown by Nyberg and Tengstål [175], who observed a small linear upward shift of the H–Pd stretching vibration with H coverage and a (somewhat lesser) linear downward shift of the D–Pd vibration. For small coverages the corresponding losses appear at 60.2 meV (= 485 cm^{-1}) with hydrogen and at 45.1 meV (= 363 cm^{-1}) with deuterium; at coverages close to 1 monolayer they shift to 63.5 eV (= 511 cm^{-1}) and to 43.5 meV (= 350 cm^{-1}), respectively [105]. The authors conclude on the importance of the dynamical indirect interactions between adjacent adsorbed H atoms for the observed frequency shift. These interactions will be the subject of section 3.1.

A very recent paper by Richter and Ho [176] communicates results on the H/Rh(100) system where also the fourfold hollow site is occupied by hydrogen. For ~ 1 monolayer H coverage the perpendicular H–Rh stretching vibration, $\omega_\perp$, appears around 82 meV (= 660 cm^{-1}) whilst the corresponding D–Rh vibration is located at 58 meV (= 467 cm^{-1}). A second (weaker) feature at 152 meV (= 1224 cm^{-1}) is identified as the first overtone of ω which points to a noticeable anharmonicity of the H–Rh interaction potential. A third weak feature at 138 meV (102 meV for D) can only be observed in the off-specular mode and is interpreted as the doubly degenerate parallel mode ($\omega_\parallel$). At smaller H coverages of around 0.4 monolayers again a frequency shift is found, the H loss $\omega_\perp$ then appears at ~ 70 meV (= 563 cm^{-1}).

Concluding the three results described here we can state that the dipole-active mode of the H–metal stretching vibration for H adsorbed in the fourfold hollow site is located between ~ 500 cm^{-1} and ~ 700 meV, depending on the coverage and the kind of metal used for adsorption.

2.8.2. The threefold hollow adsorption site

Here the experimental data are more difficult to obtain which may be due to the fact that the fcc and hcp surfaces which provide the threefold site which C_{3v} symmetry are known to be very flat and smooth. HREELS measurements have been reported so far for H/Ni(111) [177,178], H/Pd(111) [162], H/Ru(0001) [161,179], H/Rh(111) [180] and H/Pt(111) [170], whereby in all of the cases the investigators complained about the weak intensity of the H-induced vibrational losses. With Ni(111) very weak and with Rh(111) [180] no clear H-induced loss features could be detected at all which was attributed

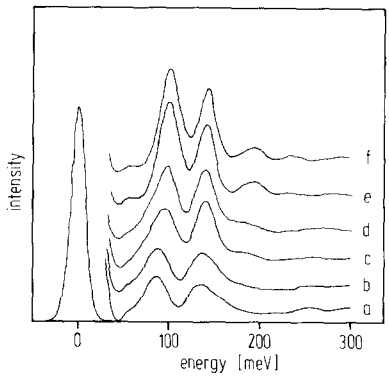


Fig. 18. Series of HREEL spectra of H_2 adsorbed on a Ru(0001) surface at $T = 115$ K showing the coverage dependence of the loss features. $\Theta/\Theta_{max} = 0.020$ (a), 0.35 (b), 0.53 (c), 0.78 (d), 0.92 (e) and 1.0 (f). The conditions were: primary energy $E_p = 5.5$ eV, angle of incidence $\sim 40°$, measurement in scattering plane, 10° off-specular. After Conrad et al. [161].

to the very small dynamic dipole moment associated with the H–Me stretching vibration. (Note that on all of these surfaces H_2 adsorbs dissociatively as atoms.)

As a first example we present the data by Conrad et al. obtained with a Ru(0001) sample [161]. At maximum coverage at 105 K the adlayer exhibits two dominant loss features, ω_1 at 102 meV ($= 821$ cm^{-1}) and ω_2 at 141 meV ($= 1135$ cm^{-1}) which were measured (for sensitivity reasons) in a 10° off-specular direction. In fig. 18 we reproduce the corresponding spectra obtained as a function of H coverage. Again, slight frequency shifts were observed as the H coverage increased. (At small coverages, the losses appear at 86 and 133 meV.) The fact that under off-specular conditions only two fundamental losses are found excludes a nonsymmetric threefold or a bridge site with high certainty, and a terminally bonded hydrogen on top of a Ru atom should lead to a much higher vibrational frequency. So the symmetrical threefold site is very likely occupied whereby, however, a distinction between fcc sites and hcp sites on the basis of the ELS measurements alone is not possible. The frequencies ω_1 and ω_2 are assigned as the modes parallel and perpendicular to the surface, respectively, that is: $\omega_1 = \omega_{\parallel}$ and $\omega_2 = \omega_{\perp}$.

As the second example the work by Baró et al. [170] on the H/Pt(111) system is chosen. Again, the weak intensity of the loss features is most striking. The spectra were recorded at ~ 90 K and at a H coverage of $\Theta_{max} = 0.7$ in and out of the specular direction. (As in the case of Ru(0001) H does not form an ordered LEED structure in the submonolayer range.) Two vibrational bands, A and B, were observed, namely, $\omega_{\perp}$ at 550 cm^{-1} (400 cm^{-1} with D_2) and $\omega_{\parallel}$, the asymmetric stretching at 1230 cm^{-1} (900 cm^{-1} with D_2). Note that here the reverse order of assignment is made. No other

frequencies could be detected and the authors concluded on the occupation of the threefold hollow site.

On Rh(111), Mate and Sormorjai [180] observed loss features at $\Theta_H = 0.4$ located at 92, 136 and 175 meV which were strongly dependent on the excitation energy (as in the case of Ru(0001) [161]). The losses were explained in terms of excitation of delocalized hydrogen to higher vibrational states (according to the model calculations by Puska et al. [51]). The results led the authors to propose a model in which the adsorbed H atoms are considered as a structureless "hydrogen fog".

With Pd(111), two fundamental losses are reported by Conrad et al. [162] under off-specular conditions, namely, ω_1 at 96 meV ($= 773\ cm^{-1}$) and ω_2 at 124 meV ($= 998\ cm^{-1}$). Higher order excitations are visible and help assigning ω_1 as the parallel and ω_2 as the perpendicular mode of hydrogen bonded in a threefold hollow site.

Summarizing the results presented above it turns out that the site with the C_{3v} symmetry causes two vibrational bands, ω_1 located between 500 and 850 cm^{-1} and ω_2 located between 950 and 1250 cm^{-1} (somewhat higher than those found with the fourfold hollow site) one of which is dipole-active.

2.8.3. The "quasi"-threefold hollow site

There are only recent reports which provide clear evidence for this site symmetry and which exhibit fairly conclusive results. Here we have to primarily mention the ELS investigations on the H/Ni(110) system by DiNardo and Plummer [181], Nishijima et al. [182,183], Jo et al. [184] and – with improved sensitivity – by Ibach et al. [173]. Work that is in progress is presently performed in Conrad's group on H/Rh(110) [171] and in our own laboratory on the H/Ru(10$\bar{1}$0) system [185].

As is evident from recent LEED structure determinations (cf. table 5) there apparently exists a special adsorption site for H on the fcc (110) and the hcp (10$\bar{1}$0) surfaces in which the H atom is located in an oblique quasi-threefold coordination with chemisorptive bonds to two top-layer atoms and one atom of the second layer of the substrate. This site has, of course, a different (and lower) symmetry than the threefold site of a hexagonal close-packed surface and, accordingly, gives rise to a richer loss spectrum. At least with Ni(110) and Pd(110) a surface reconstruction can be induced, depending on the surface temperature and the actual hydrogen coverage. This reconstruction definitely alters the surface geometry and hence the symmetry of at least one sort of H adsorption sites.

Following Ibach et al. [173] a relatively complicated picture emerges for the Ni(110)/H system. For the non-primitive 2×1-2H phases at $\Theta_H = 1$ which contains a glide symmetry plane along the [1$\bar{1}$0] direction azimuthal-dependent and angle-resolved ELS measurements were made. In specular direction, two principle losses were found in agreement with previous measurements, one

relatively strong mode at 650 cm^{-1} and a second broad feature near 1100 cm^{-1}. For off-specular scattering geometry (with the scattering plane aligned along the [001] direction) both these features remain, no additional mode is detected. With the scattering plane being turned by 90° into the [1$\bar{1}$0] direction, however, an additional mode between 860 and 910 cm^{-1} is obtained. Only the low and high frequency modes are dipole-active. The observation of *three* modes along one symmetry plane and the lack of the non-dipole active mode along the perpendicular plane require that the site symmetry is C_s with the mirror planes of the two sites being orthogonal to each other. This conclusion is entirely consistent with the structure model as derived from He diffraction [20] and LEED [129] measurements.

The pairing-row reconstruction occurring with Ni(110) above the monolayer coverage of hydrogen was also scrutinized with respect to the H adsorption site by Ibach et al. The occurrence of additional losses around 900–930 cm^{-1} and 1220–1260 cm^{-1} in the [1$\bar{1}$0] scattering plane led the authors to the assumption that besides the still existing (and only slightly modified) adsorption site a second hydrogen species is formed which should be located in a site with C_s symmetry with the mirror plane directed along the [1$\bar{1}$0] direction. This then is associated with the second "quasi"-threefold site provided by the pairing-row reconstructed Ni phase.

With Rh(110) which does not heavily reconstruct upon H adsorption there are preliminary reports by Conrad et al. [171] that at least three loss features appear at 84, 114 and 130 meV at small H coverages where the p(1 × 3) LEED phase is present. At somewhat higher coverages frequency shifts are observed and also overtones can be seen. The three fundamental frequencies are $\omega_1 = 85$ meV (684 cm^{-1}), (63 meV with D_2), $\omega_2 = 128$ meV (= 1030 cm^{-1}), (95 meV with D_2) and $\omega_3 = 141$ meV (= 1135 cm^{-1}), (105 meV with D_2). The 85 meV mode is attributed to the perpendicular vibrational mode, the others as non-degenerated parallel modes for a H atom being adsorbed in a quasi-threefold adsorption site, consistent with the LEED structure model [70,186].

Very recent investigations on the H/Ru(10$\bar{1}$0) system [185] revealed also at least three vibrational losses, a doublet around 90–100 meV (725–800 cm^{-1}) and a single loss at ~ 164 meV (~ 1320 cm^{-1}). All the features depend sensitively on the incident electron beam energy and the hydrogen coverage. Further measurements for assigning these vibrations unequivocally are underway.

With the rather scarce data material presently at hand we may deduce that the "quasi"-threefold site has its principal dipole-active frequency in the range between 650 and 800 cm^{-1} with non-degenerate parallel modes located between 800 and 1300 cm^{-1}.

2.8.4. The twofold (bridge) site

The example that we exclusively focus on is the already mentioned

W(100)/H system, for which very early ELS measurements by Probst and Piper [187] and later improved experiments by Froitzheim et al. [126] revealed loss features that were reconcilable with the occupation of a twofold adsorption site of C_{2v} symmetry. Depending on the H coverage, two losses were observed in specular direction, namely, a single loss at 155 meV around $\Theta_H \approx 0.4$, and an additional loss at 130 meV for H coverages around $\Theta_H = 1.0$. As the H surface concentration approaches 2 monolayers (the saturation density on W(100)) the 155 meV loss disappears. Since then numerous HREELS studies have been performed by Willis and his group [127,128,188], which confirmed the earlier studies but also provided a whole lot of additional information. In particular, systematic off-specular measurements revealed additional modes at 80, 160 and 260 meV, whereby the two losses are assigned as non-degenerate parallel vibrations of a H atom in a bridge configuration, and the 260 meV loss as a two-phonon satellite on the 130 meV vibration

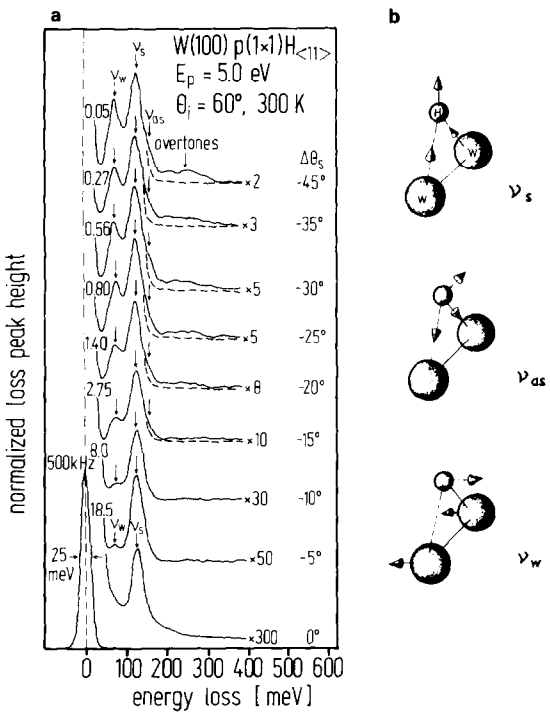


Fig. 19. (a) Angle dependence of the vibrational losses as measured by HREELS observed for a p(1×1) H layer on a W(100) surface scanning away from the specular beam towards the surface normal ($\Delta\Theta_s^0$) and along the [11] direction of the surface mesh. ν_s denotes the symmetric stretch perpendicular to the surface, ν_{as} the asymmetric stretch and ν_w the wagging mode, parallel to the surface. After Willis [123]. (b) Model of the possible vibrations ν_s, ν_{as} and ν_w of a H atom adsorbed on a W(100) surface in a bridge site geometry.

[188]. Willis has given an extensive description of the vibrational features of the H/W(100) system [110], and in order to illustrate the situation, we reproduce from his article a series of angle-dependent loss spectra along with a model of the possible surface vibrations of the adsorbed H atom (figs. 19a and 19b) on the W(100) surface.

As will be shown in more detail in section 3.6.3 a serious complication arises with this system because (i) the clean W(100) is reconstructed and (ii) H alters this and induces a new reconstruction, very likely consisting of a local pairing of adjacent W atoms in the surface. As in the case of Ni(110) the H adsorption geometry is affected and thus the surface vibrations, that is, the loss features.

Besides W(100) also the W(110) surface has been reported to exhibit bridge-bonded hydrogen, and we refer to the original literature for further details [189]. A critical review on earlier assignments of H–W vibrational losses was written by Sheppard [190].

Another example which is evidence of a bridge-type bonding of adsorbed hydrogen is the ELS work by Baró and Erley [191] on H adsorption on a Fe(110) surface. At 300 K a single vibrational band at 1060 cm^{-1} was obtained at specular conditions, off-specular measurements disclosed an additional loss at 880 cm^{-1}. The authors interpreted these features within a simple central force model and attributed them as reflecting the symmetric and asymmetry stretching modes of H in a "short"-bridge site.

However, as later dynamical LEED studies showed, rather the threefold site is occupied by H on Fe(110) [124], and the validity of the central force model in every case must be questioned.

2.8.5. The terminal adsorption site

The only examples so far for terminal adsorption of hydrogen on single crystal surfaces stem from hydrogen-on-semiconductor systems (silicon, germanium, gallium arsenide, indium arsenide, etc.). However, as shall be pointed out in more detail in section 3.7.2, we can no longer assume a simple adsorption of hydrogen but must rather consider a surface *reaction* (hydride formation) in which silane species with truly chemical Si–H bonds are formed. The evidence comes from a variety of EELS measurements, in particular by Schaefer and collaborators [192–194]. On the Si(100)-2 × 1 reconstructed surface two hydrogen phases are observed (cf. fig. 57) – a monohydride phase with a Si–H stretching frequency at 260 meV (= 2093 cm^{-1}) and a Si–H stretching frequency at 260 meV (= 2093 cm^{-1}) and a Si–H bending mode at 78 meV (= 628 cm^{-1}), and a dihydride structure with four vibrations, namely, a rocking mode at 60 meV (= 483 cm^{-1}), a wagging mode at 80.5 meV (= 648 cm^{-1}), a scissor mode at 113.5 meV (= 914 cm^{-1}) and a Si–H stretching mode at 262 meV (= 2109 cm^{-1}). On the Si(111)-2 × 1 reconstructed surface which was studied among others by Froitzheim et al. [93], the following

frequencies were observed and assigned: Again, a monohydride phase with losses at 76 meV (Si–H bending mode) and 257 meV (Si–H stretching vibration) and a peak at 100 meV ($= 805\ cm^{-1}$) which was explained as due to a H atom bound in a bridge-like configuration to two adjacent step Si atoms. Note that in all other cases the hydrogen atom(s) are bound in a terminal configuration. On Si(111)-7 × 7, Kobayashi et al. [195] found for the Si–H species the symmetric stretching vibration at 256 meV ($2057\ cm^{-1}$) and compared this value with the stretching vibrational energy of gaseous SiH in the $X^2\Pi$ ground state (which is between 2080 and 2042 cm^{-1} [196,107]). They also derived – for a harmonic potential – the force constant κ of the Si–H bond to be 0.156 atomic units and compared it with the theoretical prediction of $\kappa = 0.175$ a.u. by Appelbaum and Hamann [197]. Likewise, Herrmann and Bagus [198] have calculated the force constant of a Si–H cluster to be 0.168 a.u. using *ab-initio* Hartree–Fock LCAO methods, in excellent agreement with the experimental result.

The bending mode of the Si–H bond is not observed in dipole scattering because this mode (as pointed out earlier) does not belong to the totally symmetric representation of the point group of the Si–H surface species. We also want to point to recent (high-resolution) infrared spectroscopy work on Si(100) and Ge(100)-2 × 1 surfaces covered with hydrogen by Chabal [199]. This study shows that symmetric and asymmetric stretching modes of Si–H and Ge–H bonds can nicely be followed by this technique. From the currently available vibrational data we can construct a table of the vibrations that are possible with H atoms (molecules) adsorbed on various surfaces. This is presented in table 7.

2.9. Surface diffusion

Comparatively much less experimental and theoretical effort has been devoted to the surface diffusion of hydrogen. In the single particle picture we understand the surface diffusion as a process in which a given H atom adsorbed on a site "A" can reach a site "B" (of identical symmetry) of the surface lattice. More complicated are migration processes which comprise the *correlated* diffusion of two or several adjacent adsorbed H atoms. Of course, the diffusion of a single particle will also strongly depend on the surface coverage, i.e., depend on the number of "empty" adsorption sites through which "hopping" and diffusion steps can occur.

Although quite often neglected in adsorption studies surface diffusion is a very important processes. It influences especially the kinetics of the adsorption and desorption reaction. When dealing with surface diffusion one must distinguish lateral mobility in the physisorbed (i.e., molecular) and in the chemisorbed state.

Table 7
Surface vibrations of adsorbed hydrogen

Surface (hkl)	Observed frequencies (cm^{-1})	Observed frequencies (meV)	Mode	Coordination	References
Si(100)-2×1	2087	259	Si–H asym. stretch	Terminal	[199]
	2099	261	Si–H sym. stretch		
	2110	262	$Si–H_2$ asym. stretch	Terminal	
	2093	260	Si–H stretch	Terminal	[192,193,194]
	628	78	Si–H bending	Terminal	
Si(111)-2×1	2070	257	Si–H stretch	Terminal	[93]
	612	76	Si–H bending	Terminal	
	806	100	Si_2–H stretch	Bridge	
Si(111)-7×7	2057	256	Si–H sym. stretch	Terminal	[195]
Fe(110)	1060	132	Fe–H stretch	Twofold (short bridge)	[191]
	880	109	Fe–H asym. stretch		
Ni(100)	596	74	Ni–H sym. stretch	Fourfold hollow	[104]
Ni(110)					
[1$\bar{1}$0] scattering plane	650– 590	81– 73	Ni–H stretch symmetric and asymetric modes	Quasi-threefold	[173]
	860– 910	107–113			
	off spec.				
	1050	130			
[001] scattering plane					
	650	81			
	1100	137			
Ni(110)-1×2					
[1$\bar{1}$0] scattering plane	630– 610	78– 76			
	720 off spec.	89			
	1040	129			
	930– 900	115–112			
	1260–1220	156–151		Quasi-threefold, Two kinds of coordination (C_s symmetry)	[173]
[001] scattering plane	630– 650	78– 81			
	1040	129			
	450 off spec.	56			
	930	115			
	1240–1210	154–150			

Ni(111)	709	88	Sym. Ni–H stretch	Threefold octahedral	[178]
	1120	139	Sym. stretch	Threefold tetrahedral	
Ge(100)-2×1	1979	246	Asym. stretch	Terminal	[199]
	1991	247	Sym. stretch	Terminal	
Mo(100)	1020–1270	127–158	Sym. stretch	Twofold (bridge)	Y.J. Chabal, S.B. Christman, J.A. Prybyla and P.J. Estrup, in: Proc. 5th Intern. Conf. on Vibrations at Surfaces (1987) p. 0-1
Ru(0001)	822	102	Asym. stretch	Threefold hollow	[161]
	1136	141	Sym. stretch		
	846	105	Sym. out of plane	Threefold hollow	[179]
	1112	138	Asym. in plane		
Rh(100)	661	82	Sym. stretch	Fourfold hollow	[176]
Rh(110)	685	85	Sym. stretch	Quasi-threefold	[171]
	1031	128	Asym. stretch		
	1136	141	Asym. stretch		
Rh(111)	740	92	Excitations into hydrogen bands	Hydrogen "fog"	[180]
	1100	136			
	1410	175			
Pd(100)	511	63.5	Sym. stretch	Fourfold hollow	[105]
Pd(111)	773	96	Asym. stretch	Threefold hollow	[162]
	998	124	Sym. stretch		
W(100)	1070–1300	133–161	Sym. stretch	Twofold (bridge)	Y.J. Chabal, S.B. Christman, J.A. Prybyla and P.J. Estrup, in: Proc. 5th Intern. Conf. on Vibrations at Surfaces (1987) p. 0-1
W(100) p(1×1)	1047	130	Sym. stretch	Twofold (C_{2v} symmetry)	[123]
	644	80	Asym. stretch		
	1289	160	Asym. wagging		
Pt(111)	550	68	Sym. stretch	Threefold hollow (C_{3v} symmetry)	[170]
	1230	153	Asym. stretch		

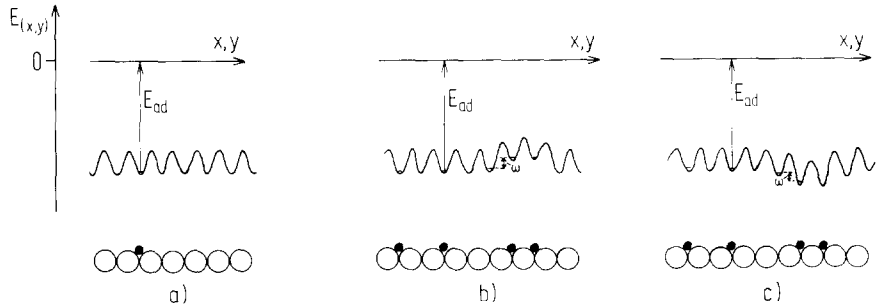


Fig. 20. Variation of the potential energy parallel to a solid surface ($x-y$ direction). (a) Clean surface with a *single* adsorbed H atom. (b) Surface covered with H atoms repelling each other and leading to a decrease of E_{ad} by ω. (c) Same as (b) except that the H atoms attract each other leading to an increase of E_{ad} by ω.

The corrugation of any periodic surface is certainly much less pronounced in the van der Waals potential minimum several ångströms away from the surface, with the consequence that the mobility in this weakly held state is usually very high. If this physisorbed state is significantly populated by H_2 molecules we commonly observe the typical "precursor" state kinetics of adsorption (cf. section 3.4) which involves a trapped but still mobile H_2 molecule travelling over large distances (many lattice constants) parallel to the surface in search of an unoccupied dissociation site.

A different situation is faced once a H_2 molecule has been dissociated into the H atoms which then are located in the substantially deeper chemisorption potential wells. As fig. 20 illustrates there is a lateral variation of the chemisorption energy by $\sim 1/10$ of the heat of adsorption (as experience shows) whereby a sinusoidal potential energy variation in $x-y$ direction can be assumed. In other words, the $E(z)$ chemisorption potential is modulated in $x-y$ direction according to the periodicity of the surface lattice. In order for an adsorbed H atom to get from one minimum to an adjacent one a certain amount of energy is required, namely, the activation energy for diffusion, E^*_{diff}, unless tunnelling is possible. In analogy with the Frenkel equation we may define a residence time τ' of a H atom in a given adsorption site as

$$\tau' = \tau'_0 \exp(E^*_{diff}/kT). \tag{14}$$

In the multi-particle description and by introducing the diffusion coefficient D this expression may be transformed to

$$D(T) = D_0 \exp(-E^*_{diff}/kT), \tag{15}$$

in which $D(T)$ is defined by Fick's law in terms of the time dependence of the concentration divided by the spatial variation of the concentration in one dimension, x (number of particles per cm^2)

$$D = -\left(\frac{\partial N}{\partial t}\right)_x \Big/ \left(\frac{\partial N}{\partial x}\right)_t. \tag{16}$$

In principle, E^*_{diff} can be obtained from measurements of D at different temperatures. Thus one task consists of determining the diffusion coefficient. This obviously requires to follow and monitor both the travelling of a boundary of adsorbed H atoms into a region of unoccupied sites as well as the dilution of a given population as a function of time.

A very helpful overview over surface diffusion in general is provided by the article of Morris et al. [78] who also give a fairly extensive listing of previous experimental diffusion measurements. Also the problem of lateral interactions is dealt with which becomes relevant for higher adsorbate concentrations. Another important review article on the field of surface diffusion of adsorbates was written by Nauvomets and Vedula [200].

Probably one of the first attempts to follow the surface diffusion of hydrogen on surfaces was undertaken in Gomer's group. They used W [201], Ni [202] and Pt [203] field emitters and watched the field emission tip half of which was covered with adsorbed hydrogen. It was found that the H_{ad} front was constantly moving and expanding even at temperatures as low as 150 K indicative of a rapid diffusion of chemisorbed hydrogen. Since then much more refined experimental techniques have been developed which, to a great extent, can still be traced back to field emission measurements, for example, the density fluctuation method [204] introduced by Gomer and his group. Also, laser-induced thermal desorption as described in refs. [155,205] has been refined to a powerful technique. The diffusion of *hydrogen* has been studied quite frequently on tungsten surfaces. Here we refer to the work by Gomer and coworkers [204,206] who followed the temperature dependence of the diffusion coefficient $D(T)$ by means of the aforementioned field emission-fluctuation spectroscopy. Particularly interesting are the results for W(110) in the low coverage regime for the three hydrogen isotopes, H, D, and T whereby these results urge one to distinguish a low temperature range $0 < T < 90$ K in which tunnelling dominates and a high temperature range $T > 100$ K where the "classical" Arrhenius form (eq. (15)) applies. For H on W(110) $D(T)$ is shown in fig. 21 for the three hydrogen isotopes. However, low T tunnelling diffusion exhibits isotope effects several orders of magnitude smaller than expected by simple rigid lattice models, and the high T behavior shows an inverse isotope effect much larger than this model predicts. As Auerbach et al. [206] showed it is possible to explain both features in terms of a single model of H–W interactions with a large separation in time scales between the motion of a H and a W atom.

In a recent study the group of White [207] uses laser-induced thermal desorption (LID) in order to map the surface diffusion of hydrogen and deuterium on a Ni(100) surface. This method is well established as several other recent investigations showed, e.g. for H (D) on Pt(111), [208], H on Ru(0001) [209] and H on Ni(100) [205,210]. Basically, a well-defined small area of a H covered surface is illuminated by a laser pulse which leads to

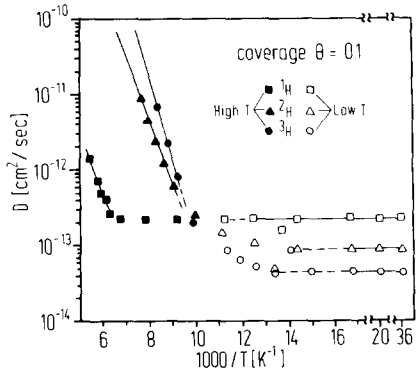


Fig. 21. Plot of the logarithm of the diffusion coefficient, D [$cm^2\ s^{-1}$], against the reciprocal temperature for the three hydrogen isotopes at low coverage ($\Theta = 0.1$) diffusing on a W(110) surface. The different behavior of the low and high temperature regime is indicated. The experimental data points are fitted by straight lines which were developed from theoretical diffusion models. After Auerbach et al. [206].

complete desorption. Thereafter the generated "empty" area is allowed to refill through diffusion from the surrounding boundary. Later laser pulses on the same spot at well defined time intervals probe the amount of refilling.

Mullins et al. [207] report on the following data for H/Ni(100): The activation energy E^*_{diff} is less for H_{ad} than D_{ad} (14.6 kJ/mol compared to 18.6 kJ/mol) with pre-exponentials of 2.5×10^{-3} $cm^2\ s^{-1}$ (H) versus 8.5×10^{-3} $cm^2\ s^{-1}$ (D). These numbers give an impression about the order of magnitude of both E^*_{diff} and D_0.

For H on W E^*_{diff} ranges around 25 kJ/mol with $D_0 = 1.8 \times 10^{-5}$ $cm^2\ s^{-1}$ [203]; for H on a Ni field emission tip $E^*_{diff} = 30$ kJ/mol was reported [201]. For comparison, the diffusion of *physisorbed* tritium on a Ni(111) surface at 15 K was activated by only 0.84 kJ/mol with a pre-exponential of 3×10^{-4} $cm^2\ s^{-1}$ [211]. As compared to other adsorbates like CO_{ad} or O_{ad} on W surfaces both the H pre-exponentials and activation energies are much smaller thus suggesting a much faster surface migration of hydrogen at a given temperature.

This is indirectly also suggested by the observation of the time required to form certain H phases with long-range periodicity. In our own measurements with H on Ni(110) or Rh(110) we found that even complicated lattice gas phases (c(2 × 6) [21] or p(1 × 3) [70,186]) are formed almost instantaneously even at temperatures around 80 or 100 K. Here it is worthwhile to point again to the concept of quantum motion which was originally proposed in our work of H on Ni(111) [50] and later pursued and refined by Nørskov and collaborators [51] to describe the motion of an adsorbed H atom or a proton, parallel to a periodically corrugated surface by means of a proton band model. At higher

vibrational excitations the narrow vibrational ground states broaden and overlap and explain the extremely rapid motion of an adsorbed H atom or proton at elevated temperatures.

3. The multi-particle interaction and collective phenomena

We may directly continue the considerations of the foregoing section as we discuss the behavior of a whole ensemble of adsorbed hydrogen molecules or atoms on a periodic surface. Two major points have to be kept in mind:
(i) The surface exhibits a lateral periodic potential, i.e., provides sort of checkerboard of adsorption sites. Only for the zero coverage limit all these sites can be assumed to be *identical* (fig. 20a).
(ii) Even two equally adsorbed particles may interact with each other whereby the strength of interaction depends on their mutual distance. This leads to a superposition of the sinusoidal potential $E(x, y)$ and the mutual interaction potential E_{H-H} (which may be attractive or repulsive). These two situations are depicted in figs. 20b and 20c. It will be shown in the following section that it is this modification of the energetic structure of a surface that largely governs the thermodynamics but also influences the kinetics of adsorption.

3.1. The lateral interaction forces

An important general experimental observation is that chemisorbed and physisorbed gases frequently form two-dimensional phases with long-range order. This order can be detected easily by diffraction techniques such as LEED. *Hydrogen phases* are no exception here although the weak scattering power of H atoms requires some considerations if LEED is used as a probe for the long-range order. Moreover, the pronounced lateral mobility of H can prevent the formation of ordered phases at elevated temperatures. The existence of an ordered overlayer with lattice spacings different from the underlying solid surface indicates a so-called "induced site heterogeneity". This means that the adsorption itself divides the initially equivalent adsorption sites into favorable and less favorable sites. Figs. 20b and 20c illustrate how this comes about: Any repulsive H–H interaction curve tends to *lower* the heat of adsorption in the direct vicinity of the H–surface complex and, vice versa, an *increase* in E_{ad} is achieved by attractive H–H interaction forces. However, the potential energy curves $E(x, y)$ usually are not a simple superposition of two interaction potentials as one might suggest. Rather, one has to treat the full quantum mechanics of the chemisorption of adjacent particles on a surface. Thinking in terms of electron wave functions it is evident that there may not only be *direct* repulsion occurring between vicinal H 1s orbitals of adsorbed atoms (so-called "direct" interactions). Rather, also

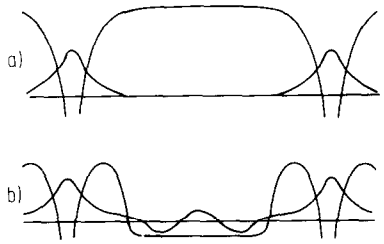


Fig. 22. Electron wave functions for two free H atoms (a) and for two atoms adsorbed on a metal surface (b). After Grimley [212].

the electronic structure of the solid underneath will be disturbed. If the solid is a metal, its s- or p-electron states are described by wave functions with periodic properties in x–y direction, i.e., parallel to the surface. It is evident that any disturbances caused by the adsorbed atom are not well localized and may be conducted forth to surface sites which are not a priori affected by direct orbital repulsions of H atoms adsorbed in these sites. Therefore, the corresponding "through-metal" interactions are called "*indirect*" *interactions*. Following Grimley [212] the origin of these interactions is illustrated in figs. 22a and 22b which show the potential energy and the ground-state wave function Ψ_{1s} for an electron in the field of two identical nuclei when these are far apart in a vacuum. Ψ_{1s} decreases rapidly in the space between the nuclei. However, if the two atoms are brought close to metal surface there is another path of interaction leading *through the metal* with the consequence that the vacuum level is depressed in the space round the two nuclei (fig. 22b). There are now wave functions (with energies close to the original atomic level) which are large on both atoms and which still oscillate in the space between them (located in the metal) instead of decreasing rapidly to zero. Thus the metal acts as a "medium" for coupling the two adsorbed atoms by propagating the wave function from one atom to the other. In other words: Two adatoms are coupled if the same electrons are used to bind them *both* to the surface.

The existence of this indirect interaction was first concluded by Koutecký in 1958 [213]. Since then a wealth of quantum-chemical calculations have been carried out in the field. We only mention the work by Grimley and Walker [214] dealing with the interaction of two hydrogen-like atoms on a metal surface, by Grimley and Torrini [215] calculating the indirect interaction between two H atoms on a W(100) surface. Other calculations consider similar pairwise interactions, and as a rough guide the magnitude of these forces can be given to be of the order of 50 meV for an adatom–adatom distance of two lattice constants ($\sim$ 6–7 Å) [216, 217]. One other important result of these calculations is that the long-range charge density (Friedel) oscillations around an adsorbed atom may lead to an oscillatory behavior of the indirect interac-

tions [214,216,218]. As an example, the work by Lau and Kohn [219] is mentioned in which a jellium and a tight-binding model was investigated and compared with the experiment. A later theoretical work by Johansson and Hjelmberg [220] communicates an expression for adatom–adatom interactions which can describe the interaction between H atoms on a jellium surface but can also predict LEED structures of H adsorbed on the (111) surfaces of Ni, Pd and Pt. Other important theoretical contributions were made to the lateral-interaction problem, e.g., by Lundqvist et al. [221] or by Muscat [222].

So far, only pairwise adatom–adatom interactions were considered. However, there is no reason not to take into account indirect interactions that comprise more than two adjacent atoms. As will be shown later (cf. section 3.6) the operation of *three-body* interaction forces can affect the formation of certain long-range adsorbate phases markedly. In an important theoretical study Einstein [223] proved the existence of these "trio"-interactions for the system O-on-W(110) which turned out to be essential also for the H/Fe(110) adsorption system [224,225]. From the present data body available it appears as if the indirect interaction forces are particularly relevant during the chemisorption of hydrogen on metals. One possible reason may be that the small H atom can get very close to the surface and induce large Friedel oscillations. Thereby attractive interaction forces can operate between H atoms that are separated even by several lattice constants. One extreme example here is the formation of a $c(2 \times 6)$ LEED phase at a H coverage of 0.33 on a Ni(110) surface [20,21] which structure implies that an adsorption site six lattice spacings, i.e. more than 20 Å, away is affected by an adsorbed H atom. Other examples are the $c(2 \times 4)$ phase on Co(10$\bar{1}$0) [69] or the 1×3 phases on Rh(110) [70,186].

Actually very little is known about the lateral interactions between H atoms or rather H_2 *molecules* physisorbed on non-metallic substrates. The observation of ordered LEED structures especially on graphite(0001) surfaces as reported by Seguin and Suzanne [83] or by Fain and coworkers [84] indicates that the occurrence of the aforementioned interactions is not limited to *metallic* substrates. In fact, in his pioneering work Koutecký [213] demonstrated that indirect interactions between adatoms could come about by an electron cloud penetrating the crystal of a non-metal.

For the sake of completeness the interested reader is referred to somewhat related recent experimental and theoretical work on molecular adsorption (physisorption) of diatomic molecules on the basal plane of graphite, especially of H_2, N_2 and CO. Here very interesting long-range ordering (and hence interaction) effects were experimentally observed with N_2 and CO at very low temperatures ($\sim$ 10 K) [226]. LEED investigations as well as neutron diffraction studies are currently performed on H_2 (D_2) on graphite (0001) by Fain and his group [227]. In section 3.6.1 we shall expand somewhat more on this subject.

In the subsequent sections we shall briefly discuss the effect of the lateral interactions on the chemisorptive binding energy of hydrogen and on the kinetics of adsorption and desorption. Also, work function effects will be considered.

3.2. The coverage dependence of the heat of adsorption

On the experimental side one of the most interesting questions is concerned with the *magnitude* and with the *sign* of the lateral interaction forces, i.e., whether they are attractive or repulsive. We shall henceforth use the symbol ω_{ij} for the interaction energy between the atom i and its neighbor j, and we define $\omega > 0$ as an attractive and $\omega < 0$ as a repulsive energy.

It has been known for a long time that with most of the adsorption systems the highest energies of adsorption are obtained at small adsorbate coverages, and a more or less pronounced decay of the heat of adsorption follows as the concentration of the adsorbate increases. Clearly, this kind of behavior would suggest the operation of *repulsive* mutual interactions between the adatoms. In a very simple case, namely the dipole–dipole interaction between positively charged adsorbed alkali atoms, these interactions can be calculated fairly accurately, and a good agreement with experimental data is obtained. However, if we deal with almost neutral adatoms held by a covalent bond any predictions are much more difficult. In addition to the most commonly observed repulsive forces in some cases also attractive interactions may dominate, at least in a certain coverage range. This then leads to an *increase* in the heat of adsorption with island formation and condensation phenomena.

Apparently, we can obtain the relevant information as to which interaction forces dominate from a measurement of the coverage (concentration) dependence of the heat of adsorption. This in turn can be achieved either by isosteric heat measurements (which require the mapping of adsorption isotherms ($\Theta(P)_T$) or isobars ($\Theta(T)_P$)), or by performing thermal desorption experiments with varying initial coverage. Most important is the isosteric heat of adsorption, q_{st}, which is defined thermodynamically as the latent heat exchanged in the phase equilibrium: gas phase (pressure P) – adsorbed phase according to

$$q_{st} = RT^2 \left(\frac{\partial(\ln P)}{\partial T} \right)_{\Theta}. \tag{17}$$

Sometimes it is necessary to distinguish the integral and the differential heats of adsorption which are related by

$$q_{diff} = \left(\frac{\partial q_{int}}{\partial n_A} \right)_T \tag{18}$$

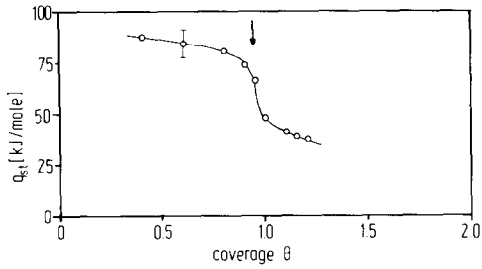


Fig. 23. Experimental verification of the coverage dependence of the heat of adsorption showing the onset of *repulsive* mutual H interaction by an arrow. The example is taken from our work on H/Rh(110) [70].

and in which expression n_A denotes the number of moles of adsorbed gas of kind A. q_{st} and q_{diff} are interrelated by

$$q_{st} = q_{diff} + RT, \tag{19}$$

the difference at 300 K being of the order of 2.5 kJ/mol.

Measurements of q_{st} as a function of coverage yield directly the variation of the binding energy with adatom concentration $E_{Me-H}(\Theta)$ and thus reflect the influence of lateral interaction energies. For CO adsorption on Pd(100) there exists a brilliant publication by Tracy and Palmberg [228,229] in which $q_{st}(\Theta_{CO})$ was determined by work function and LEED measurements. It was convincingly shown that q_{st} decreased only slightly over a fairly large coverage range but dropped sharply as the mutual distance between the CO molecules became so small that (in this case: direct) interaction forces could become effective. With H adsorption, where these direct orbital repulsions do not play a comparable role *indirect* interactions may dominate even at quite small coverages. Moreover, spurious influences of surface defects show easily up especially in the low coverage regime. As an example (besides the one already presented in fig. 12 for H on Pd(111)) we show the q_{st}–coverage dependence obtained for the system H/Rh(110) where a significant drop around $\Theta_H \approx 0.7$ suggested the sudden onset of repulsive lateral interactions [70] (fig. 23).

Even more convincing are two other examples taken from our work on H on Rh(110) [70] and H on Ru(10$\bar{1}$0) [71]. Here, thermal desorption spectroscopy was employed to map the H binding situation for different surface coverages. Both sets of thermal desorption (TD) curves reveal a sharp low temperature state denoted α_1. As parallel LEED investigations show one can directly correlate the appearance of the respective α_1 state with the beginning formation of a highly compressed H (1×1) overlayer, with mutual H–H distances as small as ~ 2 Å. Clearly, strongly repulsive interaction forces are operative in these 1×1 phases. The situation is illustrated in fig. 24 showing the H_2 TD curves at saturation from Ru(10$\bar{1}$0) (a) and Rh(110) (b). Also

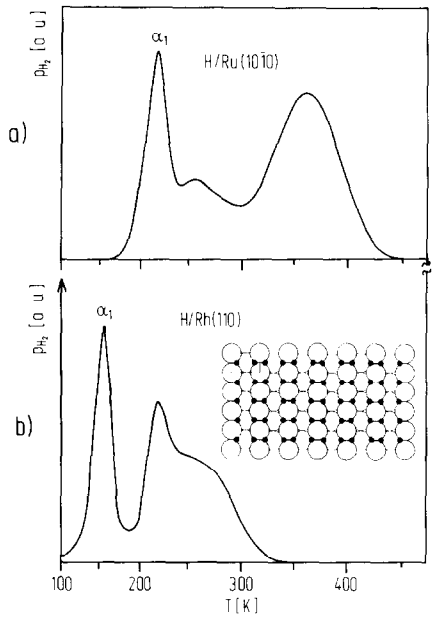


Fig. 24. Thermal desorption spectra showing the formation of a low-temperature α_1 state caused by repulsive lateral interactions between adsorbed H atoms at saturation coverage ($\Theta_H = 2.0$). (a) H/Ru(10$\bar{1}$0). (b) H/Rh(110); also shown is a structure model of the final 1×1-2H phase illustrating the close packing of the H atoms.

shown is the tentative structure model for the (1 × 1)-2H phase with the small mutual H–H distances.

Whilst most interactions turn out to be repulsive there are few examples for the occurrence also of *attractive* lateral interaction forces. This means that the system *gains* energy as the coverage increases. The example shown in fig. 25 is taken from our work on the H/Ni(110) system [155]. Whilst we find a relatively small heat of adsorption at low coverages of ~ 85 kJ/mol, q_{st} increases in a small coverage regime by ~ 12 kJ/mol. This increase is ascribed to the operation of attractive interactions between H atoms adsorbed in the troughs of the Ni(110) surface. However, the situation is somewhat more complicated since the Ni(110) face reconstructs to a "streak" phase under the respective experimental conditions [230,231]. This can and probably will also affect the energetics. This is a first example for the fact that it is by no means straightforward to prove the existence and determine the magnitude of lateral interactions. With increasing adatom surface concentration other processes may compete – lateral ordering effects with energy and entropy contributions, surface relaxations or reconstructions, overlayer–underlayer transitions, etc. All these may cost or contribute adsorption energy and therefore will affect

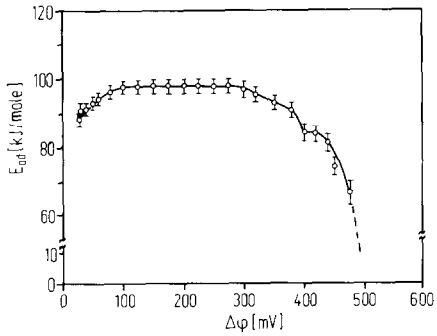


Fig. 25. Coverage dependence of the isosteric heat of adsorption for H adsorbed on a Ni(110) surface (again, $\Delta\varphi$ was taken as a coverage monitor). Note the initial increase of E_{ad} caused by attractive lateral interactions! After ref. [155].

the $E_{ad}(\Theta)$ relation. As a summary, various $E_{ad}(\Theta_H)$ dependences are shown in fig. 26.

Somewhat more revealing may be the coverage dependence of thermal desorption binding states which, in many cases, can be directly taken from a series of TPD curves obtained for various coverages. There are quite many kinetic theories which include the operation of lateral interactions ω of attractive or repulsive character, and it was shown, e.g., by King [232] that the lineshape of TPD curves was strongly modified by those interactions in that a

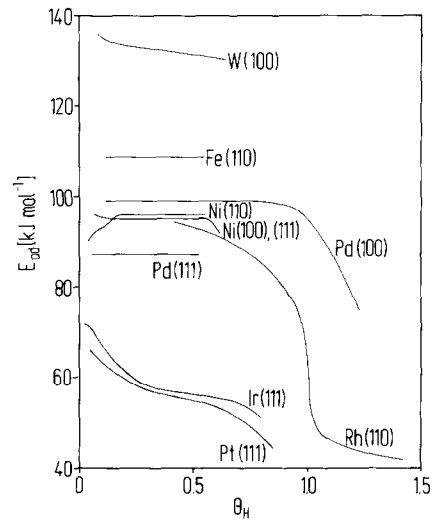


Fig. 26. Compilation of various coverage dependences of the heat of adsorption. See text for further details.

single peak structure could even produce two new states at different temperatures. Various analytical kinetic expressions have been derived which explicitly contain ω and could, in principle, be used to determine this quantity. On the other hand, by introducing ω into a model calculation, the experimental TPD lineshape could be predicted. An example here for a metal–hydrogen system is provided by Hellsing and Mällo [233] who correctly predicted the two-peak structure observed experimentally for the system H on Ni(100) [56] by assuming repulsive nearest neighbor interactions. In general, for H transition metal adsorption systems the development of a two-peak structure in TPD appears to be the rule. Besides Ni(100) also Ni(111) [50,67,155], Pt(111) [158], Ru(0001) [113,135] exhibit a shoulder state at higher H coverages which is caused by repulsive lateral interactions. Nevertheless, it is quite difficult also with a TPD analysis to correctly extract the lateral interaction parameters. Most if not all of the TPD analyses rely on the validity of simple first or second order rate equations [232], although there is ample evidence that phenomena like precursor states [234–238] or structural phase changes [231] can alter the TPD lineshapes completely.

A more elegant way to deduce sign and magnitude of lateral interactions is provided by investigations of two-dimensional phase transitions within the adlayers as will be shown further below.

3.3. The coverage dependence of the H-induced work function change

Some remarks should be added here about how the mutual adatom interactions show up in the coverage dependence of the work function change $\Delta\varphi(\Theta)$. Such a dependence is to be expected since – as pointed out in the foregoing section – the origin of these interactions is a quantum-chemical one which affects the charge distribution in the surface and in the adlayer and hence the experimentally measured surface potential. Despite a wealth of available experimental data on this subject there is still a lack of basic understanding of the work function–coverage effects, for simple reasons (cf. section 2.7.2): Again, $\Delta\varphi$ maps all structural and electronic changes of both the surface region and the bulk of the solid, and there are only a few examples in the literature where these contributions could be delineated. To complicate things even more, there are also cases in which pronounced structural transitions in the adlayer do *not* affect $\Delta\varphi$ noticeably.

In addition to any peculiarities of the $\Delta\varphi(\Theta)$ function also the *maximum attainable work function change* caused by H adsorption, $\Delta\varphi_{max}$, is an important intrinsic property of the respective system which is governed by the strength of the depolarization forces (dipole–dipole interaction), but is also affected by possible structural phase transformations within the surface region and the final location of the H adsorption site at saturation. Whenever $\Delta\varphi$ is 1 : 1 related with the surface coverage it is self-evident that, with the $\Delta\varphi$–cover-

age function being known, $\Delta\varphi$ can be used as a very convenient coverage monitor to determine thermodynamic and kinetic quantities of an adsorption system. This has been extensively utilized for the determination of adsorption isotherms [70,77,106,131,155,158,228,229]. The coverage dependence of $\Delta\varphi$ map be expressed as:

$$\Delta\varphi \propto \Theta_{\mathrm{H}}^{n}. \tag{20}$$

Depending on the coverage exponent n three cases may be distinguished: $n < 1$, $n = 1$, and $n > 1$. If only depolarization effects play a role, n is smaller than 1 and the work function rises less than linearly (often after an initial linear rise). These effects lead to the well-known Topping formula [239] which works particularly well for ionic adsorbates. If the Helmholtz equation (eq. (12)) would hold for the entire coverage range we had always $n = 1$. This case is normally fulfilled only at small coverages where no significant lateral interactions occur – frequently, for example, in the case of CO adsorption. With hydrogen, however, there may be significant deviations from linearity even at quite small coverages because of the very long-range interactions at the early stages of adsorption (cf. section 2.7.2). This then may lead to $n < 1$ (if $\omega < 0$) or sometimes to a situation in which, at least for a certain coverage interval, $n > 1$, that is, an overproportional rise of $\Delta\varphi$ with Θ. We shall now present some examples for all these three cases.

Interestingly, when scrutinizing the available hydrogen data, there is actually not a single case in which $n < 1$ over the entire coverage range. At most, an almost linear behavior is found, for example, with H on the Ni(100), (110) and (111) faces at room temperature [155]. More often one finds a relatively complicated $\Delta\varphi(\Theta)$ relation which usually depends on the surface temperature, because ordered phases with their effect on $\Delta\varphi$ can only form at sufficiently low temperatures. In fig. 27 we have compiled, for several H–metal systems, $\Delta\varphi(\Theta)$ functions. It is obvious that many curves show a left-hand curvature, that is, reveal a larger apparent dipole moment for higher coverages thus belonging to the category with $n > 1$. Thereby mostly linear sections can be distinguished in which the slope $\partial\Delta\varphi/\partial\Theta$ remains constant. Examples are H on Pd(100) [77], Rh(110) [70] or Ru(10$\bar{1}$0) [71]. In all these cases a clear correlation exists between the H coverage and the adsorption geometry. Other curves displayed in fig. 27 show a reversal of the sign of $\Delta\varphi$ at a certain coverage: H/Co(10$\bar{1}$0) and Ni(100) as well as H/Ru(10$\bar{1}$0) are examples where $\Delta\varphi(\Theta)$ runs through a maximum. Here, obviously a state is populated in which the direction of the net charge transfer is reversed. This is probably one reason why there exist so pronounced differences in the maximum work function changes observed with chemisorbed H layers on metals. A selection of experimental values of $\Delta\varphi_{\max}$ is given in table 8.

The figures vary between several tenths of an eV up to almost 1 eV. It may be worthwhile to add here that some theories, especially the effective medium

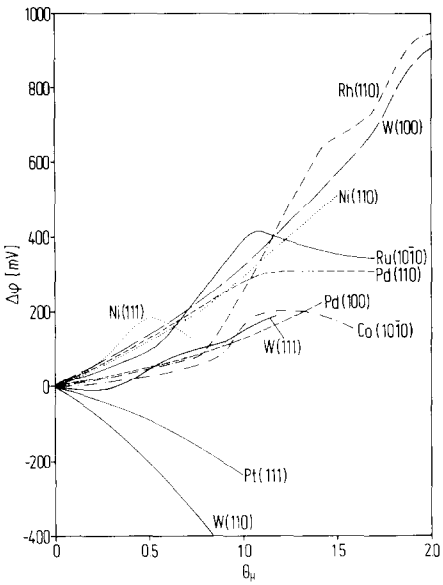


Fig. 27. Comparison of the coverage dependence of the H-induced work function change, $\Delta\varphi$, for a variety of metal single crystal surfaces. Note that in most of the cases $\Delta\varphi$ varies non-linearly with H coverage.

theory introduced by Nørskov and Lang [34], are able to predict both the sign and magnitude of a H-induced work function change. Also, Feibelman's theoretical approach (self-consistent slab calculations) [33] renders a calculation of the H-induced charge transfer possible and hence allows a prediction of the H-induced work function change. However, all these theoretical models are not sophisticated enough to fully account for the complex $\Delta\varphi(\Theta_H)$ relations displayed in fig. 27.

3.4. The coverage dependence of the H adsorption kinetics

In section 2.3 we have discussed probable mechanisms leading to hydrogen adsorption, and have introduced the initial sticking probability, s_0, as the crucial parameter that governs the initial rate of adsorption. However, any time a H_2 molecule dissociates and chemisorbs it finally occupies *two* adsorption sites. The simplest model to describe this situation has been given by Langmuir [240] which is characterized by the following assumptions: At all coverages sites with identical symmetry are occupied, which also implies that the surface is homogeneous. There are *no* lateral interactions between the adatoms, and the particles are considered immobile after having collided with

Table 8
Maximum values, $\Delta\varphi_{max}$, of H-induced work function changes of single crystal surfaces

Surface (hkl)	Temperature (K)	$\Delta\varphi_{max}$ (mV)	Remark	Reference
Fe(110)	150	−85		[159]
Fe(100)	150	+65		[159]
Fe(111)	150	+220		[159]
Co(10$\bar{1}$0)	100	+207		[69]
Ni(100)	300	+160		[155]
	120	+100 [a]	$\Delta\varphi(\Theta_H)$ runs through maximum	[56]
Ni(110)	120	+510		[21]
	120	+530		[155]
Ni(111)	300	+195		[155]
	120	+180 [a]	$\Delta\varphi(\Theta_H)$ runs through maximum	[50]
Ru(0001)	173	+25 [a]/−50	$\Delta\varphi(\Theta_H)$ runs through maximum	[113]
	273	+25 [a]/−150	$\Delta\varphi(\Theta_H)$ runs through maximum	[113]
Ru(10$\bar{1}$0)	100	+420 [a]	$\Delta\varphi(\Theta_H)$ runs through maximum	[71]
Rh(110)	85	+930		[70]
Pd(100)	120	+200		[77]
Pd(110)	250	+250		[68], V. Penka, Diplomarbeit, University of Munich (1982)
	190	+280		
	130	+320		
	300	+360		[106]
Pd(111)	300	+180		[106]
W(100)	300	+900		[253]
Ir(110)	120	+300 [a]/−140	$\Delta\varphi(\Theta_H)$ runs through maximum	[243]
Pt(110)	120	+170 [a]/−500	$\Delta\varphi(\Theta_H)$ runs through maximum	[244]
Pt(111)	150	−300		[158]
Pt(997)		+20 [a]/−360	$\Delta\varphi(\Theta_H)$ runs through an initial maximum	[131]

[a] The numbers presented give the maximum of the positive $\Delta\varphi$ value.

the surface. For completely mobile adsorption (mobile in the *chemisorbed* state) a simple second order rate expression can be derived

$$dn_{ad}/dt = s_0\, f(\Theta)\, P_{H_2}(2\pi mkT)^{-1/2} \tag{21}$$

(Hertz–Knudsen equation), with:

$$f(\Theta) = (1-\Theta)^2. \tag{22}$$

(If the layer is immobile, e.g., at very low temperatures, $f(\Theta)$ must be modified to yield (with z the number of nearest neighbor sites)):

$$f(\Theta) = z(1-\Theta)^2(z-\Theta)^{-1}. \tag{23}$$

There have been numerous experimental and theoretical articles published

which focus on the determination and description of the sticking probability–coverage function $s(\Theta)$. An excellent review here is provided by the work of Morris et al. [78].

The rate of adsorption can principally be obtained from the variation of any physical quantity with time which is directly related to the adatom concentration. The H-induced work function, $\Delta\varphi$, has already been mentioned; other quantities are, e.g., the optical absorption coefficient, the surface conductivity or the thermopower [241]. Another straightforward way to determine the coverage is direct gravimetry. We refer to the work by Kasemo and Törnqvist [242] who followed the amount of hydrogen adsorbed on Pt(111) by weighing even fractions of a H monolayer by means of a quartz microbalance. The peak intensities of Auger transitions or X-ray photoelectron spectra are quite suitable to monitor the surface coverage of heavier adsorbates such as N_2, CO or O_2. For physical reasons this fails in the case of H adsorption. Probably most commonly used, however, is the thermal desorption technique. Here, the adsorption is interrupted at time t_0 and the sample is heated such that all adsorbed adatoms leave the surface by thermal desorption. In a closed system, the resulting partial pressure increase, ΔP, can be measured, and with the gas temperature, T_g, and the system volume, V, being known, the average rate of adsorption up to time t is given by

$$\left\langle \frac{dn_{ad}}{dt} \right\rangle = \frac{1}{t}\left(\frac{V\Delta P}{kT_g} \right), \tag{24}$$

where k is Boltzmann's constant.

In common TPD spectroscopy the system is pumped and one rather obtains a transient pressure increase upon desorption. Likewise, the number of particles adsorbed at time t, N_t, is obtained by a numerical integration of the TPD curve. In this case, the sample area A and the effective pumping speed, $S_{eff} = V/\tau$, must be known (τ is a time constant):

$$N_t = \frac{S_{eff}}{AkT_g} \int P(t)\, dt. \tag{25}$$

There are numerous examples in the literature where coverages and adsorption kinetics have been determined in this way [1,155,158].

Experience shows that there is hardly an example for a clear Langmuir adsorption kinetics. A second-order adsorption was reported for H on Rh(111) by Yates et al. [243] and for H on Ir(111) by Engstrom et al. [244]. Several other more previous reports have either not been performed on single crystal surfaces or under not very well defined experimental conditions and have to be viewed at with some care.

Relatively often, however, the hydrogen adsorption seems to obey a first-order rate expression, i.e.,

$$f(\Theta) = 1 - \Theta, \tag{26}$$

which is expected for molecular (associative) adsorption only. Examples here are the systems H on Ni(111) at 300 K [155], H on Ir(110)-1 × 2 (the β_2 state) [245], H on Pt(110)-1 × 2 [244], or H on Ru(0001) (the negatively "N" polarized state) [113]. In none of these cases, however, was there any indication for a molecular H_2 adsorption. A possible explanation for the first order behavior would be that the adsorption proceeds via an island growth mechanism so that the uncovered surface area would be proportional to $1\text{-}\Theta$. Another possibility could be that the dissociation is not rate-determining, but rather the trapping of the molecule. This was, e.g., proposed for H adsorption on Ru(0001) into the "N" sites by Feulner and Menzel [113].

So it appears that deviations from the simple Langmuir type of adsorption are the rule, for a number of reasons. Again, the influence of adatom–adatom interactions can be very pronounced. One immediately realizes that, e.g., island formation caused by attractive interaction forces creates two surface areas with different particle densities in which the sticking could be different, and deviations from eq. (22) occur. Kinetic models can be set up to describe the rate of adsorption in terms of the lateral interaction forces. Here, the quasi-chemical approximation is often used [246]. However, an attempt to quantitatively model the adsorption *kinetics* merely on the basis of lateral interactions frequently fails, because the existence of weakly held trapping states, the so-called precursor states, modifies the adsorption kinetics considerably. The reason is that the trapped particles are usually very mobile; they diffuse across the surface and can very effectively search for an empty adsorption site. In this case, one observes the typical convex $s(\Theta)$ dependence in which s does almost not drop until medium coverages but sharply falls as the population of the adsorption sites has proceeded such that the molecule can no longer find an empty site while diffusing across the surface in the precursor. The influence of these precursor states on the kinetics of adsorption was first realized by Ehrlich and by Kisliuk [247], and later reinvestigated by King [235], Schönhammer [237], Gorte and Schmidt [236] or Steinbrüchel [248]. Frequently analytical rate expressions were derived which – by assuming a simple lateral interaction energy ω as well as probability functions for the precursor population – could quite well predict the experimentally observed $s(\Theta)$ curves. An extensive consideration of this matter can be found in the article by Morris, Bowker and King [78]. Here we confine ourselves to the presentation of some experimental examples that have been reported in the literature for H adsorption. A clear evidence for the existence of a hydrogen precursor state for a chemisorbed system was obtained for the Pd(100)/H system by Behm et al. [77]. This is shown in fig. 28. Other investigations proved the importance of these mobile precursor states also for H on Ni(110) at temperatures below 200 K [249], for H on Pd(110) [68], H on Ru(10$\bar{1}$0) [71] and Co(10$\bar{1}$0) [69], as well as on Rh(110) [70] and H on W(100) [1,250–255], one of the most frequently studied systems. In summary, it appears as if

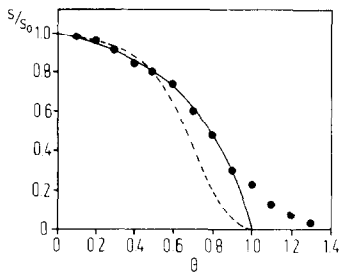


Fig. 28. Variation of the relative sticking probability, s/s_0, with hydrogen coverage for a Pd(100) surface at $T = 170$ K. Full circles: experimental data. Dashed line: second-order precursor model. Full line: first-order precursor model. After Behm et al. [77].

particularly the crystallographically "rough" metal surface planes would be very suited to provide a precursor state adsorption kinetics.

Apart from the relatively distinct precursor state influence all processes which contribute to the long-range order within the surface region may more or less disturb the simple adsorption kinetics of eq. (21). Among these are order–order and order–disorder phase transitions and particularly, reconstructive phase transitions involving the *substrate* surface layer. Whilst the effect of the long-range order of a merely adsorbed H layer on $s(\Theta)$ could not so clearly be established experimentally the influence of a H-induced surface reconstruction can give rise to pronounced alterations of this function which are relatively easy to observe. An example is provided by the H-on-Ni(110) system [231]). Depending on the adsorption temperature the surface under hydrogen remains stable up to $\Theta_H = 1.0$ and then suddenly transforms to a 1×2 reconstructed *pairing row* arrangement (this behavior is observed for $T_{ad} < 220$ K) or, for $T_{ad} > 220$ K, it reconstructs immediately to a streaky ("Str") (perhaps *missing row*) phase. Fig. 29 shows how this affects the exposure dependence of the coverage (the slope of this curve contains the sticking coefficient s). In fig. 29 $\Delta\varphi$ was taken as a coverage monitor. Obviously, the phase transformation $1 \times 1 \rightarrow 1 \times 2$ at $\Theta = 1.0$ gives rise to a plateau in the $\Delta\varphi$–exposure dependence. An evaluation of the true relation: coverage against exposure, however, does not yield this plateau anymore (which apparently is only a work function effect) but still exhibits a change of the sticking coefficient at $\Theta_H = 1.0$ which is more obvious from the $s(\Theta)$ plot of fig. 30. We deduce from this figure that the phase transformation $1 \times 1 \rightarrow 1 \times 2$ provides "new" adsorption sites for hydrogen on which the sticking can take place more efficiently than on the almost saturated non-reconstructed Ni face at $\Theta \approx 1.0$.

We just add that the dependence of $s(\Theta)$ on the surface crystallography is not a peculiar property of H adsorption. Dramatic effects of this kind were also reported for oxygen adsorption on Pt(100) surfaces (which are recon-

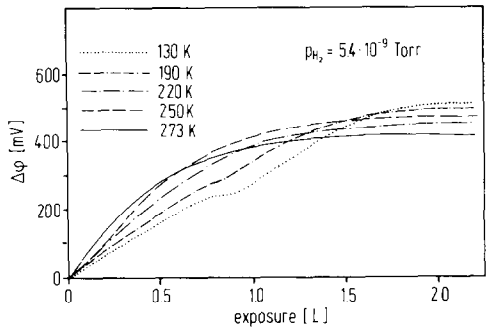


Fig. 29. Exposure dependence of the work function change, $\Delta\varphi$, produced by H adsorption on a Ni(110) surface, shown for various surface temperatures. The slope of the curves contains the sticking probability. Note that the plateau around 0.7 L exposure indicating the phase transition $2\times1 \rightarrow 1\times2$ occurs only for temperatures below 220 K!

structed in the clean state, with s_0 being small, and unreconstructed in the case of CO adsorption – with s_0 being at least an order of magnitude greater) [256].

Another very recent observation is concerned with H_2/D_2 isotope effects influencing the sticking probability at elevated H coverages on Rh(110) surfaces. The experimental observations which are illustrated in fig. 31 point to a pronounced dependence of $s(\Theta)$ on the long-range order of the adsorbate layer. H as well as D form among others a 1×2-3H phase on Rh(110) at $\Theta_H = 1.5$. Interestingly, D_2 adsorbs faster (higher s) into this phase than H_2, and one can only speculate on the reasons for this behavior [103]. Undoubtedly, however, the better ordering of the D_{ad} phases as compared to the H_{ad} phases improves the efficiency of the sticking.

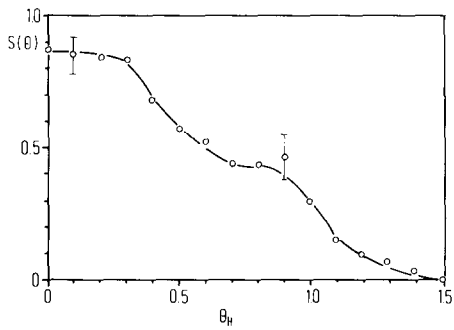


Fig. 30. Coverage dependence of the sticking probability of H on a Ni(110) surface at $T = 120$ K. Note the plateau around $\Theta_H = 0.8$ which is caused by the onset of the 1×2 surface reconstruction.

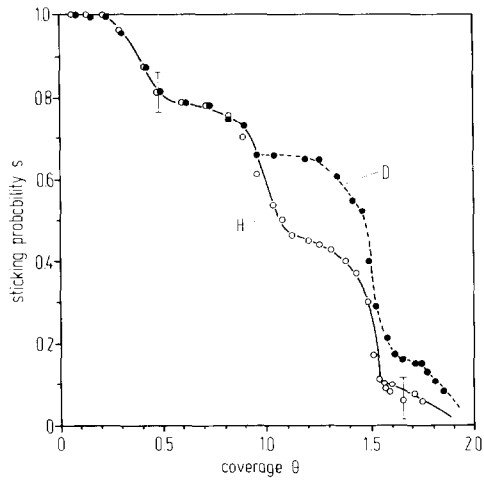


Fig. 31. Coverage dependence of the absolute sticking probability of H (D) adsorbing on a Rh(110) surface at 85 K. Full circles, dashed line: $s(\Theta_D)$. Open circles, full line; $s(\Theta_H)$. Most striking is the difference between H and D in the coverage range from 1 to 1.5 monolayers. After ref. [103].

3.5. Thermal desorption and the coverage dependence of the desorption kinetics

The reverse process to sticking and chemisorption is the desorption. Once being trapped in any potential of the surface the particle (H_2 molecule or H atom) can only leave the surface again if it has gained sufficient energy to reach the zero energy level. This is equivalent to breaking the bond to the surface. In case of an existing activation barrier in the *adsorption* path, ΔE_{ad}^{*}, this must additionally be surmounted. The energetics of this process is schematically illustrated in fig. 32.

There are various ways in which the required energy can be supplied to the bound particle, namely, by simply heating the substrate (whereby phonon coupling is responsible for the energy transport) – this is the *thermal* desorption or temperature programmed desorption (TPD) – or by particle (mostly: electron) impact (EID) and by illuminating the surface with electromagnetic radiation, e.g., laser light, whereby again thermal effects must be distinguished from true photon-induced desorption mechanisms in which the quanta excite the adsorbed particle directly. It cannot be the intention of this article to enter any discussion of the (certainly very interesting) mechanistic processes that are involved in the excitation of a single bound adatom. Instead, we refer to the excellent review articles which cover this field [257–259]. It is just mentioned that, e.g., the laser-induced desorption is a technique which is presently employed in a number of laboratories [258]. Nevertheless we have to state that there is a general lack of experimental data as compared to the research in the

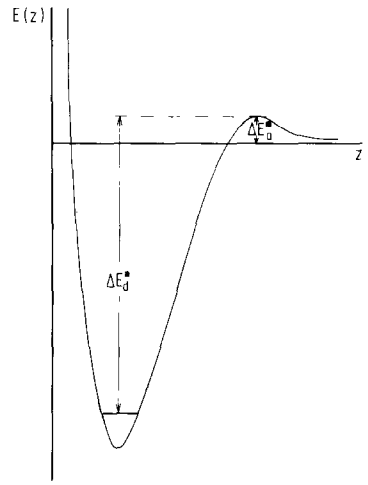


Fig. 32. Schematic one-dimensional potential energy diagram for desorption from an activated adsorbed state. ΔE_a^* denotes the activation energy for adsorption.

adsorption kinetics or dynamics. Also, theoretical investigations are scarce. Again, we mention the work of Kreuzer and Gortel [73,260] and the references given in their work. This is why we confine ourselves to the process of *thermal* desorption and here particularly on the kinetics of desorption of either H_2 molecules or H atoms from metal surfaces. Although the thermal excitation process may be complicated as well, it is by far the most commonly employed experimental technique and therefore a huge data body exists. A description of thermal desorption spectroscopy in general is given in numerous articles [257,259] to which we refer for any further details. Nevertheless, it is deemed useful to present some specific examples for H_2 desorption spectra and to expand on the analysis of the respective desorption kinetics.

For dissociative adsorption (which is the rule with H on transition metals) there is ample evidence that in most of the desorption reactions the rate-limiting step is the *recombination* of separated adsorbed H atoms to form the molecular transition state, $H^{\ddagger}_{2(ad)}$. Normally this exhibits such a small binding energy to the substrate that it desorbs immediately after its formation at the temperatures usually employed in the TPD experiment: The underlying reaction:

$$H_{ad} + H_{ad} \underset{k_{ad}}{\overset{k_d}{\rightleftharpoons}} H^{\ddagger}_{2(ad)} \xrightarrow{k'_d} H_{2(g)} \tag{27}$$

can be described by the rate expression:

$$d[H_{2(g)}]/dt = k_d[H_{ad}]^2 \tag{28}$$

(apparent second-order desorption kinetics), whereby it is tacitly assumed that $k_{ad} \ll k_d$ and $k'_d \gg k_d$ under the conditions of the experiment (high surface temperatures). k_d then can be expressed as a normal rate constant:

$$k_d = \nu_2(\Theta) \exp(-\Delta E_d^*(\Theta)/kT), \tag{29}$$

in which $\nu_2(\Theta)$ denotes the pre-exponential factor or "frequency' factor and $\Delta E_d^*(\Theta)$ the activation barrier (which is the depth of the chemisorption potential plus an eventual activation barrier for adsorption, ΔE_{ad}^*). Usually both ν and ΔE_d^* depend on the adsorbate concentration, Θ. According to eq. (28) $\nu_2(\Theta)$ has the dimension [$cm^2/s \cdot$ H-atom] for a true second-order rate process.

A more quantitative interpretation of the frequency factor may be obtained within the framework of transition state theory (TST) [261] whereby refinements of this theory regarding the thermal desorption have been made by Garrison et al. [262]. It can be shown that, for the zero coverage limit, the pre-exponential factor is obtained by equating the rates of adsorption and desorption, r_{ad} and r_{des} (which corresponds to the adsorption–desorption equilibrium):

$$\nu_2 = s_0 \bar{c} \frac{n_g}{n_{ad}^2} \exp\left(\frac{\Delta E_d^*}{kT}\right), \tag{30}$$

in which $\bar{c}$ denotes the mean velocity [cm s^{-1}] and n_g the number of H_2 gas molecules per unit volume [cm^{-3}]. n_{ad} stands for the number of H atoms adsorbed per unit area [cm^{-2}] of the surface. Using statistical mechanics the ratio n_g/n_{ad} for the equilibrium

$$H_{2(g)} + 2* \rightarrow 2\ H_{ad} \tag{31}$$

may be written as

$$\frac{n_g}{n_{ad}^2} = \frac{f_g f_{s,0}^2}{f_{ad}^2 f_{s,ad}^2} \exp\left(-\frac{\Delta E_{ad}}{kT}\right) \tag{32}$$

for the case of non-activated adsorption ($\triangleq \Delta E_d^* = E_{ad}$) and the limit of small coverages and non-interacting particles. f_g is the molecular partition function per cm^2 of H_2 gas $(2\pi mkT/h^3)^{3/2}$ and f_{ad} that for adsorbed H atoms per cm^2 of the surface area. f_{ad} is given by the total number of adsorption sites provided by the surface, N_s, times the complete partition function of the single adsorbed H atom. $f_{s,0}$ is the molecular partition function of the bare surface (except for contributions from the motions of the adsorbate). The partition functions $f_{s,0}$, $f_{s,ad}$ and f_{ad} can be combined to yield:

$$f'_{ad} = f_{ad} f_{s,ad}/f_{s,0} \tag{33}$$

and we obtain, by relating eqs. (30), (32) and (33) for the pre-exponential factor

$$\nu_2 = s_0 \bar{c} f_g / f_{ad}'^2 . \tag{34}$$

Thus, for the zero coverage limit ν_2 can be expressed by the initial sticking probability, s_0 and the partition function of the gas molecule, f_g, and the adsorbed phase, f_{ad}. f_{ad} is equal to $2\pi mkT/h^2$ if the adatoms are fully mobile and translationally excited, and it equals N_s, the number of available sites, if the adsorbate is immobile. Normally, the partition function of the substrate surface is not affected by the adsorption, i.e., $f_{s,0} = f_{s,ad}$ and $f_{ad}' = f_{ad}$, respectively. This assumption, however, may be considerably wrong, if, e.g., the substrate reconstructs under hydrogen. Then, soft phonon modes may become excited and $f_{s,0} \neq f_{s,ad}$. Such a change of the dynamical characteristics of the surface in the adsorbate–surface complex would lead to $f_{s,ad} \gg f_{s,0}$ and thus cause a large value for f_{ad}'. While there are several H–metal systems which undergo a H-induced surface reconstruction so far a careful coverage-dependent analysis of frequency factors has not been carried out. The problem is that the accuracy of ν determinations is usually very low, and only pronounced changes of ν will probably show up in the TD spectra. Nevertheless, for the H/Ni(110) system the existence of soft phonon modes was recently reported by Ibach and coworkers [263].

Determinations of *absolute* values for ν should be possible by a consequent application of TST. The rate constant for the second order (dissociative) desorption which passes an activated complex (cf. eq. (27)) reads

$$k_d = \kappa_d \frac{kT}{h} \frac{f_d^{\ddagger}}{f_{ad}'^2} \exp\left(- \frac{\Delta E_d^*}{kT} \right), \tag{35}$$

whereby κ_d is the transmission coefficient. A comparison with eq. (29) reveals immediately

$$\nu_2 = \kappa_d \frac{kT}{h} \frac{f_d^{\ddagger}}{f_{ad}'^2} . \tag{36}$$

It has been established that with hydrogen adsorption/desorption κ_d is of the order of 0.1, whereas with CO adsorption $\kappa_d \approx 1.0$ [264]. $f_d^{\ddagger}$, the partition function of the merely loosely bound activated complex $H_{2(ad)}^{\ddagger}$, can be split up further into the two-dimensional translation of this complex parallel to the surface as described by $f_{2D,trans}$ and the internal modes of the gaseous molecule, f_i (which comprise rotational and vibrational partition functions):

$$\nu_2 = \kappa_d \frac{kT}{h} \frac{f_{2D,trans} f_i}{f_{ad}'^2} . \tag{37}$$

As pointed out by Madix et al. [265] this yields for hydrogen

$$\nu_2 = 2.2 \times 10^{28} (f'_{ad})^{-2} \left[\mathrm{cm^2/s \cdot H\text{-}atom}\right]. \quad (38)$$

In table 9 we present a selection of frequency factors that have been obtained with hydrogen desorption. Apparently, most of the numbers range

Table 9
Compilations of some 2nd order frequency factors, ν_d, for hydrogen desorption

Metal surface (hkl)	Frequency factor, ν_d ($cm^2\ s^{-1}$ H-atom^{-1})	References
Fe(100)	1.5×10^{-2}	J. Benziger & R.J. Madix, Surface Sci. 94 (1980) 119
Ni(100)	3×10^{0}	[56]
	2.5×10^{-1}	J. Lapujoulade & K.S. Neil, Surface Sci. 35 (1972) 288
	8×10^{-2}	[155]
	5×10^{-2}	S. Johnson & R.J. Madix, Surface Sci. 108 (1981) 77
Ni(111)	2×10^{-1}	[155]
	2.3×10^{-2}	[50]
Mo(110)	10^{-2}	M. Mahnig & L.D. Schmidt, Z. Phys. Chem. 80 (1972) 71
Mo(100)	5×10^{-2} (β_2 state)	H.R. Han & L.D. Schmidt, J. Phys. Chem. 75 (1971) 227
Ru(0001)	3×10^{0} (state 1)	[113]
	5×10^{-2} (state 2)	[113]
	4×10^{-3}	[135]
Rh(111)	1.2×10^{-3}	[243]
Pd(100)	1×10^{-2}	[77]
Pd(110)	8×10^{-2}	[106]
W(100)	4.2×10^{-2} (β_2)	[1,255,268]
W(110)	10^{-2} (β_1)	[268]
	1.4×10^{-2} (β_2)	[268]
Ir(111)	2×10^{-6}	[244]
Ir(110)	1.5×10^{-2} (β_2)	[245]
	10^{-7} (β_1)	[245]
Pt(111)	10^{-3}	[23]
	3×10^{-9}	[158]
Pt(110)-1×2	$\sim 10^{-4}$ (β_2)	[244]
	3×10^{-4} (β_1)	[244]

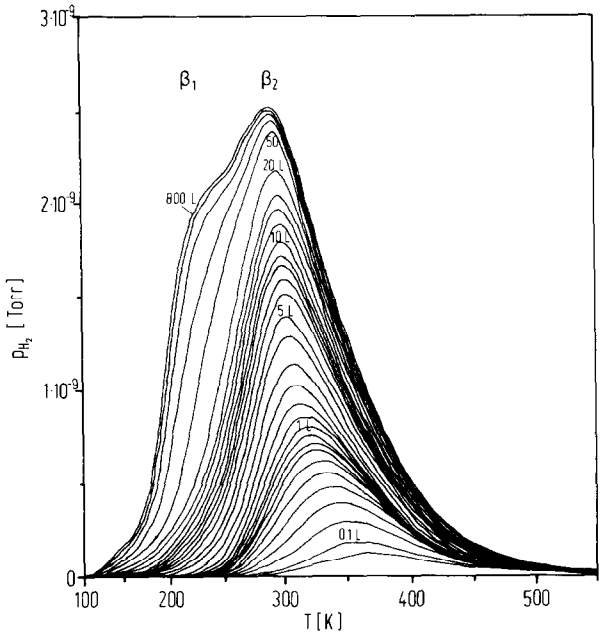


Fig. 33. Family of thermal desorption curves of the H/Pt(111) system for various exposures ranging from 0.05 to 800 L. The heating rate β was 10 K s^{-1}.

between 10^{-1} and 10^{-3} $cm^2/s \cdot$ H-atom. In terms of TST this would suggest a fairly mobile adsorption, in accordance with our previous statements.

Experimentally, a second-order rate process is characterized by a continuous shift of the TPD maximum towards lower temperatures with increasing coverage. A second-order desorption reaction has been reported so far for a variety of metal–hydrogen systems, for example for the high-energy binding states of H on Ni(111) [50,67] and on Ni(100) [56], on Pd(110) [68] and (111) [266], H desorbing from Rh(111) [243], Pt(100) [267], Fe(110) [159], Ru(0001) [113] or W(110) [268]. An example for a typical second order TPD spectrum is given in fig. 33 which was obtained with H on Pt(111) [158]. However, there are quite many cases in which true second-order desorption kinetics is difficult to establish. It may turn out that the TPD data can be interpreted also in terms of a first-order rate process. An almost "classical" example is provided by the system H on Pt(111) which was first investigated by Lang et al. [269] who claimed that there was no adsorption at all. Later reinvestigations by Lu and Rye [270], by Christmann et al. [158], by Collins and Spicer [271] or McCabe and Schmidt [272] clearly revealed that H chemisorption was indeed occurring, however, there were considerable discrepancies with respect to the desorption kinetics, the value of the frequency factor and the height of the

activation energy for desorption. Whilst the authors of ref. [158] claimed a second-order desorption and a very low pre-exponential [265] as well as a low activation energy for desorption McCabe and Schmidt considered a first-order process with a much higher frequency factor and a higher activation energy ΔE_d^*. This problem persisted until 1981 when molecular beam work in the group of Comsa revealed strong evidence for a higher activation energy for desorption which in turn makes a first-order order pre-exponential more likely [273]. It also turned out that the concentration of surface defects was crucially influencing the adsorption/desorption kinetics [60].

Whereas a second-order desorption is often observed for the high-energy binding states a first-order desorption kinetics has occasionally been reported, when additional low-energy binding states become populated. Examples are the β_1 state for H on Ni(110) [230,231], the α_1 and α_2 states for Rh(110) [70], and the α_1 and α_2 states for H on Pd(110) [68]. This first-order kinetics

$$d\left[H_{2(g)}\right]/dt = k_d'\left[H_{2(ad)}^{\ddagger}\right], \tag{39}$$

with

$$k_d' = \nu_1(\Theta)\exp(-\Delta E_d^*/kT) \tag{40}$$

and

$$\nu_1 = \kappa_d \frac{kT}{h}\frac{f_d^{\ddagger}}{f_{ad}'}\ [s^{-1}] \tag{41}$$

would suggest an *associative* desorption mechanism in which the desorptive removal of an already formed $H_{2(ad)}^{\ddagger}$ entity would at least formally be rate-limiting ($k_d' \ll k_d$ in eq. (27)).

With many of the first-order systems cited above, however, there is no other evidence for such a mechanism and one must caution that either coverage-dependent lateral interaction forces or structural phase changes within the adsorbed layer during the temperature program are responsible for those "pseudo-first-order" kinetics. Indeed, for the system Rh(110)/H [70] a careful inspection of the TPD spectra revealed small peak shifts indicating a possible slight fractional-order desorption kinetics. This could for example be caused by preferential desorption from the ends of linear H chains which are formed along the troughs of that (110) surface. Furthermore, with Pd(110)/H and Ni(110)/H clearly surface reconstruction phenomena and/or population of subsurface sites [275] are accompanying the adsorption which likewise modify the desorption kinetics.

Other complications, as mentioned before, may arise from the existence of precursor states in the desorption path which, according to the "detailed balancing", should modify the thermal desorption traces considerably, as long as the temperatures are low enough to ensure a noticeable population of these states. Shanabarger [276,277] studied the adsorption/desorption on iron and

nickel films and interpreted his kinetic results in terms of a precursor state model, that is, he regarded the formation of a molecular precursor state as the rate-limiting step in the desorption channel. Later King [235] and Gorte and Schmidt [236] subjected the precursor influence to a more rigorous treatment and showed that the desorption rate must be significantly modified. More recent work on the same subject by Alnot and King [278], Cassuto and King [238] and Alnot and Cassuto [279] led to an analytical expression for the desorption rate which contains only probability functions for adsorption and desorption from the precursor and for the surface migration in the precursor. The advantage of the formula by Cassuto and King [238] is the fairly general applicability. For more details about how precursor states can alter the desorption kinetics we recommend the review paper by Morris et al. [78] for further reading.

In principle, a *state selective* desorption spectroscopy is required to disentangle the details of the desorption process, for example, the various atomic or molecular excitations (translation, rotation, vibration).

An important contribution to the desorption dynamics of hydrogen from surfaces was, among others, published by Kubiak et al. [274] who studied the rotational and vibrational distribution of H_2 and D_2 recombinatively desorbing from clean Cu(110) and (111) surfaces following atomic permeation through the Cu crystals at $T_s = 850$ K and 960 K. Multiphonon excitation (ionization) was combined with time-of-flight mass spectroscopy. The rotational distributions of the desorbing H_2 (D_2) molecules were found to be non-Boltzmann and showed mean rotational energies which where 80%–90% of the surface temperature, T_s, regardless of the isotope and of the surface orientation. In contrast, the vibrational population ratio as given by the fraction of molecules with quantum numbers $v_1 : v_0$ was found approximately a factor of 100 greater than the ratio predicted by Boltzmann statistics at T_s. The conclusion was that detailed balance arguments applied to the vibrational distributions measured in recombinative *de*sorption are not suited to predict correctly the dissociative *ad*sorption probability as a function of vibration: It is concluded that these two processes are dynamically different at least for the Cu(110) and Cu(111)/H systems.

As in the case of the adsorption (cf. section 3.4) the lateral interaction forces can modify the desorption kinetics to a great extent. This is illustrated by fig. 34 which is taken from Adams' work [246] and shows how the magnitude of the nearest neighbor repulsive energy ω can really alter a thermal desorption spectrum. Goymour and King [280] used the quasi-chemical approximation to model the effect of repulsive lateral interactions on second-order thermal desorption. The presence of these interactions lowers the heat of adsorption (and hence E_d') as the coverage increases, and the frequency factor takes on a complicated coverage dependence which, among others, contains the interaction energy, ω and its coverage dependence.

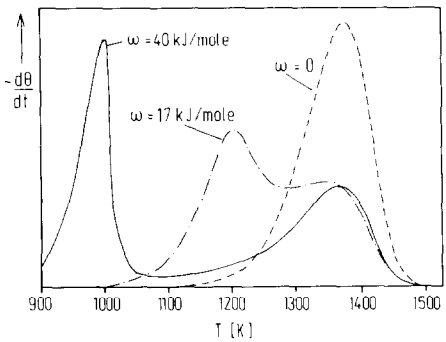


Fig. 34. Computed thermal desorption spectra showing the influence of lateral nearest neighbor repulsive interactions. Assumed was first-order desorption with $E_d = 318$ kJ/mol (dashed curve). Reproduced are TD traces for $\omega = 17$ kJ/mol (— — —) and $\omega = 40$ kJ mol (———). After Adams [246].

However, for the sake of all the complications mentioned above it seems to be not very wise to rely on TPD analysis *alone* to extract the information about sign and magnitude of lateral interactions. More promising is certainly an investigation of the long-range order effects which can be observed quite frequently with interacting particles on a surface.

3.6. Two-dimensional phase transitions in adsorbed hydrogen systems

Before we enter a detailed description of long-range order effects occurring with adsorbed hydrogen layers it may be useful to premise some remarks about the matter of two-dimensional phase transitions in general. Two-dimensional phase transition phenomena have attracted the interest of physicists for a long time, and there exists a vast amount of literature on that subject [281–286,288–305]. We want to draw particular attention to a very recent review on that subject by Bauer [306]. It is well-known since the early days of LEED that the formation of adsorbate phases with long-range periodicity is the rule rather than the exception, and such phases have been reported by Davisson and Germer with adsorbed oxygen on nickel surfaces in their first LEED experiments as early as 1927 [287]. LEED (as any surface-sensitive diffraction method) is especially suited to detect phases with long-range order even in cases where the order is not very well developed.

The *position* of the adsorbate-induced diffraction maxima is determined by the mutual distances of the scatterers, whereas the diffraction *intensities* depend among others on the degree of long-range order within the adsorbate layer. If we simply take the atomic scattering factor of an adsorbed particle as independent of the state of the surrounding adsorption sites (kinematic approximation, single scattering events) and describe the contribution from

thermal vibrations by an effective Debye–Waller factor, $\exp(-2W)$, which is independent of the degree of order of the surface layer we can express the LEED intensity of an "extra" spot:

$$I = I_p\, F(E_p, \varphi, \psi, \Theta)\, G(\text{order}) \exp(-2W), \tag{42}$$

in which I_p denotes the intensity of the primary LEED beam. E_p is the primary energy (eV), φ, ψ are the azimutal and polar angles of the impinging electron beam with respect to the surface, respectively, and Θ is the adsorbate coverage. With the geometrical and energetical parameters kept constant the "extra" intensity can be essentially regarded as a function of the long-range order alone

$$I \propto G(\text{order}) \exp(-2W) \tag{43}$$

and thus of the temperature of the adsorbate layer. This "configurational" part of the LEED intensity can be obtained by a summation over the individual scattering contributions of all adatoms within the LEED coherence zone consisting of M^2 adsorption sites:

$$I_{\text{conf}} = \frac{C}{M^4} \sum_r \sum_{r'} \langle f_r f_{n'} \rangle \exp(\mathrm{i}\Delta \boldsymbol{k}(\boldsymbol{r} - \boldsymbol{r}')), \tag{44}$$

in which $\Delta \boldsymbol{k} = \boldsymbol{k} - \boldsymbol{k}'$ is the difference between the wave vectors of the scattered and the primary electrons and $\boldsymbol{r}$ the location of the adsorption site of scattering factor f_r. The brackets $\langle\ \rangle$ denote the ensemble average. The expression (44) can be regarded as the basis for all measurements in which the temperature dependence of the LEED intensities is taken to map the order of an adlayer at a given temperature. Correlations between LEED intensities and ordered adlayers were stated in various early studies [288,289]. From the variety of later LEED studies we simply refer to the work by Lagally et al. [290] which contains more recent developments. A theoretical approach was made among others by Doyen et al. for a square lattice [291]. These authors considered order–disorder phase transition phenomena in adsorbed layers in terms of a lattice gas model.

A lattice gas model is a simple and frequently used basis for describing adsorbate phases with long-range order. The lattice gas approximation assumes bonding of the adatoms in specific sites on the periodic substrate. These sites can be empty or singly occupied, and the interaction between the adatoms is only a function of their mutual distance. In this model the local geometry is *not* affected by the adsorption itself, and the phase diagram (e.g., the temperature–coverage–order dependence) is only determined by the lateral adatom interactions ω.

Returning to our description given in section 3.1 (fig. 20) we recall that the potential energy $U(x, y)$ is lowered in the case of repulsive and increased in the case of attractive mutual interactions. At $T = 0$ K all adatoms will be

located in the potential minima of fig. 20c, and therefore form a phase with perfect long-range order. However, increasing temperature (that is: thermal energy) will give more and more atoms the ability to migrate from one site to another, and disorder will gradually be introduced until the temperature has reached a value where any long-range order does no longer exist: An order–disorder phase transition has occurred. It is just mentioned here that there is a general thermodynamic classification of these phase transitions into first-order and continuous transitions. For more details we refer to the literature [292]. With LEED such a phase transition shows up by the disappearance of the adsorbate-induced "extra" spots and by a broadening of their half-width, respectively. In other words, the function $I(T)$ drops to zero. There are many cases in which the strength (and the sign) of mutual interactions depend on the *concentration* of the adatoms, Θ. Then it is often observed (depending on the coverage Θ) that *different* ordered phases are stable at a given temperature – the system exhibits order–order transitions. Again the LEED "extra" intensity I can be measured as a function of the adsorbate coverage Θ at constant temperatures, viz. $I(\Theta)_T$, to reveal those transitions.

The reason for the interest in 2D-phase transitions is that they give insight into the nature and strength of the adatom–adatom interactions and also provide examples and test cases for theoretical models of two-dimensional critical phenomena [293–304].

A relatively simple theoretical means to model the temperature-dependent order of an adsorbed ensemble is provided by the Monte Carlo simulation and we refer to the monograph by Binder [299] for further details. For our purposes the work by Ertl and Küppers [303] is particularly interesting. They consider a surface lattice of n^2 lattice points and adsorb Θn^2 particles on that surface. These particles are allowed to diffuse, that is, to change sites. The system changes from state 1 with energy E_1 to a state 2 of energy E_2 with the probability $P = \exp(-(E_2 - E_1)/kT)$ if $E_2 > E_1$. Furthermore, $p = 1$, if $E_1 > E_2$. The pairwise mutual interaction energy is a parameter in the calculation and governs the total energy of the system and hence the probability of a site change. $E(x, y)$ is the energy of a particle bound on a site $A(x, y)$ and is obtained by a summation over all the interaction energies ω_{jk} (in terms of thermal energy kT) with all nearest and next nearest neighbors:

$$E(x, y) = \frac{1}{kT} \sum_{jk} \omega_{jk}. \tag{45}$$

Beginning with a statistical distribution of the particles the Monte Carlo procedure is applied 10^5 to 10^6 times and the equilibrium distribution of the particles is analyzed in terms of the interaction energy. Although Küppers' work dealt with oxygen adsorption on Pd(110) it is quite straightforward to adapt the technique also to H-on-metal systems and to include more than just a single interaction energy parameter.

There are various models in statistical mechanics which predict the degree of long-range order, that is the configuration of the adatoms, as a function of the temperature, for a fixed set of lateral interaction energies [292]. For strictly pairwise ω and for a square lattice the two-dimensional Ising model [304] predicts the relation between the critical temperature, T_c, of the ordered phase and the interaction energy ω as

$$\omega = 1.76kT_c. \tag{46}$$

Other surface geometries (triangular lattice, rectangular lattice) are expected to yield very similar results.

Besides the Ising model there are a variety of other statistical models such as the x–y model, the 3-state Potts or the 4-state Potts model, which allow a classification of the systems with respect to the so-called "universality classes". Again, reference is being made to relevant literature only [283].

The essence here simply is that more or less detailed information can be obtained on the sign and magnitude of the mutual interaction energies simply from measuring critical temperatures of ordered phases and their critical exponents (the latter are required in order to classify the respective system to a given universality class). Examples for this with regard to H_2 chemisorption will be given further below.

However, in order to determine, e.g., the critical exponent of an order–disorder transition very careful measurements of the degree of order as a function of the temperature must be performed. So far only for very few systems the critical exponent was obtained by LEED measurements, one example being the O/Ni(111) system [296,305]. For chemisorbed *hydrogen* phases and their transitions critical exponents could not be measured yet because of the usually very low transition temperatures and inherent experimental problems of the LEED method to image the degree of order (for example, finite size effects, short-range order effects, etc.).

There is often another severe complication observed with chemisorbed H phases as already pointed out before. Owing to its comparatively small size the chemisorbed H atom is usually located very close to the substrate atoms and frequently induces relaxation or even reconstruction processes. Then the whole phase transition becomes much more complicated since the lattice geometry itself becomes a function of the adatom coverage, and, for example, the simple lattice gas model no longer applies. This will be emphasized in conjunction with the discussion of the chemisorbed H phases and their transitions. In the following we shall give selected examples for an ordered *molecular* hydrogen phase and particularly for some chemisorbed (H_{ad}) phases.

3.6.1. Two-dimensional phase transitions in physisorbed H_2 systems

Physisorbed systems such as rare gases (Ar, Kr, Xe) on single crystal surfaces exhibit a whole variety of interesting phases and phase transitions,

and a lot of work has been directed so far to the determination of the phase diagrams of, e.g., argon physisorbed on graphite surfaces in the temperature range from 0 to ~ 40 K. In particular, specific heat measurements have turned out to be very suited to analyze the phase diagrams of physisorbed systems [307]. "Classical" phase diagrams exhibit a liquid–gas coexistence range, a solid–liquid–gas triple point and a critical point, like e.g., neon on graphite [308], whereas others reveal only commensurate phases but no liquid–gas coexistence range or a critical point, such as H_2 on graphite. Again, just with H_2 and D_2, respectively, fairly large quantum effects are expected in that repulsive contributions may be added to the interaction potential due to zero-point energy motions. With physisorbed diatomic molecules (H_2, N_2, CO) interesting additional effects come into play, namely, *orientational ordering*. This means that, with the molecular axis being oriented parallel to the surface plane, a variety of mutual azimuthal orientations of the molecules with respect to each other are possible thus reflecting different mutual interactions between the adsorbed molecules.

Generally, such rich phase transition phenomena are always expected when adsorbate–adsorbate interactions dominate the adsorbate–substrate forces. In this case, e.g., the spacing of the adsorbed particles will only be modified to a greater or lesser degree, thereby causing an incommensurate interfacial structure with periodic strains induced by the substrate. A theoretical treatment of these effects and their consequences on orientational ordering in adsorbed monolayers was performed by McTague and Novaco [209]. Experimental work has been reported several times, among others by Fain [226,227] for nitrogen and carbon monoxide on graphite (0001) surfaces. Here, the aforementioned interesting long-range ordering and interaction effects were observed at very low temperatures (~ 10 K). According to Harris and Berlinsky [310] who applied means field theory (MFT) for point quadrupoles confined to a triangular lattice of sites (as provided by C(0001)) and subject to an axially symmetric crystal field the theory predicts that only four orientationally ordered structures should occur which differ from one another by the relative orientation of the molecular axes. These possibilities are:

(a) 2-in herringbone,
(b) pinwheel,
(c) 2-out herringbone, and
(d) ferro structure

and are illustrated in fig. 35. For example, a strong negative crystal field causes the 2-in herringbone phase, whereas a weak field gives rise to the pinwheel structure. For physisorbed N_2 on graphite, for example, the 2-in herringbone phase was experimentally verified [226]. In the particular case of physisorbed hydrogen on graphite Fain et al. investigated among others the question as to whether H_2 was orientationally ordered or disordered by means of LEED and concluded that H_2 was orientationally disordered above ~ 10–15

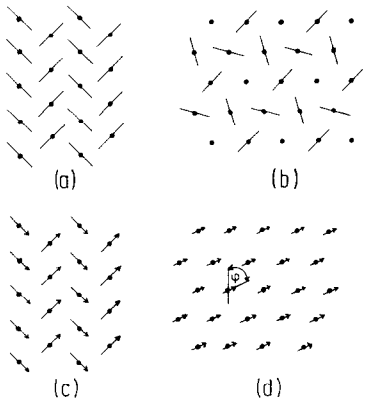


Fig. 35. Schematic drawing of the four phases formed by diatomic molecules with different ordering of their molecular axis as predicted by Harris and Berlinsky [310]: (a) 2-in herringbone, (b) pinwheel, (c) 2-out herringbone and (d) ferro structure whereby the polar angle φ can take any value.

K. The structure and stability of the various physisorbed hydrogen phases can be taken from the respective phase diagram which was reported independently by various authors [227,311]. Here we refer to the work by Freimuth and Wiechert [311] who studied the H_2 and D_2 on grafoil system by means of heat capacity measurements and reproduce their D_2/C(0001) phase diagram as fig. 36. As one of the prominent features a commensurate $(\sqrt{3} \times \sqrt{3})R30°$ D_2

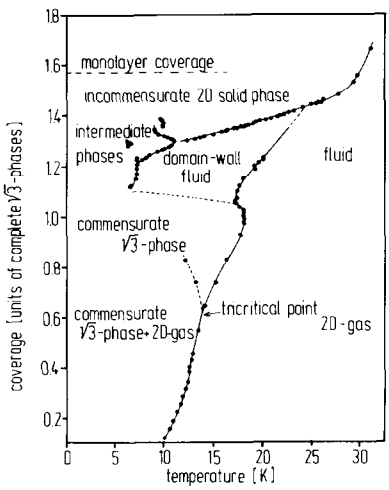


Fig. 36. Phase diagram of physisorbed deuterium on graphite (0001) showing the possible two-dimensional phases. After Freimuth and Wiechert [311].

LEED phase exists which was already reported by Seguin and Suzanne for hydrogen on C(0001) [83]. In the low and medium coverage range this phase coexists with a two-dimensional (2D) gas and exhibits a tricritical point. At higher coverages ($\Theta > 0.6$ monolayers) the system undergoes a commensurate–incommensurate (C–IC) phase transition with several unusual features [311]. The transition is spread out over quite a large coverage range.

There are few recent studies also of molecular H_2 adsorption on metal substrates (Ag, Cu, Al for example [86,87,312] which were briefly mentioned before. However, in no case was there any report on $H_{2(ad)}$ phases with long-range order, neither of their stability range nor of eventual phase transformations. It is nevertheless felt that this area may become a matter of growing interest in the near future and could lead to a better understanding not only of through-bond- but also of the van der Waals type of interaction.

3.6.2. Phases and phase transitions of chemisorbed hydrogen

Compared to the physisorption systems with their relatively weak adsorbate–substrate interactions and their correspondingly strong lateral interactions the *chemisorbed* systems are characterized by strong adatom–substrate forces and only weak adatom–adatom interactions, which are only about 1/10 of the heat of chemisorption.

Like any other chemisorbed species also hydrogen atoms on metal surfaces may form a variety of phases with long-range order, and we classify these phases into those not involving noticeable changes of the substrate equilibrium atom positions (H phases *without* surface reconstruction) and those that definitely exhibit at least additional ordering features arising from reconstruction processes (H phases *with* reconstruction). Recently, a brief summary was given about phase transitions in chemisorbed hydrogen layers from the experimental point of view [313,314]. Likewise, there are several theoretical articles which are concerned with the prediction or explanation of chemisorbed hydrogen phases [222,315,316]. Before we enter a description of the phase *transition* phenomena first a listing of the currently established ordered H_{ad} phases will be given. In table 10 we present this listing for those ordered phases which are believed not to involve a noticeable surface reconstruction. (A more or less pronounced *relaxation*, i.e., a change of the lattice atom distance *perpendicular* to the surface can, however, not be ruled out. Instead, just for the most "open" surfaces this relaxation is rather most likely occurring as recent structure analyses showed.) Also given in table 10 are the nominal H_{ad} coverages leading to the respective ordered phases, and the corresponding critical temperatures beyond which no long-range order can exist anymore.

For the majority of the systems listed in table 10 a complete dynamical LEED analysis to determine the absolute location of the H atom in the respective phase has not been carried out so far. However, for a more detailed theoretical treatment of the phase stabilities, and of phase transformations

Table 10
Ordered chemisorbed H phases without surface reconstruction

Surface (*hkl*)	H phase	H coverage (ML)	Critical temperature (K)	References
Fe(110)	p(2×1)	0.5	245	[159,224]
	3×1-2H	0.67	265	
Co(10$\bar{1}$0)	c(2×4)	0.5	~ 270	[69]
	2×1-p2mg	1.0	~ 290	
Ni(111)	c(2×2)-2H	0.5	270	[50]
Ni(110)	2×3-1D	0.33		[21]
	c(2×6)	0.33		[20]
	c(2×4)	0.50		[21]
	c(2×6)	0.67		[20,21]
	c(2×6)	0.83		[20]
	2×1-p2mg	1.0	≤ 200	[20,21]
Ni(100)	1×1	1.0		K.H. Rieder & H. Wilsch, Surface Sci. 131 (1983) 245
Ru(10$\bar{1}$0)	split 1×2	1.0		[71]
	c(2×2)-3H	1.5		
	1×1	2.0		
Rh(110)	p(1×3)	0.33	140	[70,186,388]
	p(1×2)	0.5	180	
	1×3-2H	0.67	220	
	1×2-2H	1.5	200	
	1×1	2.0	~ 180	
Pd(100)	c(2×2)	0.50	260	[77]
Pd(111)	p($\sqrt{3}$ × $\sqrt{3}$)R30°	0.33	85	[326–328]
	c($\sqrt{3}$ × $\sqrt{3}$)R30°	0.67	105	
Pd(110)	2×1-p2mg	1.0		[68]
Pt(111)	1×1	1.0		I.P. Batra, J.A. Barker & D.J. Auerbach, J. Vacuum Sci. Technol. 2 (1984) 943
W(110)	2×2	0.75	30–160	V.V. Goncher, O.V. Kanash, A.G. Nauvomets & A.G. Fedorus, Pis'ma Zh. Eksperim. Teor. Fiz. 28 (1978) 358
	p(1×1)	1		

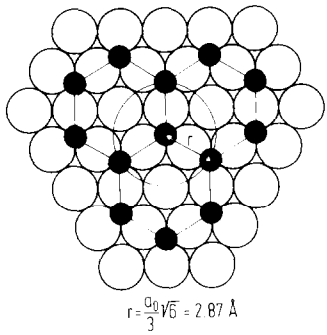


Fig. 37. Structure model of the c(2×2)-H phase ("honeycomb" structure) formed at $\Theta_H = 0.5$ on Ni(111). Dark circles represent H atoms, open circles the Ni atoms.

such as order–disorder transitions, the knowledge of the respective adsorption geometry is mandatory. Hydrogen phases for which this information is available have already been itemized in table 5. In order to illustrate the procedure for obtaining the pase diagram experimentally we shall present some examples, namely, the (probably best known) Ni(111)/H system and the Pd(100)/H system. Then the Fe(110)/H system shall be described in order to illustrate how one can distill out the relevant (and sometimes complex) information about the mutual interaction forces by comparing the experimental data with theoretical predicted phase diagrams. Also, the recent investigations on H/Pd(111) shall be briefly considered.

At room temperature hydrogen does not form an ordered overlayer on a Ni(111) surface [155]. However, upon cooling to below ~ 270 K faint but sharp LEED "extra" spots of a c(2 × 2) structure become visible [50,327] which exhibit a characteristically lower intensity than the substrate LEED reflections ($0.01 < I_{c(2\times2)}/I_{1\times1} < 0.1$) indicative of a "true" H phase with long-range order. A dynamical LEED analysis in conjunction with thermal desorption mass spectrometry reveals the following structural details: The H atoms occupy threefold sites of a hexagonal (so-called "honeycomb") structure which is shown in fig. 37. The Ni–Ni layer distance is 1.15 Å (±0.05) and the H–Ni atom distance is 1.84 Å [50]. From later vibrational loss measurements, however, Plummer and coworkers [178] concluded on a somewhat smaller H–Ni distance. The c(2 × 2) phase contains 2 H atoms per unit cell and thus corresponds to a coverage of $\Theta = 0.5$. It is the only ordered structure that can be observed in the temperature range between 70 and 280 K. The phase transition of the c(2 × 2)-2H phase was extensively investigated by means of LEED. Fig. 38 shows a sequence of $I_{1/2}(T)$ curves of a fractional-order beam for various H coverages around $\Theta_H = 0.5 \pm \Delta$, whereby $\Delta \approx 0.2$. For a clear order–disorder transition, a first gradually and thereafter sharply declining curve should be obtained without any inflection point. However,

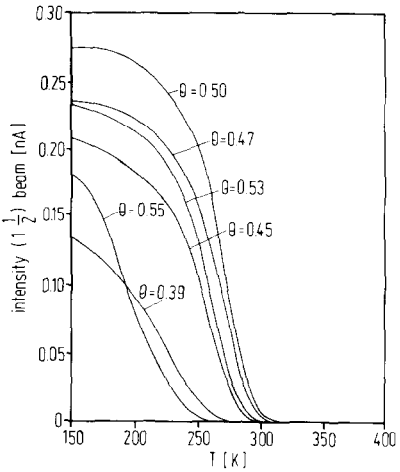


Fig. 38. Temperature dependence of the fractional-order LEED beams of the c(2×2)-2H structure on Ni(111). Parameter of the curves is the coverage.

owing to short-range order and finite size effects there is always a small intensity tail observed beyond the transition temperature, and the curves exhibit an inflection point. This can be taken as a good indicator for the order–disorder transition temperature. From fig. 38 it is quite clear that this transition temperature is highest for the structure with the least number of imperfections ($\Delta = 0$, $\Theta_H = 0.5$) and strongly decreases with disorder being introduced. In the T–Θ_H plane we can construct the phase diagram which is shown in fig. 39 and clearly reveals an asymmetry with respect to $\Theta_H = 0.5$. It is much wider on the low coverage- than on the high-coverage side which can simply be explained by the amount of disorder introduced by a mere vacancy

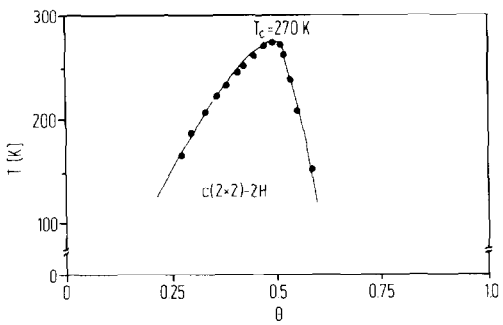


Fig. 39. Phase diagram for the c(2×2)-2H structure on a Ni(111) surface as derived from the inflection points of the curves shown in fig. 38. The critical temperature T_c is indicated.

(which leaves the overall honeycomb lattice intact) and by an interstitial H atom (which, since it exerts repulsive interactions on its neighbors, destroys a fair amount of order). The order–disorder problem of a honeycomb lattice, its classification to a certain universality class (Ising or 4-state Potts), predictions of critical exponents, etc. have been considered theoretically by a number of authors [302,315,318], and specifically for the H-on-Ni(111) system by Nagai et al. [318] utilizing position-space renormalization-group theory with prefacing transformation. They compared two sets of interactions:
(A) nearest-neighbor exclusion, second-neighbor repulsion and third-neighbor attraction (as previously proposed by Domany, Schick and Walker [319]), and (B) nearest-neighbor exclusion, second and third neighbor repulsion and further long-range interactions. Whereas both sets (A) and (B) reveal phase diagrams in agreement with fig. 38 the authors exclude set (A) on the basis of a comparison of the experimentally observed [155] and theoretically predicted coverage dependence of the heat of H adsorption. Another theoretical treatment of the H/Ni(111) phase transition phenomena was performed by Roelofs et al. [320] who claimed that, in addition to the repulsive interactions, very long-range attractive forces prevail.

Another quite interesting H chemisorption system with long-range order is provided by H/Pd(100). The experimental features are described at length in refs. [77] and [321], here we concentrate only on some details.

Below 260 K hydrogen forms a c(2 × 2) structure on Pd(100), again with low-intensity "extra" spots. A model of the c(2 × 2) phase is presented in fig. 40. It is therein assumed that the H atoms occupy the fourfold hollow sites.

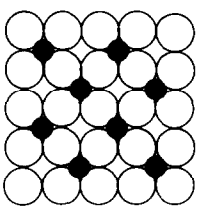


Fig. 40. Structure model of the c(2×2)-H phase formed on a Pd(100) surface. The dark circles represent the H atoms.

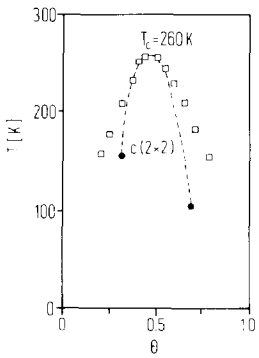


Fig. 41. Experimental data points and theoretical phase diagram (broken line) of the c(2×2)-H structure on Pd(100). Note that the theoretical diagram is considerably narrower according to Binder and Landau [322].

This adsorption site was later confirmed in a vibrational loss spectroscopy study by Nyberg and Tengstål [105,175]. The phase diagram which was obtained from LEED intensity–temperature dependences of the fractional-order beams is shown in fig. 41. The critical temperature is around 260 K at $\Theta_H = 0.5$, and the phase diagram is almost symmetric with respect to $\Theta_H = 0.5$, but not completely. Moreover, it exhibits an appreciable width, as compared to theoretical predictions [322,323] which means that islands with long-range order are already formed at fairly low hydrogen coverages. This is always indicative of attractive interaction forces which lead to clustering already at a low coverage stage.

Since the Pd(100)/H system exhibits a lattice of square symmetry there has been a great interest of theoretical groups in modelling the experimental phase diagram and to predict the correct interaction energies. From their LEED experiments Behm et al. [77] gave an estimation of the lateral interaction energies. On the basis of an Ising model with repulsive nearest-neighbor interactions only and with $T_c = 260$ K relation (46) would yield 3.81 kJ/mol for ω_1. For non-vanishing next-nearest-neighbor attractions ω_2 the interaction parameters are approximately related to T_c through [324].

$$\exp\left(\frac{\omega_1}{2kT_c}\right) = 2\exp\left(-\frac{\omega_2}{2kT_c}\right) + \exp\left(-\frac{\omega_2}{kT_c}\right), \tag{47}$$

whereby ω_2 stabilizes the long-range order. If we take $|\omega_1| \approx 0.5\omega_1$ we obtain from eq. (47) for $\omega_1 \approx 1.0kT_c \approx 2.13$ kJ/mol and for $\omega_2 = 1.07$ kJ/mol which is in agreement with Monte Carlo calculations performed by Binder and Landau [322] for that system.

On the basis of their calculations Binder and Landau [323] determined the complete phase diagram for $\omega_2 = -0.5\omega_1$, and their result is reproduced in fig. 41 as a dotted line. Evidently, the theoretical range of existence for the $c(2 \times 2)$ structure is somewhat narrower than observed experimentally. A likely explanation would be that $|\omega_2| > 0.5|\omega_1|$ which could lead to an increased range of stability of the ordered phase. Also, three-particle interactions possibly play a role which had been postulated for the system O/W(110) by Einstein [223] as pointed out in the beginning of this section.

The systems dealt with so far only exhibited a single ordered H phase, at least in the experimentally accessible temperature range. There are several other H chemisorption systems with more than a single ordered H phase which offer the possibility of order–order transitions. Among these systems the Fe(110)/H and the Pd(111)/H systems have found experimental and theoretical interest, and we shall briefly describe some of the most prominent features of the systems. The adsorption of hydrogen on Fe(110) leads to two well-ordered H structures at temperatures below 250 K [159], namely, a $p(2 \times 1)$ and a (3×1) phase. A later dynamical LEED analysis revealed the threefold adsorption site in the long bridge as the location of the chemisorbed H atoms

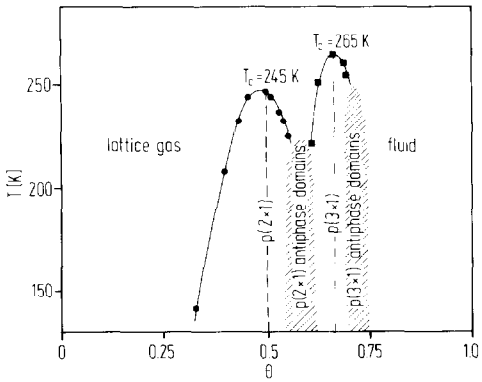


Fig. 42. Experimental phase diagram of the Fe(110)/H system. After Imbihl et al. [224].

as was already shown in fig. 11 [124]. The phase transition phenomena were subjected to a detailed study by Imbihl [325] and Imbihl et al. [224]. LEED $I(T)$ and angular beam profile measurements led to the experimental phase diagram shown in fig. 42 with two zones in which antiphase domains coexist between the two ordered structures. The critical temperatures are 245 K for the 2×1 and 265 K for the 3×1 phase and are thus of the same order of magnitude as observed for Ni(111) or Pd(100). Clearly, the phase diagram is asymmetric with respect to $\Theta_H = 0.5$. Most significantly, an expected low-coverage p(3×1) phase is missing. This is certainly a consequence of the operation of distinct lateral interaction forces, and it was the goal of several theoretical calculations to disentangle these interaction energies. Kinzel et al. [225a] and Selke et al. [225b] have performed phase diagram calculations for the Fe(110)/H system based on Monte Carlo and transfer matrix methods. They assumed a lattice gas model and used three pairwise interactions, ω_1, ω_2 and ω_3 which are indicated in the inset of fig. 43. The strong asymmetry of the experimental phase diagram suggests also the operation of trio interactions, ω_t. Kinzel et al. [225a] conclude ω_2 and ω_3 to be repulsive, and ω_1 weakly repulsive as well. In order to verify the large asymmetry in the phase diagram, in particular to predict the 3×1 phase at $\Theta_H = 0.67$ and the absence of the p(3×1) phase at $\Theta_H = 0.33$, a rather strong *attractive* trio interaction must exist. In fig. 43 we present the theoretical phase diagram which agrees well with the experiment (cf. fig. 42). The choice of the interaction parameters was $\omega_1 = -7.9$ kJ/mol, $\omega_2 = -1.3$ kJ/mol, $\omega_3 = -2.1$ kJ/mol and $\omega_t = 3.35$ kJ/mol [225]. Also, the critical temperatures could be reproduced quite well. In summary, this investigation of the phase transition phenomena of the Fe(110)/H system provides a very good example for the possibility to deduce reliable interaction energies even for a complicated adsorption system, which exhibits several ordered phases. This detailed information could probably never be obtained from, e.g., a thermal desorption spectroscopy analysis.

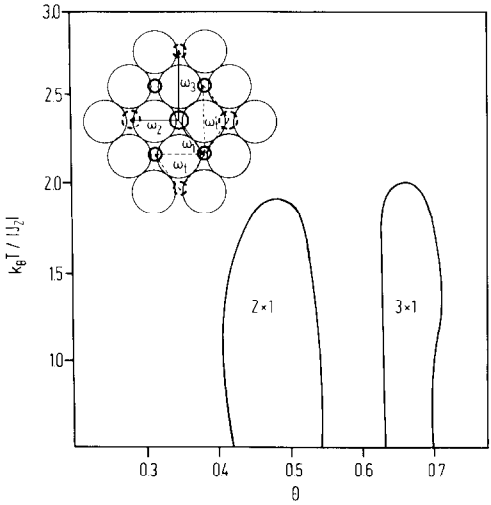


Fig. 43. Theoretical phase diagram of the Fe(110)/H system as calculated by Kinzel et al. [225]. A structure model indicating the different types of H–H interaction ω_1, ω_2, ω_3 and ω_t is also presented.

As a final example we consider here H on Pd(111) where another complication comes into play, namely, the principal ability of the hydrogen atoms to penetrate the near surface region of the Pd crystal. Hints that these "subsurface" phenomena could be important were obtained by Plummer and his group by means of angle resolved photoemission [145]. In 1985, Felter and Stulen reported on the observation of two $(\sqrt{3} \times \sqrt{3})R30°$ phases formed at ~ 80 K by hydrogen on Pd(111), one being formed at $\Theta_H = 1/3$ and the other, a $(c(\sqrt{3} \times \sqrt{3}))R30°$ structure, at $\Theta_H = 2/3$ [326–328]. In the fractional-order LEED-intensity–exposure dependence shown in fig. 44 clearly two maxima can be distinguished which parallel the development of the maximum degree of order of the respective two structures. Again, the weak intensity of the "extra" spots suggests a non-reconstructed "true" hydrogen phase. As compared to Ni(111)/H, Fe(110)/H or Pd(100)/H the critical temperature of these phases is rather low, namely, around 85 K for the $p\sqrt{3}$ structure and 105 K for the $c\sqrt{3}$ phase. The structure models of these two phases (along with the honeycomb structure formed by H on Ni(111) for comparison) are displayed in fig. 45, and it can easily be seen that the difference between the $\sqrt{3}$ and the $c(2 \times 2)$ symmetry is simply caused by a by 30° different rotation angle of the lattice vector of the H unit mesh with respect to the metal surface vector. It is also interesting to note that on Ni(111) no "true" 2×2 phases at $\Theta_H = 0.25$ nor any non-primitive 2×2 phase corresponding to $\Theta_H = 0.75$ exists. There may be simple geometrical

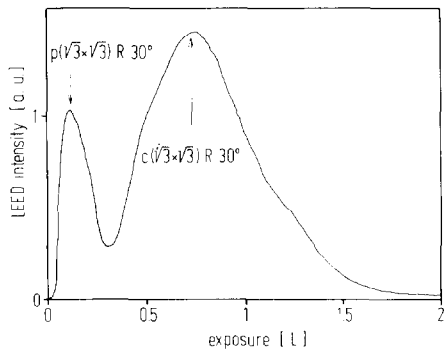


Fig. 44. LEED intensity of a fractional-order beam of the p($\sqrt{3} \times \sqrt{3}$) and c($\sqrt{3} \times \sqrt{3}$) structure of H on Pd(111) at $T = 85$ K, as a function of hydrogen exposure. After Felter et al. [326].

reasons, namely, the somewhat larger lattice constant of Pd ($a_{0,\mathrm{Pd}}/a_{0,\mathrm{Ni}} = 1.10$). This brings along that the c$\sqrt{3}$ structure with $\Theta_{\mathrm{H}} = 2/3$ has almost the same H density as the c(2 × 2) structure found with Ni(111), namely, 10.2×10^{14} cm^{-2} compared to 9.3×10^{14} cm^{-2}. One other important point could be the enhanced solubility of H in the Pd surface as compared to Ni. The phase transitions occurring with the Pd(111)/H system have therefore been subjected to detailed theoretical calculations by Daw and coworkers using the embedded atom method (EAM) [35,327]. It is convincingly shown therein that the ordered phases both involve hydrogen adsorbed in *subsurface* sites. The successful prediction of the ordering behavior represents a good advance in the ability to model lateral hydrogen interactions on surface. In detail, the EAM

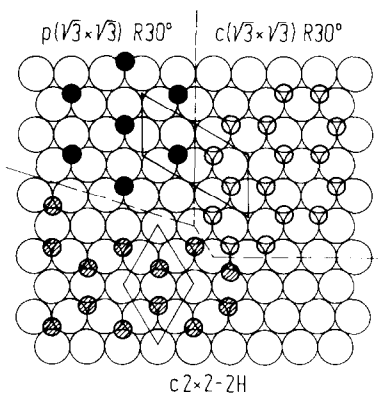


Fig. 45. Comparison of the three experimentally observed H structures on hexagonal Pd (p($\sqrt{3} \times \sqrt{3}$)R30° and c($\sqrt{3} \times \sqrt{3}$)R30°) and Ni (c(2×2)) surfaces. A true p(2×2) phase at $\Theta_{\mathrm{H}} = 0.25$ has not been found with Ni(111).

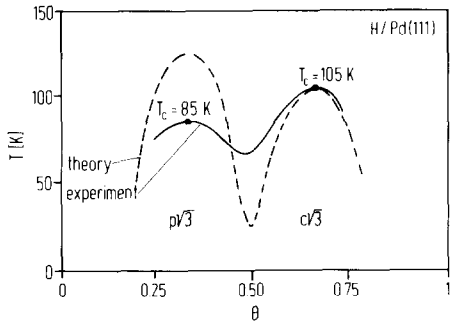


Fig. 46. Experimental (full line) and theoretical (broken line) phase diagram of H on Pd(111) with the critical temperatures for the two $\sqrt{3}$ phases being indicated. After Felter et al. [326–328].

calculations show that the H atom rests in the threefold hollow sites of the Pd(111) surface *as well as* in octahedral sites between the first and second Pd layer. The H–H interactions predicted theoretically are strongly repulsive and prevent nearest-neighbor occupation in the low coverage phases. Calculations of the classical diffusion barriers for hydrogen motion on a surface and between surface and subsurface sites reveal that the equilibrium is reached very rapidly, that is, within the time scale of the experiment. This makes it very difficult to reach an unambiguous distinction between two types of sites experimentally, the more so since almost all surface-sensitive techniques probe only the entire surface region, that is, the first four or five surface layers. Felter et al. have performed Monte Carlo calculations to model the experimental phase diagram of the Pd(111)/H system, and the experimental data as well as the result of the MC calculations are presented in fig. 46. It appears that the regions and coverages of the two $\sqrt{3}$ phases are correctly predicted but that the critical temperature of the p$\sqrt{3}$ phase comes out 40 K too high ($T_{c,\mathrm{theory}}$ = 125 K). The authors underline that the inclusion of many-body contributions to the interaction energy is crucial to the determination of the critical temperatures, and they emphasize that any theoretical treatment of an order–disorder transition of hydrogen must account for these many-body interactions.

3.6.3. Phases and phase transitions involving the substrate (reconstruction)

Next, ordered hydrogen phases on those surfaces shall be considered that undergo a H-induced reconstruction. The idea that a reconstruction is induced by an adsorbate dates back to 1968 [329]. Again, a description of some of these systems has been given recently [313,330,331]. Whenever reconstruction phenomena are observed, the question of most obvious interest is to explain the nature of the microscopic interactions responsible for these phase transformations. Unfortunately, most of the theoretical work done so far has been

directed to the clean surfaces and the major conclusion was usually that the driving mechanism is an electronic instability of the surface. Even slight changes in the density of states induced by an adsorbate lead to a redistribution of electronic charge with the consequence that lower energy states become populated [332]. A different approach to describe the reconstruction phenomenon is in terms of lattice dynamics, whereby lattice distortions are conducted forth over many lattice constants. In other words, soft phonon modes are created [333] which lead to a structural instability. Evidence for this was recently obtained by Ibach and coworkers [173,263] with the H/Ni(110) system by means of ELS measurements. Both possible reconstruction mechanisms may cause a long-range reconstruction which could be observed already at fairly small adsorbate coverages. If long-range lattice strain or quenching of intrinsic electronic surface states is involved even traces of impurity atoms (carbon, sulfur, oxygen) may stabilize one or the other lattice geometry, and it is not always easy to delineate these impurity effects and the true adsorbate influence on the equilibrium structure of the surface.

Probably of much different nature is the short-range reconstruction which requires a fairly concentrated adsorbate layer, that is, a "critical" coverage in order for the reconstruction to occur. Moreover, a certain geometrical configuration of the adsorbate (surface complex) may additionally be necessary. It appears as if the occupation of metal electronic charge to build up the chemisorptive bond leads to a local weakening of metal–metal bonds (an extreme case here is certainly the formation of a surface *compound* with a well-defined stoichiometry). This latter type of reconstruction can frequently be found with adsorbed hydrogen systems [331]. Energetically, it is accounted for by the interaction term $\Sigma p_j E'_{\mathrm{Me-Me}}$ in eq. (11). According to the statements made before also the systems reconstructing under hydrogen can be divided into two categories (a) and (b).

(a) The "short-range" type which involves displacements of substrate atoms only in the direct vicinity of the adsorbed H atom. Among the H adsorption systems that fit into this category are the 1×2 phases formed with Ni(110), Co($10\bar{1}0$), Pd(110) or Rh(110) [331].

(b) The "long-range" type with the largely delocalized reconstruction in which only few nuclei make the substrate structure "flip" over to a new equilibrium configuration. Examples are provided by the "streak" phase of the Ni(110)/H system and probably also by the famous H/W(100) and H/Mo(100) adsorption systems.

The other perhaps more common classification of reconstructions distinguishes "displacive" and "reconstructive" processes. The displacive reconstruction does not involve mass transport; all pairing models (local or long-range) fit into this category. The reconstructive type of reconstruction does require transport of matter; therefore these processes (e.g., the missing-row reconstruction) are usually thermally activated in order to initiate diffusion

Table 11
H phases which involve displacements of surface atoms (surface reconstruction)

Surface (*hkl*)	LEED phase	H coverage (ML)	Type of reconstruction	References
Co(10$\bar{1}$0)	1×2	>1.0 (1.5)	Pairing row (?)	[69]
Ni(110)	1×2	>1.0 (1.5)	Pairing row + second layer buckling	[156,344]
	"Streak"	> 0.1	Local pairing row, 5×2	[347]
Ru(10$\bar{1}$0)	c(2×2)	1.5	Buckling (?)	[71]
Pd(110)	1×2	>1.0 (1.5)	Pairing row	[377]
Mo(100)	$\sqrt{2}\times\sqrt{2}$ + complex str.	≥ 0.5	Local pairing (?)	[369,370]
W(100)	$\sqrt{2}\times\sqrt{2}$ + complex str.	≥ 0.5	Local pairing + buckling	[356,360,361,365]
W(110)	1×2	0.5	Lateral shift of atoms	[372]

processes. A listing of the currently known H-induced reconstructions is given in table 11. It is felt thereby that more careful inspections of other H adsorption systems in the future might make this list even longer.

As far as phase *transitions* are concerned the theoretical treatment of these transitions occurring on reconstructed surfaces is extremely difficult and has been attempted so far only for the H/Ni(110) [334] and H/W(100) systems [335–337]. It must be borne in mind that the lattice gas approximation does no longer hold and a much more refined statistical mechanical treatment must be employed to describe an eventual order–order or order–disorder phase transformation. For each stable phase also the coordinates of the substrate lattice atoms must be principally taken as variable parameters. An interesting question comes to one's mind as to whether at all further phase transitions occur during adsorption of hydrogen on a surface which has already undergone a H-induced reconstruction at a small hydrogen coverage. Chemically this is equivalent to following various stages of the reaction path of a metal transforming to a metal–hydrogen compound, a hydride. The H/Ni(110) system could be reckoned among that group because of the 2D → 1D order transition at 220 K whereby a well-ordered reconstructed 1 × 2 phase transforms spontaneously into a likewise reconstructed but merely one-dimensionally ordered "streak" phase. However, neither in the 1 × 2 nor in the "streak" phase the position of the *H atoms* is known with certainty.

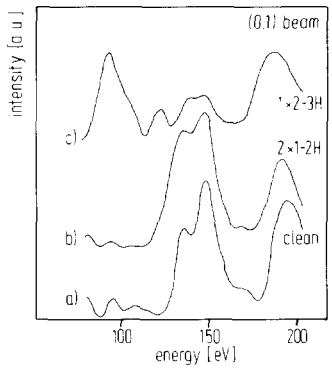


Fig. 47. LEED-intensity–voltage curves of the (0,1) beam of a Ni(110) surface at 120 K: (a) clean surface, (b) non-reconstructed 2×1-2H phase, (c) reconstructed 1×2-3H phase. Note that curve (c) differs from (a) and (b) thus indicating the reconstruction.

We shall now select two typical examples, namely, the already frequently mentioned H/Ni(110) and H/W(100) systems, and give a somewhat more extensive description here.

The H on Ni(110) system is one of the most extensively studied adsorption systems. The formation of a H-induced 1×2 LEED structure has been reported many times [115–118,155,156,338–341] and even in the early LEED papers the possibility of a surface reconstruction was discussed [115–118,156, 338]. From the recent He diffraction [20,342,343], LEED [21,129,230,231,344] and other investigations [147 (UPS), 157, 345, 346 (RBS, $\Delta\varphi$), 347 (STM), 173,181–184,348 (ELS)] a fairly complete picture arises. At temperatures below ~ 200 K a variety of H lattice gas structures forms, the final phase being a 2×1-2H structure at $\Theta_H = 1.0$. Yet, the H uptake does not stop here, rather the H adsorption continues and another 0.5 ML becomes adsorbed up to saturation at $\Theta_H = 1.5$ ML. The LEED pattern then exhibits a sharp 1×2 structure with *intense* "extra" spots. During this phase transition $2\times1 \rightarrow 1\times2$ (which was already mentioned in the context of the sticking probability, cf. section 3.4) the (1×1) LEED $I(V)$ curves change dramatically as is illustrated in fig. 47. This is indicative of a surface reconstruction. A recent dynamical LEED structure analysis clearly showed that the 1×2 surface is reconstructed in a pairing-row (PR) configuration with second layer buckling [344]. The PR reconstruction was originally proposed by Taylor and Estrup [156] and is schematically shown in fig. 48.

The kinetics of the $2\times1 \rightarrow 1\times2$ transition could be followed by the Video–LEED method [349]: The LEED intensity of the $(-1,\frac{1}{2})$ beam of the 2×1 phase and that of the $(\frac{1}{2},-1)$ beam of the (reconstructed) 1×2 phase were simultaneously measured as a function of H_2 exposure. This is illustrated by means of fig. 49. A careful inspection of the respective curve also reveals

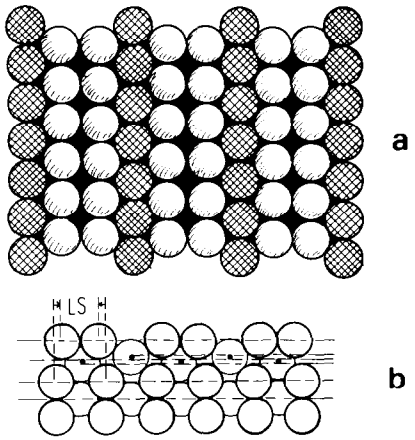


Fig. 48. (a) Schematic structure model of the pairing-row (PR) reconstruction of the Ni(110) surface (top view). (b) Side view showing the second-layer buckling (LS = layer shift). After Kleinle et al. [344].

that the 2×1 phase reaches its maximum ~ 0.1 L *before* the 1×2 intensity begins to rise – it appears as if there were a short "ignition" period and critical supersaturation to make the 2×1 phase "flip over" to the reconstructed 1×2 phase.

We recall that it is not until a critical coverage of 1 ML of adsorbed H is reached that the Ni(110)-1×2 reconstruction takes place. The hydrogen concentration of the 2×1-2H phase has the surface Ni atoms sufficiently decoupled from their second-layer neighbors that the new PR equilibrium

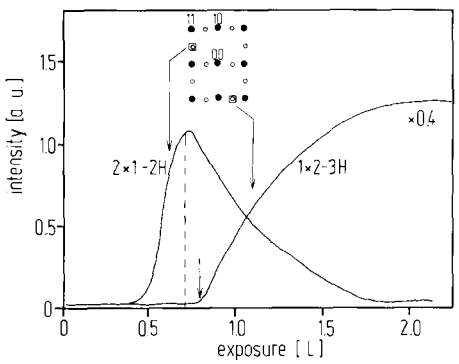


Fig. 49. LEED intensity of the $(\frac{1}{2},1)$ beam of the 2×1-2H structure and of the $(-1,-\frac{1}{2})$ beam of the (reconstructed) 1×2 structure of a Ni(110) surface at 120 K, as a function of H exposure as measured with the Video–LEED method. After ref. [330].

positions can be reached without a noticeable activation energy – the 1×2 reconstruction occurs spontaneously even at 100 K adsorption temperature.

Whilst this phase transition certainly belongs to the "short-range" reconstruction category, there is another peculiar property of the H/Ni(110) system, namely, the appearance of the streak structure ("Str") at temperatures above 220 K. Whilst in previous studies [155,115,338] this streak structure (which tends to form a streaky 1×2 phase after prolonged exposures at room temperature) was regarded as the only H-induced Ni(110) surface phase it was Taylor and Estrup [156] who first did low temperature H adsorption studies and recognized that the low-T 1×2 phase was actually *different* from the "streaky" 1×2 high-T phase. In the context of our considerations here it is only necessary to point out that the features of the "Str" phase appear already after short H_2 exposure, at coverages around 0.2 ML. Recent ion channeling work by Pfandzelter and Varelas [350] suggested a missing-row (MR) type rearrangement of the Ni surface atoms in the "Str" phase, but there exists neither a reliable LEED structural analysis of the Ni surface nor is the position of the H atoms known. On the basis of statistical model calculations Rys and Müser [351] investigated the $1 \times 2 \rightarrow$ "Str" phase transformation and could obtain equilibrium structures which were in line with the experiment. A very recent scanning tunneling microscopy study by Kuk et al. [347] focussed on the structure of the "streak" phase. It was argued that the streaky 1×2 phase is primarily a (5×2) Ni reconstruction with small domain size along the [001] direction. The top-layer Ni atoms are paired along the [001] direction by a 0.5 Å lateral displacement to form tetramers and every fifth row is missing. A structure model as proposed by Kuk et al. is presented in fig. 50b while the unreconstructed surface is shown in fig. 50a. Other structure models have been developed, for example, a 1D-disordered pairing-row phase is discussed (fig. 50c) as well as a missing-row (MR) configuration. The fact that an activation energy is required for the "Str" phase to form could be taken as some hint for the MR model since a merely displacive reconstruction such as the PR is normally not activated, simply because there is no transport of matter involved. The immediate formation of the streaks as the Ni surface is exposed to H_2 prompts us to classify the "Str" phase among the type of "long-range" reconstruction mentioned before. It should be kept in mind, however, that the "Str" phase can also be produced by heating the "low temperature" 1×2 phase to beyond 220 K which then abruptly decomposes to the "Str" phase [230,231] whereby ~ 0.4 to 0.5 ML of H are released into the gas phase. Only upon prolonged H_2 exposure this loss of hydrogen can gradually be compensated for whereby the streaks concentrate into streaky 1×2 spots as mentioned before. Careful Rutherford backscattering measurements in the group of Norton yielded the corresponding H coverages to range between 1 and 1.5 ML, probably much closer to 1.5 ML than 1 1 ML [157,345].

Whereas this short description of the Ni(110)/H system could have left the

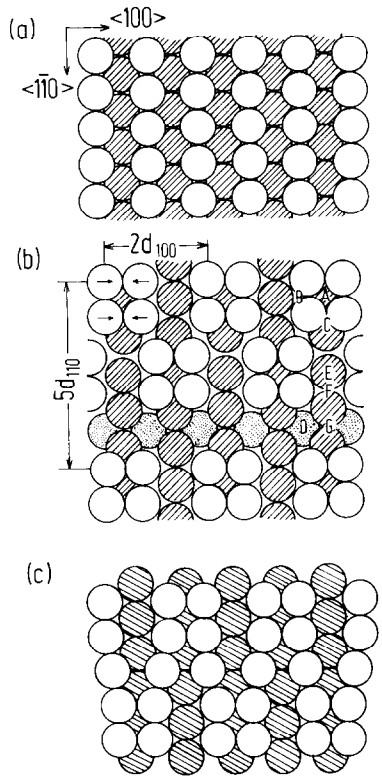


Fig. 50. (a) Clean unreconstructed surface. (b) (5×2) structure model proposed for the reconstructed "streak" phase formed by H on Ni(110) at room temperature; after Kuk et al. [347]. (c) Disordered pairing-row model; after ref. [230].

impression that things are very complicated there is indeed one other H adsorption system which also fits into the group of long-range reconstructions and which exhibits even more complicated features, namely, the H/W(100) system. For the sake of completeness we present a list of references [126–128,141,251–255,268,352–367] referring to experimental and theoretical work on that interesting and most frequently studied adsorption system. Besides the already mentioned H thermal desorption and binding energy studies by Tamm and Schmidt [1,255,268] there exist also early structural investigations, particularly by Estrup and his group [352–354,359,360] which stimulated a whole lot of further work. As additional data became available the complexity of that system was realized and attributed to the occurrence of displacive structural transformations of the substrate. The main results are briefly presented: From LEED studies a clean surface reconstruction (to form a $\sqrt{2} \times \sqrt{2}$ structure) became evident when the crystal was cooled to below 300

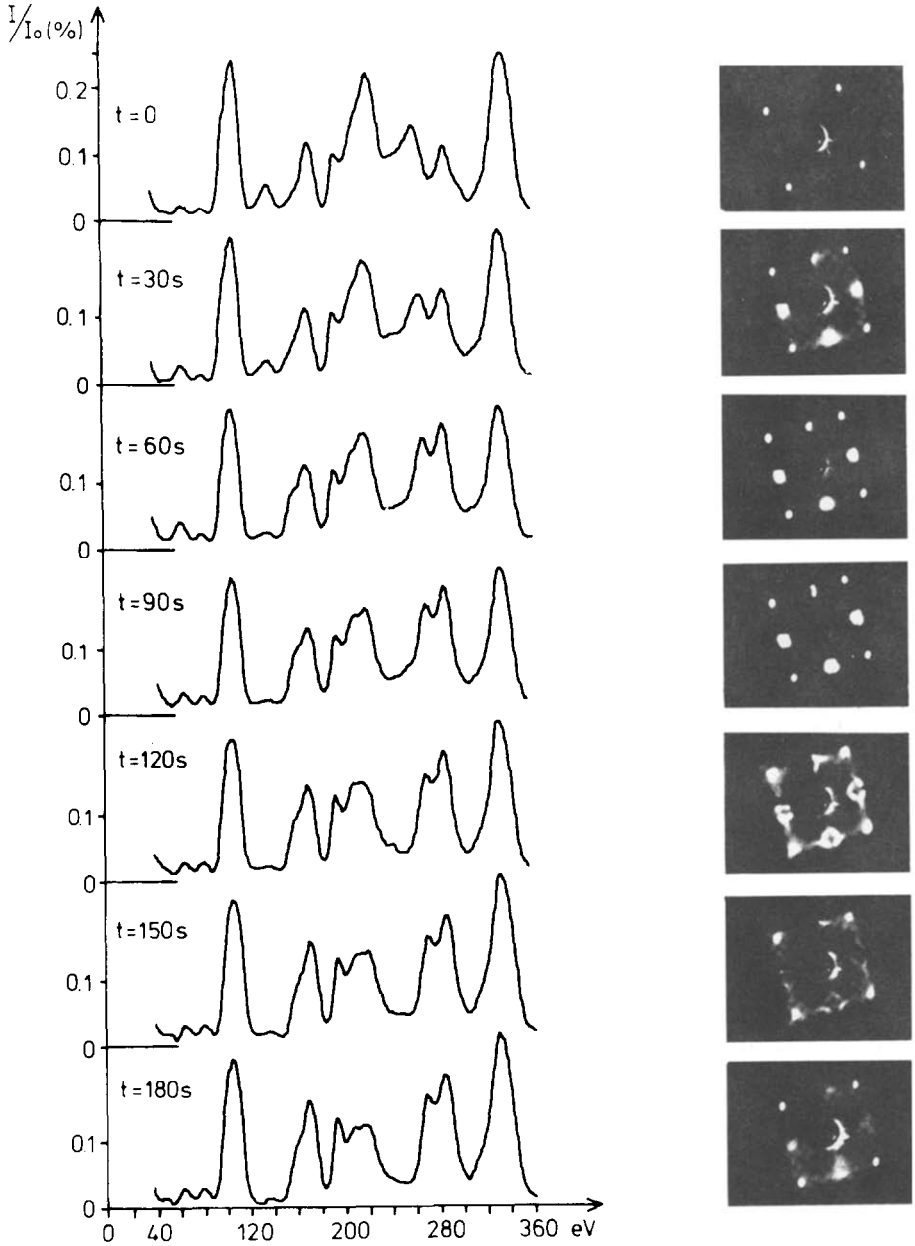


Fig. 51. Sequence of LEED patterns (left-hand side) and corresponding $I(V)$ curves of the substrate beams of a W(100) surface, as a function of H exposure time. After Heinz and Müller [361]. Reproduced with permission.

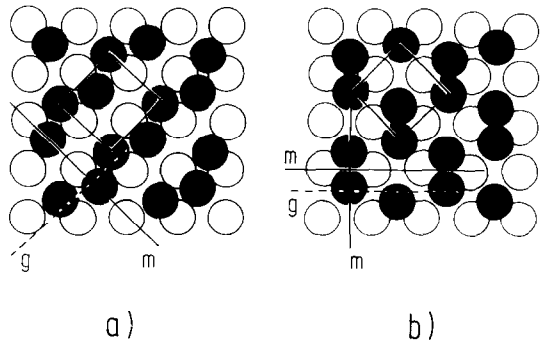


Fig. 52. Structure models of the reconstruction of the W(100) surface as proposed by Barker and Estrup [352]: (a) without hydrogen: $c(\sqrt{2} \times \sqrt{2})_{\langle 11 \rangle}$; (b) with hydrogen: $c(\sqrt{2} \times \sqrt{2})_{\langle 10 \rangle}$. The dark circles represent the W surface atoms. The unit cells are also indicated.

K [359] which was thoroughly investigated by Debe and King [356], Barker et al. [360], Heinz and Müller [361], Heilmann et al. [362] and Hulpke and Mann [363] using mainly LEED but also ion scattering techniques [363]. Particularly reliable are the fast LEED measurements using the Video technique by Heinz and Müller. In fig. 51 taken from their work we comprise the LEED features that develop upon adsorption of hydrogen. It shows the complexity of the LEED structures. Latest since Debe and King's work [356] there is no doubt that also the H exposed W(100) undergoes a reconstruction which is of a similar type than that of the clean surface – only, the order–disorder phase transition is shifted towards higher temperatures. This reconstruction leading to a $(\sqrt{2} \times \sqrt{2})$R45° structure (frequently also denoted as $c(2 \times 2)$) consists very likely of a periodic displacement of adjacent W atoms, i.e. a horizontal pairing. Two possible cases of the $\sqrt{2} \times \sqrt{2}$ structure are illustrated in the structure model of fig. 52 taken from Barker and Estrup's work [352], whereby actually the orientation of one domain is shown only. It is argued by the authors that fig. 52a represents the clean reconstructed surface, whilst fig. 52b is valid for a hydrogen coverage of $\sim 1/3$ of a ML. The displacements of the W atoms are only less than 0.2 Å and have been exaggerated in fig. 52 for clarity. The two structures shall be henceforth denoted as $c(\sqrt{2} \times \sqrt{2})_{\langle 11 \rangle}$ (a) and $c(\sqrt{2} \times \sqrt{2})_{\langle 10 \rangle}$ (b), in agreement with Willis' notation [358].

There are interesting symmetry properties of these structures, particularly of the case (a) which was originally proposed by Debe and King for a clean reconstructed W(100) surface [356]. Whilst normally both possible domain orientations are energetically equivalent and should therefore show up in a LEED pattern with equal intensity the sample used by Debe and King gave a preference for one domain only, and the authors were able to deduce the existence of a mirror plane (m) and a glide plane (g) in the structure as indicated in fig. 52a. These symmetry properties denoted as P2mg for a $\sqrt{2}$

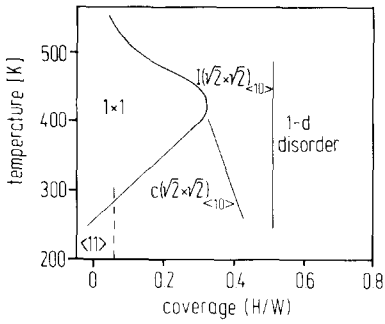


Fig. 53. Phase diagram of the W(100)/H system showing the H-induced commensurate → incommensurate phase boundary and the $(\sqrt{2} \times \sqrt{2})_{\langle 11 \rangle}$ displacive commensurate → disordered (1×1) transition line. After Willis [358].

structure determine the direction of the displacements unequivocally, and later LEED structure analyses [360] have confirmed this model. Lattice dynamical calculations also revealed a preference for this kind of substrate atom displacement [364]. As pointed out before also the H-induced surface phases clearly involve a W surface reconstruction, based on the evidence obtained from LEED [360–362], ion scattering [252,355] and high-resolution electron energy loss measurements [126–128]. There are strong similarities between the two $\sqrt{2}$ structures, but also certain differences which rule out the possibility that the H-induced structure and the "clean $(\sqrt{2} \times \sqrt{2})$R45°" are identical. These differences consist of subtle changes of the LEED $I(V)$ curves and a change of symmetry at a certain H coverage. From these arguments Barker and Estrup have chosen model (b) which exhibits an additional mirror plane and belongs to the symmetry group C2mm [352]. Heilmann et al. [365] have concluded from their LEED data that also a vertical component of the atomic displacement must be considered, in other words, a slight layer buckling should be taken into account.

At higher H coverages, more complicated H-induced structures are formed (cf. fig. 51), and we simply present the phase diagram for the H/W(100) system in fig. 53 as it was published by Willis [358]. He states that the W(100) system represents an example of the magnetic spin model of the 2-dXY class with cubic anisotopy, whose critical exponents are nonuniversal. Fore more details on this subject we refer to Schick's overview [283].

Evidently, with increasing H coverage, the $(\sqrt{2} \times \sqrt{2})$-H phase suffers a commensurate–incommensurate (C–IC) phase transition which is caused by the spontaneous formation of domain wall dislocations (domain "lines"). This process can also be explained in terms of a "soft phonon" model. Recently it was proposed by Didham et al. on the basis of ELS data that two lattice vibrational modes play a role in the transformation: $\sqrt{2} \times \sqrt{2} \rightarrow 1 \times 1$, the

Rayleigh branch with displacements normal to the surface and a phonon branch with displacements parallel to the surface [368]. The phase diagram shows that the incommensurate phase is stable up to temperatures at which the H atoms desorb. It is also worth to mention that no superstructure is formed by H below 200 K. There is evidence that the periodic lattice distortions are hindered under these conditions. A somewhat restricted mobility of the H adsorbate could be responsible for that behavior. Interestingly, the transformation $\sqrt{2} \times \sqrt{2} \rightarrow 1 \times 1$ is the same for the clean and the H covered surface. It is apparently a continuous phase transition in which the magnitude of the displacements and their long-range correlations *gradually* decrease with increasing temperature.

The role of hydrogen consists in a stabilizing effect in that each adsorbed H atom sort of clamps two W atoms underneath together. In this respect, some similarities exist with the PR reconstruction of the Ni(110) surface. At saturation, two H atoms are adsorbed per W surface atom, i.e., $\Theta_H = 2.0$ [251], with the hydrogen being likely adsorbed in bridge positions. Again, as in the Ni(110) case, the formation of local H–W complexes is suggested.

Generally, quite a number of theoretical works exist in which the origin of the reconstruction phenomena of the W(100) system is investigated. A very good status report was recently given by Inglesfield [366] which is recommended for further reading. There is wide agreement that the transformations are a result of a lowering of the surface electronic energy, but the detailed mechanism is still not explored [335,367]. Under discussion is a model in which a Fermi surface instability leads to the formation of charge density waves (CDW) which are coupled to a periodic lattice distortion. It is also possible that a kind of Jahn–Teller surface lattice distortion occurs as known from cluster or complex chemistry. In this case a local bond model would be a more appropriate description.

In conclusion, it is felt that the W(100) surface with and without adsorbed hydrogen still could provide further generations of scientists with interesting problems.

We just want to add here that the H/Mo(100) system reveals a very similar (although not identical) behavior, namely, $\sqrt{2} \times \sqrt{2}$ superstructures and a sequence of more complex LEED patterns as hydrogen becomes adsorbed. We refer to the original literature [254,353,356a,359,369–371]. Again, the work done in Estrup's group deserves most attention [369,370].

Recently, a H-induced reconstruction was also reported for W(110), the "densest" tungsten plane [372]. Chung et al. concluded from their LEED experiments that a lateral shift of the topmost W atoms occurred at a critical coverage of hydrogen of about 0.5 ML, since at this coverage suddenly one of the mirror planes disappears. This reconstruction is between 85 and 300 K independent of temperature. It consists of a lateral shift of the top layer of W atoms, as a whole, parallel to the $[1\bar{1}0]$ direction.

As a brief summary, examples for various types of H-driven reconstruction have been presented and it is emphasized at this point that there is a need for more theoretical work in order to understand the tendency of H atoms to induce reconstructions of metal surfaces.

3.6.4. Phase transformations involving an overlayer–underlayer transition of hydrogen

It may be worthwhile to add some brief remarks on a problem that has been coming up during the last two or three years, namely, the eventual population of so-called "subsurface" binding states. This subsurface state may be regarded as an intermediate between a chemisorbed and a dissolved hydrogen species and was proposed among others by Behm et al. [275] for hydrogen interacting with a Pd(110) surface. The evidence for this subsurface hydrogen came from combined TDS and LEED experiments, in conjunction with proposals for potential energy diagrams given for Pd surfaces previously by Comsa et al. [75]. In particular the TD features allowed a fairly detailed evaluation of the potential energy of a H atom adsorbed *on* the surface and *in* the surface. This diagram is reproduced in fig. 54 and basically consists of a superposition of the potential energy curves of the dissociative chemisorption

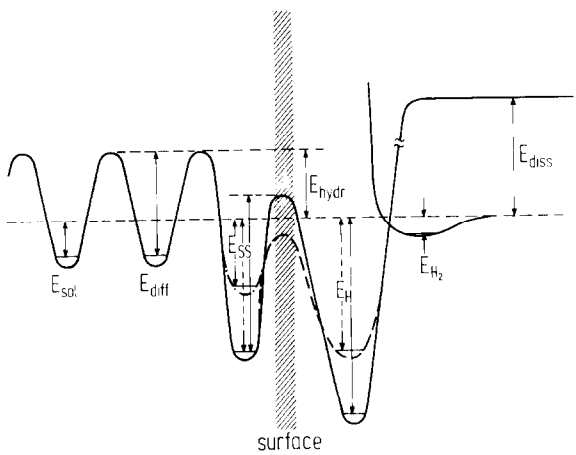


Fig. 54. Schematic potential energy diagram of the Pd(110)/H system showing the chemisorption, subsurface site population and bulk site population of hydrogen. The symbols have the following meaning: E_{sol}, heat of solution; E_{diff}, activation energy for bulk diffusion; E_{ss}, activation energy for desorption from the H subsurface sites; E_{hydr}, activation energy for hydrogenation; E_H, activation energy for desorption from the chemisorbed state; E_{H_2}, adsorption energy of the physisorbed H_2; E_{diss}, heat of dissociation of the H_2 molecule. The dashed line indicates the potential energy situation after the surface has been reconstructed by 1.5 monolayers of hydrogen. Clearly, the activation barrier for population of subsurface sites is removed by reconstruction.

(cf. fig. 6) and of the dissolved hydrogen in the Pd bulk, whereby the respective bulk energies were taken from earlier work by Auer and Grabke [373] and Lynch and Flanagan [374]. Later, Behm's results were confirmed by the He diffraction measurements by Rieder et al. [375]. It is worth to mention that quite a similar potential energy diagram was arrived at by Engel and Kuipers [376] in their molecular beam study of the H_2–D_2 isotope exchange reaction on a Pd(111) surface. Behm et al. pointed to the important influence of the surface (1×2) reconstruction of the Pd(110) surface at $\Theta_H \approx 1.5$ ML which may lead to structural changes of the surface region of the Pd crystal so that further uptake of H may easily become possible which in turn causes the reconstruction to proceed. We want to add here that the pairing-row reconstruction mechanism was confirmed recently by ion scattering studies of Niehus et al. [377].

Actually the first report that proposed the population of "subsurface" sites came from Plummer's group [144]. Eberhardt et al. found in their photoemission measurements evidence for a (temperature dependent) occupation of subsurface sites with the (111) surfaces of Ni, Pd and Pt. They observed the usual H 1s resonance features at -8 eV in the photoelectron energy distribution curve at low temperatures (~ 100 K). However, as the H-covered sample was heated the H became "invisible" in the UP spectra, also the surface state near the Fermi edge of, e.g., Pd(111) was not affected anymore by adsorbed H. Eberhardt et al. concluded therefore on a subsurface H species formed under these conditions and on a conversion of regularly chemisorbed H to this subsurface hydrogen upon heating to room temperature. In a later study dealing with the H/Pd(111) system [145] they specified their statements somewhat since it could be shown that at least for H/Ni(111) [50] and for Pt(111) [378] no such subsurface H species was detected by other methods (LEED, ion channeling). They argue that also subsurface H should produce a H 1s–metal bonding level split off from the Pd band which should be detectable somewhere between 7 to 9 eV below E_F. Measurements on Pd hydride by Schlapbach and Burger [379] indeed exhibit a weak extra emission around 8 eV below E_F. Various explanations are offered to understand this invisible form of adsorbed hydrogen [145] which are, however, not very satisfactory and call for more detailed experimental information. We remind here that the phase diagram calculations by Foiles and Daw [328] for H/Pd(111) clearly revealed the necessity for assuming the population of subsurface sites even and particularly at *low temperatures* when the ordered three subsurface hydrogen structures form. This again confirms that subsurface hydrogen and the "invisible" form of hydrogen are not necessarily correlated with one another. Recently there have been reports which state the existence of subsurface hydrogen also for some other metal surfaces.

With Cu(111) and Cu(100) again Greuter et al. [380] reported on the occurrence of "invisible" H and attributed it to a subsurface species. We recall

here that the involvement of interior sites in the bonding of hydrogen to surfaces was also proposed by Comsa and his group for Pd(100), Cu(100) and Cu(111) surfaces [75,76], whereby the evidence came from measurements of the velocity distribution of desorbing species.

A controversy came up recently with the system H on Ru(0001). In their first extensive study [113] Feulner and Menzel observed two differently polarized H states and discussed the possibility of subsurface site population. In a somewhat later report on the same system Yates et al. [381] studied the coadsorption of H_2 and D_2 by thermal desorption and interpreted their results (an inverse kinetic isotope effect in the desorption of H_2, HD and D_2) in terms of a preferential occupation of Ru subsurface sites by hydrogen following low temperature adsorption (which occurs into the threefold surface sites as suggested by a HREELS study by Barteau et al. [179]). In the hcp lattice, two kinds of subsurface sites are possible, namely, a tetrahedral and an octahedral site, and this later site is likely occupied by subsurface hydrogen as suggested by Yates et al. Again, as in the case of Pd(110)/H, the potential energy diagram of the Ru surface region interacting with hydrogen is discussed and it is concluded that in going from the vacuum side to the bulk metal a first deep potential populated by *chemisorbed* H is passed followed by a shallower endothermic subsurface potential which is then transforming to the endothermic bulk H sites. All potential minima are separated by appreciable activation barriers. Within this model it was possible to explain the temperature dependent conversion from H_{ad} to H_{subs}. There was a response to Yates' work by Feulner et al. [382] who scrutinized the question of the overlayer–underlayer transition by means of very low energy electron reflectivity (VLEED) measurements and came to the conclusion that the above-mentioned transition could be ruled out, since the "invisible" (with respect to UPS) form of H was clearly visible with the VLEED technique. Replies and comments on this interesting question were published very recently [383,384] and a final conclusion is not yet reached.

As another interesting example the Cu(110)/H system should be mentioned here which was investigated in Rieder's group using He diffraction and LEED [385]. They report on a H-induced subsurface *reconstruction* of a Cu(110) surface: *Room temperature* adsorption of atomic hydrogen initially produces a "special" 1×2 phase (denoted as $(1 \times 2)^*$) which can be seen by LEED (which method is known to probe several surface layers). However, this phase remained *invisible* with the atom diffraction; neither He nor Ne gave any additional diffraction intensities. The conclusion therefore was that H moved to subsurface sites and there induced a reconstruction of deeper layers thereby giving rise to the LEED features. Upon further H uptake the (1×2) periodicity extends to the surface and then becomes also visible with He diffraction. Low temperature phases in which H adsorbs *on* the surface (among others a 1×3 phase is formed) transform around 240 K into the reconstructed phases.

Whilst the *direct* evidence for subsurface hydrogen is actually scarce there are some theoretical publications which could, at least partially, support the experimental assumptions; we refer to the work by Lagos [2] and Lagos and Schuller [386] in which it was shown that the self-trapping energy of an interstitial impurity (e.g., a H atom) increases substantially when it approaches the surface, taking at the first layer of interstitial sites a value which is considerably larger than that of the bulk. Other more recent calculations by Jacobsen and Nørskov [287] using the effective medium theory focussed on the specific problem of chemisorption and incorporation of H on the Cu(110) surface, whereby the H-induced relaxations of the Cu lattice and their role in stabilizing hydrogen just under the surface was given particular attention. It is found that H indeed induces local lattice distortions even in the chemisorbed state, and these relaxations are largest when H is located just under the surface. However, the relaxations are not large enough to give this subsurface site more stability than the chemisorption site, and thus Nørskov's calculations do not support Rieder's interpretation of the observed effects [385]. Clearly, also in the Cu(110)/H system there is a H-induced surface reconstruction of the 1×2 type occurring, but it is not yet clear if it consists of the missing-row (MR) on the pairing-row (PR) type, or is actually a novel type of reconstruction involving subsurface layers as proposed by Rieder.

Nørskov's calculations however support very much a view that the author of this article has gained from many experimental observations and that has been put forth recently [331], namely, that whereever and whenever the small H atom adsorbs on a surface a strong perturbation of the local electronic structure occurs which also causes a local strain on the crystal lattice of the substrate. In the easiest case just layer relaxation effects may be the consequence, but as the perturbation increases, local reconstructions (as in the case of H on Rh(110) [70,388]) may be induced or even PR or MR reconstructions, in the most complicated case accompanied by a steady uptake (dissolution, permeation) of hydrogen leading to a progressive reconstruction and final hydride formation (as observed with Pd(110)).

3.7. Surface reactions of hydrogen

The final remarks of the foregoing section may be carried over to the general problem of possible chemical reactions that may run on a surface with hydrogen being one essential reactant. Although it is beyond the scope of this article to enter the field of surface reactions in greater detail some supplementary remarks may nevertheless be useful here, particular on the H_2–D_2 isotope reaction.

The simplest surface reaction that one can think of is the recombination of two isolated chemisorbed H atoms to form a H_2 molecule. As we have learnt in section 3.5 this is quite often the rate-limiting step in thermal desorption.

Other more complicated reactions are the H_2–D_2 isotope exchange, the water formation or the ammonia synthesis reaction. Also, hydrogenation and hydrogenolysis reactions of hydrocarbons or carbon monoxide come to one's mind which are of fundamental interest in the field of heterogeneous catalysis. However, it would be outside the scope of this article if we would expand on this matter which is dealt with in many reviews and monographs [389–392], and accordingly we shall only concentrate on those reactions in which (besides the surface) only hydrogen (deuterium) is involved.

There is, however, another class of reactions that involves hydrogen, namely, all chemical processes in which the hydrogen reacts *with the surface* material itself, that is to say, hydride formation reactions. Again, this is a matter of widespread interest in view of reactor technology, metallurgy and energy storage technology, and there exists a great many special reports on that subject as mentioned in the introduction [3,4,7,14,17]. We have learnt in section 3.6.3 that the surface *reconstruction* may be regarded as a preceding step towards the surface compound formation, and the systems H on Ni(110) [231] and Pd(110) [68] are good examples here although the characterization of a surface hydride of Ni or Pd is certainly not straightforward. However, if a *volatile* hydride species is formed it is fairly simple to prove the existence and to characterize the stoichiometry and structure of such a compound. An excellent example here is the reaction between hydrogen and *silicon* to form various silane species which we shall briefly deal with in this section. Also, certain clean-off reactions are known in which H reacts with surface contaminations such as carbon or oxygen to methane or water at elevated temperatures. A common feature of these reactions is that the reaction *rate* is strongly dependent on the chemical state of the hydrogen: Very low rates are found with molecular hydrogen whereas very efficient and fast reaction cycles are obtained with predissociated atomic hydrogen.

3.7.1. The H_2–D_2 exchange reaction

A somewhat more complicated reaction than the simple desorption is the hydrogen–deuterium isotope exchange which has already been mentioned as a simple method to probe the dissociative or associative character of H_2 adsorption, whereby H_2 (or D_2) dissociation is indicated by isotope scrambling, i.e., HD formation. Besides this "fingerprint" technique there are numerous reports in which the mechanism, kinetics and energetics of the isotope exchange reaction have been investigated. A valuable review in this field was given some years ago by Engel and Ertl [393]. Several questions have been addressed therein, for example, the kinetics of the H_2–D_2 exchange and its relation to the adsorption and desorption kinetics of the pure reactants, H_2 and D_2. Also, the origin of an eventual activation energy for the exchange was critically discussed. The actual reaction mechanism was studied using mainly molecular beam techniques, and the influence of the surface structure on the mechanism

of the isotope exchange was subjected to a detailed study. At equilibrium conditions and at temperatures above ~ 300 K the relative amounts of hydrogen, hydrogen deuteride and deuterium formed on active transition metal surfaces are related by 1 : 2 : 1 [394] (this was already pointed out in section 2.6, cf. eq. (9)). One may therefore deduce that both the reaction mechanism and the rate constant for their formation are very similar for the three species. The adsorption and net reaction as described by eqs. (7)–(9) do, however, not allow any conclusion about the reaction *mechanism*. Whenever surface reactions are dealt with, *two* principally different mechanisms can be involved: The Langmuir–Hinshelwood (LH) and the Eley–Rideal (ER) [395–398] mechanism.

In the LH model both species H and D react from the *adsorbed* state, according to the scheme:

(1) dissociative adsorption of *both* species

$$H_{2(g)} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} 2\,H_{ad}, \tag{48}$$

$$D_{2(g)} \underset{k_{-2}}{\overset{k_2}{\rightleftharpoons}} 2\,D_{ad}; \tag{49}$$

(2) reaction on the surface (equivalent to a recombination step)

$$H_{ad} + D_{ad} \underset{k_{-3}}{\overset{k_3}{\rightleftharpoons}} HD_{ad}; \tag{50}$$

(3) desorption of the molecular entity into the gas phase

$$HD_{ad} \underset{k_{-4}}{\overset{k_4}{\rightleftharpoons}} HD_g; \tag{51}$$

whereby steps (2) and (3) are, of course, also occurring with the homonuclear species.

Contrary to the LH reaction the ER mechanism reads:

(1) dissociative adsorption of *one* species only (same as eq. (48))

$$H_{2(g)} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} 2\,H_{ad}; \tag{48a}$$

(2) reaction of the second species from the gaseous state with a *preadsorbed* atom

$$D_{2(g)} + H_{ad} \underset{k_{-5}}{\overset{k_5}{\rightleftharpoons}} HD_{ad} + D_{ad}, \tag{52}$$

which step obviously requires the activation and dissociation of a D_2 molecule at the very moment of its impact on the surface;
(3) the desorption of HD, in complete analogy to step (51).

It is noted that these fairly simple reaction schemes may become any more complicated if molecular precursor states are admitted for population and reaction. Surfaces like Pd which provide a reservoir of dissolved or "subsurface" hydrogen are expected to exhibit an even more complex reaction behavior as was convincingly shown by Engel and Kuipers for H_2 and D_2 interacting with a Pd(111) surface [376]. It is a long-standing question if the H_2–D_2 exchange obeys the LH or the ER mechanism, and until now there is no general answer possible. From their molecular beam work Engel and Kuipers concluded on the LH mechanism for the system H_2/D_2–Pd(111) [376] whilst Bernasek and Somorjai [399] assumed an ER reaction type for a Pt(111) surface at elevated temperatures. A sharp distinction between the two mechanism for a fast reaction is not at all easy; however, as Engel and Ertl [393] have pointed out it should be possible with modulated molecular beam experiments.

A simple kinetic consideration of the reaction scheme (eqs. (48)–(52)) yields the rate expressions:

$$\mathrm{d}[\mathrm{H_{ad}}]/\mathrm{d}t = 2k_1\left[\mathrm{H_{2(g)}}\right] - k_{-1}[\mathrm{H_{ad}}]^2, \tag{53}$$

$$\mathrm{d}[\mathrm{D_{ad}}]/\mathrm{d}t = 2k_2\left[\mathrm{D_{2(g)}}\right] - k_{-2}[\mathrm{D_{ad}}]^2, \tag{54}$$

$$\mathrm{d}[\mathrm{HD_{ad}}]/\mathrm{d}t = k_3[\mathrm{H_{ad}}][\mathrm{D_{ad}}] - k_{-3}[\mathrm{HD_{ad}}], \tag{55}$$
(LH mechanism)

$$\mathrm{d}[\mathrm{HD_{ad}}]/\mathrm{d}t = k_5[\mathrm{H_{ad}}]\left[\mathrm{D_{2(g)}}\right] - K_{-5}[\mathrm{HD_{ad}}][\mathrm{D_{ad}}], \tag{56}$$
(ER mechanism)

$$\mathrm{d}\left[\mathrm{HD_g}\right]/\mathrm{d}t = k_4[\mathrm{HD_{ad}}] - k_{-4}\left[\mathrm{HD_g}\right]. \tag{57}$$

The kinetic equations for the decisive steps can be simplified to yield for the LH mechanism:

$$\mathrm{d}[\mathrm{HD_{ad}}]/\mathrm{d}t = k_3[\mathrm{H_{ad}}][\mathrm{D_{ad}}], \tag{58}$$

and for the EH mechanism:

$$\mathrm{d}[\mathrm{HD_{ad}}]/\mathrm{d}t = k_5[\mathrm{H_{ad}}]\left[\mathrm{D_{2(g)}}\right], \tag{59}$$

if we neglect in each case the reverse reactions ($k_{-3} \ll k_3$ and $k_{-5} \ll k_5$). $D_{2,g}$ is proportional to the deuterium gas pressure, and we obtain:

$$\mathrm{d}\left[\mathrm{HD_g}\right]/\mathrm{d}t = k_5'[\mathrm{H_{ad}}]\ P_{\mathrm{D_2}}. \tag{60}$$

This means that the rate of formation of HD should be proportional to the D_2 pressure (if H is preadsorbed) for the ER mechanism whereas this rate should be linearly correlated with both the H and D surface coverage, if the LH mechanism holds. Unfortunately, of course also H_{ad} and D_{ad} depend (through the adsorption rate) on the respective gas pressures, so that a clear distinction

cannot be made in this way. With the adsorption–desorption equilibrium being established we have

$$\frac{d[HD_{ad}]}{dt} = \frac{d[D_{ad}]}{dt} = 0, \tag{61}$$

and we can express $[H_{ad}]$ and $[D_{ad}]$, respectively, by $[H_{ad}] = ((k_1/k_{-1})[H_{2(g)}])^{1/2}$ and $[D_{ad}] = ((k_2/k_{-2})[D_{2(g)}])^{1/2}$. We then obtain for the LH mechanism (eq. (58)):

$$\frac{d[HD]}{dt} = k_3\left(\frac{k_1}{k_{-1}}\frac{k_2}{k_{-2}}[H_{2(g)}][D_{2(g)}]\right)^{1/2}, \tag{62}$$

and for the ER mechanism (eq. (59)):

$$\frac{d[HD]}{dt} = k_5'\left(\frac{k_1}{k_{-1}}[H_{2(g)}]\right)^{1/2} P_{D_2}. \tag{63}$$

Since $H_{2,g}$ and $D_{2,g}$ are directly related to the respective gas pressures we can rewrite eq. (62)

$$d[HD]/dt = k_3 k_1' k_2' P_{H_2}^{1/2} P_{D_2}^{1/2}, \tag{64}$$

and eq. (63) accordingly

$$d[HD]/dt = k_1' k_5' P_{H_2}^{1/2} P_{D_2}. \tag{65}$$

The rate constants can be expressed using eq. (21) and eq. (29) to yield (non-activated adsorption assumed):

$$\frac{d[HD]}{dt} = \nu_{HD}\left(\frac{s_0^H s_0^D P_{H_2} P_{D_2}}{\nu_H \nu_D 2\pi kT m_H^{1/2} m_D^{1/2}}\right)^{1/2} \exp\left(-\frac{\Delta E_H^* + \Delta E_D^* - 2\Delta E_{HD}^*}{2kT}\right). \tag{66}$$

In this equation ΔE_{HD}^* is the activation energy required for the formation of HD and ν_{HD} the corresponding pre-exponential factor. ΔE_{HD}^* is also equal to the activation energy to remove HD from the bottom of the potential well according to the rate of desorption:

$$d[HD]/dt = \nu_{HD} \exp(-\Delta E_{HD}^*/kT). \tag{67}$$

This means that there should be a net activation energy for the HD production left only if $\Delta E_H^* + \Delta E_D^* \neq 2\Delta E_{HD}^*$. Since these energies are usually quite similar, a significant activation energy is not to be expected. One could only think of different surface diffusion rates of H_{ad} and D_{ad} caused by different diffusion coefficients and activation energies which might introduce a small net activation barrier for the HD production. A more important result of eq. (66) is that the square root of the initial sticking probabilities for H_2 and D_2 enters the rate expression. This means that high initial sticking coefficients will

cause high HD production rates and vice versa. This again elucidates the role of "active centres" in catalysis which can be regarded as sites at which dissociation and chemisorption occurs readily. This result is completely in line with isotope exchange measurements using the molecular beam technique by Bernasek and Somorjai [399] and Gale et al. [59] mentioned earlier (cf. section 2.3) which emphasized the importance of step sites in the dissociation and reaction of H_2 and D_2 for Pt surfaces, a question that was also addressed by Wachs and Madix [400].

Eq. (66) has been derived for the limit of small H_2 and D_2 coverages. Accordingly, it contains the *initial* values of the sticking probabilities, frequency factors and activation energies. It may be extrapolated to higher coverages by considering the corresponding coverage-dependent parameters. We do not evaluate a complete expression for the ER mechanism since the elementary steps involved are certainly more complicated, in particular the vibrational activation of the molecular bond of the diatomic gas phase molecule which collides with the preadsorbed atom. We recall here that the gas phase reaction: $H_2 + D_2 \rightarrow 2HD$ is activated by 176 kJ/mol [401,402]. Very likely, a mobile precursor may be involved which would increase the reaction efficiency substantially. If the life time of a trapped molecule is estimated to be of the order of 10^{-8} s the time allowed for a reaction to occur is several orders of magnitude greater than for a single collision event. A strong hint that some sort of ER mechanism is operating was obtained for Pt(111) [158] from the following experiment: The surface was saturated with adsorbed H and then exposed to D_2 gas. Surprisingly, the TD spectrum showed distinct amounts of HD desorption in all binding states. This can only be reconciled with an exchange between the chemisorbed H_{ad} and a non-adsorbed D_2 molecule (which, however, might have been trapped in a mobile precursor state).

As far as experimental investigations are concerned basically two types of studies can be distinguished. In the first place we mention studies of the dynamical behavior of an adsorption/reaction system whereby the general isotope exchange activity of a given surface is examined and its dependence on time, partial pressures of the reactants, surface structure, etc. Very revealing information here is obtained from molecular beam measurements [59,376,393, 399], especially on the mechanism of the reaction. On the other hand, turn-over rates are frequently determined in a flow system using steady-state partial pressures of H_2 and D_2, as a function of the sample temperature. This type of experiments yields the activation energy for reaction and pre-exponential factors, as a function of reactant pressure, surface temperature and structure [131,158,403–408]. Of particular interest is the question whether or not there is a distinct activation energy required for the H_2–D_2 exchange reaction to take place. As was emphasized by Engel [409] there is not need to assume such activation barrier if the reaction runs via the LH mechanism;

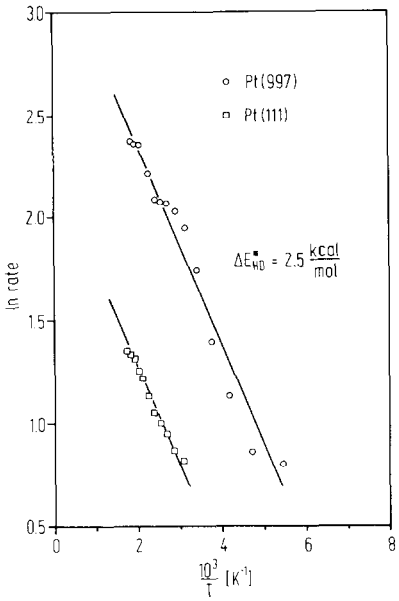


Fig. 55. Plot of the logarithm of the HD formation rate versus the reciprocal temperature for a flat (open circles) and a stepped (open squares) Pt(111) surface. Note that in both cases the same activation energy results whilst the pre-exponentials are different. After ref. [131].

rather, the desorption of the product species HD is thermally activated with the usual activation energies known from H_2 (D_2) adsorption/desorption experiments. Nevertheless there are several reports that a distinct activation energy has to be surmounted for HD to form. Christmann and Ertl report on the same $\Delta E^* = 10.5$ kJ/mol for a stepped (997) and a flat Pt(111) surface, whereby the actual rate was a factor of 10 higher on the stepped surface [131]. This is illustrated in fig. 55. Schober [407] arrived at $\Delta E^* = \sim 40$ kJ/mol at low and 15 kJ/mol at high temperatures for the isotope exchange on Ni(111). Again we point to the eventual importance to the influence of precursor states when discussing these data. Also, eventual surface impurities (carbon) can cause artificial temperature dependences which might then be misinterpreted in terms of activation energies for isotope exchange reactions. An example is provided by Robota et al. [406] who studied the H_2–D_2 isotope exchange over Ni(111) and (110) surfaces under steady-state conditions as a function of the surface temperature. An anomaly around the Curie temperature, T_C, was found in the first experiments which could be ascribed to a so-called "magneto-catalytic effect". It turned out, however, that this anomaly was actually caused by spurious amounts of surface carbon which, right at T_C, diffused into the bulk and thus changed the reactivity of the surface. Subsequent measurements with carefully cleaned surfaces did not show the anomaly anymore.

3.7.2. The reactive interaction between hydrogen and semiconductor surfaces

As a typical example of a hydrogen reaction that can easily lead to an attack of the solid surface we focus on the reaction of hydrogen with silicon. This section then may also provide the reader with some typical features of the general interaction between hydrogen and a semiconductor. Usually a noticeable interaction can only take place if the hydrogen is offered in the *atomic* form, then, however, this interaction is quite strong. The reason is that molecular dihydrogen does not dissociate spontaneously on low-index semiconductor surfaces. In order to obtain reactive *atomic* hydrogen, in most cases a hot W filament is placed in front of the sample (as was already mentioned in section 2.5) which produces at least a mixture of molecular and atomic hydrogen. Sometimes, reactive atomic hydrogen can be formed from molecularly adsorbed H_2 in small amounts on highly coordinated step and kink sites as they are formed by cleavage of silicon crystals in vacuo. This was proven by Henzler and Froitzheim and his group by means of electron energy loss spectroscopy [93,410]. Also, the thermal decomposition of adsorbed water on Si surfaces may lead to reactive atomic hydrogen as was demonstrated by Ibach et al. [411] for a Si(100) surface and by Schaefer et al. for Si(100) and Si(111) [192]. The investigation of H-covered Si surfaces dates back to 1974 when Ibach and Rowe studied the reconstructed Si(111)-7×7 surface by LEED and UV-photoelectron spectroscopy [412]. They found that the 7×7 LEED "extra" spots disappeared only after prolonged exposure of hydrogen and reported the development of a single H-induced UPS peak 11.3 eV below the vacuum level E_{vac}. The observed features were explained in terms of the formation of silicon monohydride SiH on the surface. Sakurai and Hagstrum [413] and Appelbaum et al. [414] applied ion neutralization spectroscopy (INS) to this system and found two peaks at 10 and 12 eV below E_{vac} in agreement with later studies by Butz et al. [415]. At the same time a variety of theoretical studies on the Si/H system was performed, for example, by Pandey [416], Appelbaum and Hamann [197], or by Ho et al. [417]. In the theories it was assumed that the dangling bonds of the Si(111) surface atoms became saturated by adsorbed hydrogen thereby causing two "extra" peaks in the local density of states about 4–5 and 7 eV below the valence band maximum. Later Wagner et al. [418] concluded on the basis of their electron energy loss spectroscopy data at 300 K that the Si surface atoms could readily be hydrogenated to form dihydride (SiH_2) and trihydride (SiH_3) species. In a more recent investigation, Kobayashi et al. [195] using EELS studied again the Si(111)-7×7 surface and reported the existence of four different adsorbed states: In the very early stage of adsorption ($\Theta \leq 0.3$) the SiH species is predominantly formed by a covalent interaction between the H 1s orbital and the dangling bonds of the Si(111) surface atoms whereby each such atom contributes one dangling bond. The situation is depicted in fig. 56. The predominant loss feature in the EELS vibrational loss spectra is a peak at 2057 cm^{-1} that occurs after exposure of

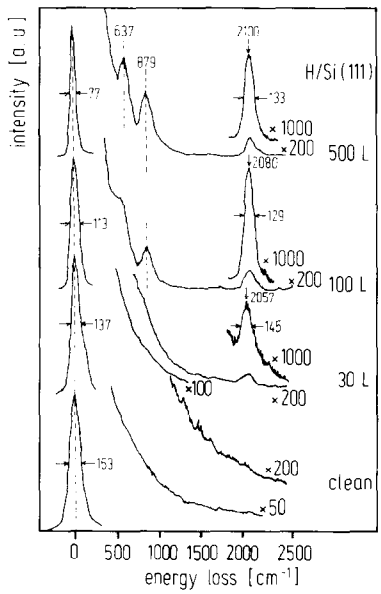


Fig. 56. Series of HREEL spectra obtained from H interaction with a Si(111) surface and showing the various H-induced vibrational losses, as a function of H exposure. See text for further details. After Kobayashi et al. [195].

30 L of hydrogen. Higher exposures lead to two new loss features at 637 and 879 cm^{-1} (rocking and bending modes of SiH) and an energy shift and intensity increase of the ν_{Si-H} peak which occurs at 2100 cm^{-1} after 500 L H_2 exposure at 300 K. All loss peaks disappear upon heating to ~ 900 K indicating the thermal desorption of SiH_x compounds. Higher H_2 exposures induce a Si–Si bond breaking accompanied by a successive hydrogenation, SiH_4 and, possibly, SiH_3 which desorb after their formation. Owing to this progressive corrosion more and more Si atoms become surface atoms and can react to either SiH, SiH_2 or SiH_3 complexes. Upon heating to ~ 650–750 K progressive decomposition of SiH_2 and SiH_3 complexes occurs; SiH species are formed and H_2 desorbs. It must be emphasized here that the dipole selection rule which on metal surfaces specifically samples vibrations with a dynamical dipole moment perpendicular to the surface seems not to apply to semiconductors since here the screening is not complete [168,419]. Correspondingly, a greater variety of loss features can result.

The (2×1) reconstructed Si(100) surface and its interaction with hydrogen was studied by Schaefer et al. [193,194] by means of EELS. Fig. 57 is taken from their summary work [192] and shows the various possible vibrations of Si monohydride and dihydride. Depending on the H_2 exposure and surface temperature, two stable H phases are found. At small exposures only the

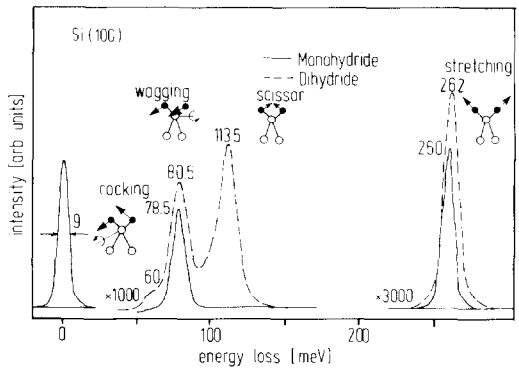


Fig. 57. Principal vibrational loss features obtained with the H/Si(100) system by Schaefer et al. [192]. The diagram shows the monohydride losses (full line) and the dihydride losses (dashed line) with the respective vibrational modes indicated.

monohydride SiH is formed. The two modes of this 2×1 SiH phase are due to Si–H stretching (2080 cm^{-1}) and bending modes (624 cm^{-1}). Higher exposures (> 1000 L) lead to the dihydride SiH_2 with four losses (dashed curve in fig. 57) at 480 cm^{-1} (rocking mode), 644 cm^{-1} (wagging mode), 908 cm^{-1} (scissor mode) and 2096 cm^{-1} (stretching mode). More details can be taken from the original work. Schaefer [420] also reports on H interaction with the (2×1) reconstructed Si(111) surface and observed basically the quenching of all clean surface features (electronic surface state transitions and surface phonons). Instead the Si–H wagging mode appears at 609 cm^{-1} and the stretching mode at 2057 cm^{-1}. The author also points to the possibility of using the adsorption of atomic hydrogen on semiconductor surfaces as a tool to mark the local atomic structure of the semiconductor surface, since both the intensity as well as the frequency position of the characteristic vibrational modes react sensitively upon the local configuration of the adsorption complex. This statement holds not only for pure Si surfaces but also for alloys of the type Ge_xSi_{1-x} with (100) orientation and III–V semiconductors such as GaAs and InP [421]. There are more recent investigations of H interaction with Si single crystal surfaces [422–424]. Muller et al. [425] have studied the coexistence region fo the SiH and SiH_2 phase formed with the Si(111)-7×7 surface using UV and X-ray photoemission spectroscopy. They provide evidence that the adsorption sites for the SiH_2 states are not produced by corrosion of the Si surface. Johansson et al. [426] have focussed on the electronic structure of the chemisorbed H phase (SiH compound formation) on the Si(100)-2×1 surface. They observe, with angle-resolved UV photoemission, two strong hydrogen-induced surface states or resonances (4.8 and 5.6 eV below E_F at the J′ point of the surface Brillouin zone) and another surface

state 1.8 eV below E_F at J′ which is associated with the dimer bond. The authors point to the good agreement between their experimental results and recent band structure calculations [427].

There is generally a growing interest in the investigation of H–semiconductor systems since adsorbed H can easily modify the device characteristics: Defect sites may become passivated, or the chemical interaction between Si and a metal deposit layer can become perturbed by adsorbed H interlayers [428]. Besides the aforementioned hydrogenation of silicon which is apparently easily possible with atomic hydrogen, there have been a few studies performed also with other semiconductors, for example GaAs(100) and $(\bar{1}\bar{1}\bar{1})$ [429]. It is well-known that many of the clean polar surfaces of GaAs exhibit a variety of surface reconstructions. In this situation the bonding of atomic hydrogen to these surfaces can greatly modify the surface electronic structure since H interacts with the dangling bonds of the semiconductor which then causes modifications of the surface structure. In this respect, the semiconductor behaves somewhat similar to the metal surfaces that reconstruct under hydrogen (cf. section 3.6.3). The main difference, however, is that with the semiconductors *directed* bonds are saturated (which leads to purely covalent bonds) whereas with metals a more delocalized and only predominantly covalent H bond is formed. Bringans and Bachrach [429] report on dramatic changes of the surface structure of GaAs as determined by LEED after exposure of $> 10^6$ L H_2. Angle-resolved UV photoemission and core level spectroscopy were used to characterize the H covered surface. Among others a H-induced surface state ~ 5 eV below the valence band energy was found. The authors conclude that on both the (100) and $(\bar{1}\bar{1}\bar{1})$ surfaces adsorbed H layers are formed which lead to surface properties that are independent of the composition of the initial surfaces. From the XPS data it appears as if the H atoms are bonded mainly to the arsenic atoms.

All in all, there is strong evidence that – whenever atomic hydrogen is available – Si and other semiconductor surfaces adsorb this species readily with high sticking probability. The work on Si(111)-7 × 7 by Schulze and Henzler [92] and Schulze [430] so far is the only work in which thermal desorption and Si–H binding energy measurements were reported. The binding energies given in that work lay between 2.2 eV/mol at small coverages ($\Theta_H \approx 0.05$) and 3.1 eV at $\Theta_H = 1.0$ and are thus considerably larger as the typical adsorption energies found with metals. However, since the H was offered in the atomic state, this is expected. Interestingly, the value of a single Me–H bond, for example Ni–H (~ 2.6 eV) is of the same order of magnitude as that found with Si–H ($2.2 < E_{Si-H} < 3.1$ eV). In this respect the chemical interaction between hydrogen on the one hand and metals and semiconductors on the other seems quite comparable. The major difference is that there is a large activation barrier for H_2 dissociation on a semiconductor surface, very likely caused by the absence of d-electrons right at the Fermi edge.

4. Model theories to describe the chemisorption of hydrogen

4.1. General remarks

In the foregoing sections we have learnt about the considerable progress that has been achieved in the experimental handling of hydrogen adsorption. On various occasions we have emphasized the inseparable role of the *theory* in the fundamental understanding of the H–surface interaction phenomena. In order to mimic the physical and chemical state of a given adsorption system (including the changes of the solid that may accompany the adsorption) one can, however, hardly rely on a single theoretical description alone. Rather one must consider several kinds of complementary theories (which may use entirely different formalisms). Only, if all these descriptions independent of each other reach similar results one can believe in a real physical understanding of the respective process.

In this sense, *electronic theories* yield information about the electron distribution and energy levels of the gaseous molecules to be adsorbed, of the solid and of the respective adsorbate system which consists of the adsorbed particles and one or more atomic layers of the substrate underneath. It is possible with an appropriate electronic theory to predict, for example, heats of adsorption at $T = 0$ K. There exist certain relations to two other kinds of theories, namely to the theory of the dynamics and kinetics of the gas–solid interaction (which, among others, treats the particle scattering, absolute rates of adsorption and desorption, parameters of activation, sticking and dissociation probabilities, etc.), and to the body of statistical theories. Quantum statistics or statistical thermodynamics furnish information about the occupation of energy levels at $T > 0$ K, about the role of the surface concentration of the adsorbate, and about thermodynamic functions such as free energy, enthalpy or entropy. Finally, two more disciplines contribute significantly to the understanding of the gas–solid interaction, namely, the phenomenological (chemical) thermodynamics and the quantum mechanics, the latter one providing knowledge about lattice vibrations, rotational and vibrational properties of adsorbed particles (curvature of interaction potentials, force constants, etc.).

In this chapter it is, of course, simply impossible to enter into the particulars of all of these theories, nor will it be possible to present any mathematical formalisms. Instead, we shall restrict ourselves to a qualitative description of some promising concepts for electronic model theories which have proven in the past to successfully describe and predict the interaction of hydrogen with surfaces, especially transition metal surfaces. Some of these theories can even model the H_2 dissociation step in that they calculate the interaction potential of the incoming H_2 molecule as a function of the distance to the surface (cf. section 2.5, fig. 7).

Generally, an appropriate electronic theory should predict the following properties of the H–metal interaction system in satisfactory agreement with the experiment: The ground state binding energy as a function of the separation of the H atom (molecule) from the substrate, the charge density, the induced dipole moment, the geometrical parameters (absolute adsorption site, bond lengths, bond angles), ground state vibrations, lateral interactions, etc.

There exists a vast number of reports on all kinds of theory, and we refer to some monographs [431,432] or review articles [433–436] for further reading. These articles contain details of the theoretical description, computational techniques, comparison with other theoretical models, etc.

4.2. Classification of chemisorption theories

One may distinguish basically two groups of model electronic theories:
(i) band structure models including, e.g., pseudo-potential methods, X_α band theory, ab-initio Hartree–Fock (HF) calculations;
(ii) cluster models, where again several "sub"models exist, such as all-electron (AE) ab-initio calculations without or with multiconfigurational interaction (CI), the SCF X_α approach, or various pseudo-potential cluster methods.

This classification of chemisorption theories corresponds somewhat to the distinction between solid-state physics (where the infinite solid with its electronic band structure is regarded) and molecular physics (where independent molecules are considered mainly using the molecular orbital picture). Evidently, a handling of the chemisorption problem can greatly benefit from an appropriate combination of these two theoretical approaches. We may subdivide the theories into semi-empirical (or, in a sense, "simple") and into first-principle methods. "Simple" means here that, in addition to the always adopted independent particle picture, additional approximations are often introduced (e.g., lack of self-consistency, use of semi-empirical parameters, neglection of electron correlation effects, etc.). On the other hand, we consider those calculations as "first-principle" ones that are performed self-consistently and include all the Coulombic interactions between the electrons, the nuclei and, of course, between the electrons and the nuclei as well as local or non-local exchange potential. The problem is always the inclusion of the local perturbation introduced by the adsorbate. In the layer (solid state) methods this is accomplished, for example, with the concept of the Green's function formalism, whereas in the cluster models this perturbation has to be calculated explicitly. At this point we have already left the physical *models* to describe the chemisorption problems and have moved to a consideration of the theoretical *procedures* or *formalisms* to tackle the mathematical problems when attempting appropriate solutions to the physical models. A whole variety of such formalisms exists, and we list here only some more general routes that

have served as a starting point for many other more refined "theories". These are:
(i) the semi-empirical models, among which we mention the simple Hückel (HMO) theory, related CNDO methods (complete neglect of different overlap), the extended Hückel methods (EHT), the modified Anderson models or the parametrized LCAO methods (linear combination of atomic orbitals);
(ii) the density functional methods which may be subdivided into the jellium models, the effective medium (EMT) theory, the embedded atom (EAM) method, the linear augmented plane wave (LAPW) method, etc.

Some of these concepts shall now be explained in somewhat more detail.

4.2.1. The "solid state' methods

At a first glance, the solid-state methods seem particularly well suited to model metal surfaces and/or surface–adsorbate complexes. Usually they treat the substrate problem in two different ways: They mimic the crystal either as a semi-infinite solid or as a slab with finite thickness. Sometimes it may be useful for symmetry and computational reasons to assume an adsorbed layer on both sides of the slab. Within this solid-state description we may further distinguish two situations: First, a *complete* layer of adsorbate on a solid crystal can be considered, and second, only a *single* adsorbate atom on such a surface is treated. This latter situation is in a sense the simplest one in that any adatom interactions of finite-size effects do not play a role.

A very early use of this adsorbed layer model was made by Koutecký [437] who considered the dissociative chemisorption of hydrogen on the (111) surface of a semiconductor crystallizing in the diamond lattice, whereby a terminal adsorption was assumed. Koutecký used the one-electron LCAO approximation with sp^3 hybrid orbitals and 1s orbitals of the semiconductor and the H atom, respectively. The H 1s wave function then coupled to the dangling bond hybrid orbital of the atom underneath. The result was that there is not only a tendency for the dangling bond and its surface state to vanish but also *new* surface states between the valence band and the conduction band are generated consisting of a bonding and an antibonding level. The H/Si(111) system (which may be regarded as a prototype of Koutecký's model calculation) has been theoretically investigated by Appelbaum and Hamann [197,438]; the calculated density of occupied states exhibits maxima at 4 and 7 eV below the top of the valence band, in accordance with measurements by Sakurai and Hagstrum [413]. These states could indeed represent the bonding states predicted earlier by Koutecký. In the context of self-consistent band models we must also mention the attempts of Feibelman and his group [33,439] and Hamann et al. [440] to model surfaces and adsorbate layers within their theory of "self-consistent surface linearized augmented plane wave" (SLAPW) which, among others, yields very successfully the local density of states LDOS near the Fermi level. Generally, all charge distribution effects are nicely predicted (cf. section 4.3.2).

Whilst the latter calculations were performed self-consistently there are also numerous examples of non-self-consistent layer calculations which shall be dealt with further below (cf. section 4.3.1).

Within the band structure approach the adsorption of a single H atom can likewise be treated by LCAO methods or by those theories which make use of a simplified semi-empirical Anderson model [441]. These have much in common with the tight-binding types of model theories which are extensively described elsewhere [442,443].

Usually, the one-electron approximation is chosen and in some cases, a certain self-consistency is reached in a semi-empirical way. Since a whole variety of calculations is based on the Anderson approach we shall present some examples in section 4.3.1.

All the LCAO–MO procedures suffer from a neglect of electron correlation effects. There are possibilities to include them in an extended Anderson model [444]. Another possibility consists in adopting the Heitler–London valence-band concept to chemisorption which, however, overemphasizes the electron–electron interaction. Schrieffer and collaborators [445] have chosen this route but it must be admitted that the computational effort is huge.

The other quite powerful handling of the single-particle chemisorption is provided by first-principle self-consistent calculations utilizing the local density function formalism (LDF) [446]. Here, the adsorption of a single H atom (or H_2 molecule) (represented by a positive point charge) on the surface of a jellium substrate is calculated in which the ion cores of the substrate atoms are replaced by a uniform positive background charge. The LDF approximation includes Hartree, exchange and correlation effects and can very well be treated within the jellium model. Approximate total energies of states, work function changes, etc. have been quite successfully predicted for H adsorption, in particular by Lundqvist and collaborators [447]. The simple jellium model can be significantly improved if the ion-core pseudo-potentials are introduced as a first-order perturbation on the jellium background.

4.2.2. The cluster methods

Again, the attempts to describe the chemisorption problem by modeling the bonding situation of a H atom on a *finite cluster* of substrate atoms can be classified into different categories, among which are one-electron theories, self-consistent approximations, for example, LCAO techniques, and theories taking into account the full electron correlation. Depending on the computational effort, H atoms on smaller or larger clusters can be handled, whereby, of course, a more rigorous treatment can only be made for a smaller cluster. There are certain advantages of the cluster approach which mainly consist in a reduced computational effort in order to reach the same level of sophistication as for the corresponding semi-infinite treatment. Localized effects can easily be described, and one can take advantage of concepts and methods which

were already developed in theoretical chemistry. Of course, there are also shortcomings of the cluster approach. The results may strongly and sometimes in a complex way depend on the size and geometry of the cluster, convergence may be difficult to obtain and must be tested in each case. The proper description of long-range effects can make it necessary to use a cluster that cannot be handled mathematically anymore.

Since the H chemisorption bond exhibits a fairly localized character there is, however, some justification for modeling this bond by a cluster of metal atoms. An especially valuable approach is the already mentioned SCF X_α treatment first suggested by Slater and Johnson [448].

It is based on the assumption that the electrons in a large molecule or in a solid can be successfully described by the "muffin-tin" potential approximation. This is characterized by a Coulomb-like potential near the positive ion core and by a constant potential in the intermediate range between the cores. The electron exchange interaction is taken care of by a statistical exchange potential which is derived from the charge density of the valence electrons. In detail, the following assumptions are made: (i) The cluster is divided into three potential regimes: A spherical potential close to individual atoms, a constant potential between the atoms, and again a spherical potential outside a sphere that surrounds the atoms and interatomic ranges. (ii) The Schrödinger equation can be solved exactly for the various spherically symmetric regimes. At the boundaries where the spherical wave functions overlap the approximation of multiple scattering waves is used. The empirical parameter α to describe the electron interaction has given this method its name.

A problem encountered is the choice of the radii of the respective spheres which influence the results in a sensitive manner. It is worth to mention that the SCF X_α method has been used to calculate the adsorption of a H atom on a Pd_4 cluster [449].

For small clusters more sophisticated theoretical methods can be used. A very reliable approach here is SCF–LCAO calculations using a large basis set of electron configurations including excited states, for example the MRD-CI method (multi-reference double-excitation configuration interaction) [450]. Due to the computational requirement this method is particularly suited for simple low-Z elements such as Li or Be; we refer to the work of Koutecký's group [451,452]. It is shown therein that the H chemisorption reacts very sensitively upon the geometry chosen for the respective metal cluster. Sometimes a rhombic configuration leads to lower energies than a tetrahedral configuration. By modifying the theoretical procedure H chemisorption even on Ge [453] or Pd clusters [454] can be calculated successfully with the Hartree–Fock CI methods. On the same line are calculations by Melius et al. [455] who considered a trapezoidal H_2Ni_2 cluster and compared it with an H_2Cu_2 cluster. They could convincingly show that the d-electrons are required to sort of "conduct" the H_2 molecule to the dissociation site whilst the chemical bonding is provided by the Ni sp electrons.

We also want to point to recent related work by Siegbahn and his group who treat, e.g., H chemisorption on Ni clusters of different size [456]. The calculations are performed in various stages of different complexity. At the lowest level the chemisorption is treated in the SCF approximation with frozen 3d orbitals. The next level is to relax the 3d orbitals, and the relaxed 3d results are then analyzed for covalency effects. Up to this stage small clusters are considered and treated at the all-electron level. At the most sophisticated level the metal atoms are described as one effective core potential (ECP) atoms and the correlation contribution is added. Clusters up to 50 Ni atoms are used to model the surface. For the chemisorption energies, the results for the largest clusters are in remarkable agreement with the best available experimental results. The calculated barriers for surface migration of H_{ad} ($\sim$ 22 kJ/mol) are likewise close to the experiment.

Self-consistent field calculations for clusters of up to 15 Be atoms and a H atom have been performed by Cox and Bauschlicher [457]. The most interesting result of their work was that the H chemisorption induces a significant relaxation on the (0001) surface of beryllium, in that a shortening of the nearest surface bonds in the Be substrate by 1% or 2% occurs.

Embedding of clusters into metallic surfaces is a frequently chosen route to circumvent the disadvantages of pure cluster models. We refer to the work by Cremaschi and Whitten [458,459]. They consider the chemisorption of H_2 on Ti(0001) by means of an ab-initio configuration–interaction theory for the surface region. The theory [460] is based on a self-consistent-field cluster model for the lattice, treated only approximately, from which then an N-electron subspace for a local region is defined by a unitary localization transformation. This local subspace includes the H_2 adsorbate and is treated accurately within the fixed field of the surrounding lattice. Dissociation of H_2 occurs above the surface but more stable threefold sites lie closer to the surface at $\sim$ 1.3 Å. The bonding is predominantly provided by the 4s electrons of the metal and leads to hydridic hydrogens and a polarized lattice electronic distribution.

4.3. Examples for theoretical treatments of H chemisorption

4.3.1. LCAO methods, the Newns–Anderson model and other related concepts

Looking back probably one of the first attempts to model the H chemisorption came from Toya [30]. He investigated the energy of a system which consisted of a single H atom and a metal with N conduction electrons. The procedure started from the wave function of $N+1$ electrons given by a linear combination of three kinds of Slater determinants: (i) the Slater determinants of $N+1$ Bloch wave functions of metal electrons (according to a Me^-–H^+ configuration), (ii) the Slater determinants of N metal electrons and the 1s

electron (Me–H configuration) and (iii) the Slater determinants of $N-1$ Bloch wave functions and two H 1s functions (Me^+–H^- configuration).

In the LCAO treatment the coefficients of the linear combination were obtained by the variation or by the SCF method. Toya concluded on the existence of two principally different types of chemisorption, which he called "r-type" and "s-type" adsorption. The r-type is the resonance state between the M–H and M^-–H^+ as well as the M^+–H^- configuration with the proton located *outside* the metal. The heat of adsorption was calculated for Ni and Cu to be ~ 3 eV/atom with an equilibrium distance ~ 1 Å outside the metal surface. The H atom in the r-state is negatively polarized with approximately $0.02e_0$ charge on it at the initial stage of adsorption, in good agreement with the experiment. The s-type adsorption, on the other hand, is characterized by sort of a bulk H atom whereby an ionization occurs with the proton being dissolved in the metal whilst the 1s electron is fed into the conduction band of the metal. The r-type adsorption predominates for Ni and Cu in that order at the initial stages of adsorption whereas the s-type of adsorption was believed to prevail at higher coverages, owing to a strong decrease of the heat of r-type adsorption with coverage. According to the larger work function and radius of platinum the heat of r-type adsorption as well as the polarization of H on a Pt surface was calculated to be smaller, and the heat of s-type adsorption closer to that of the r-type so that the s-type might predominate except at the very early stage of adsorption. It is interesting to comment here that there are indeed remarkable differences between a Ni(111) and a Pt(111) surface interacting with hydrogen: The heat of adsorption is on Pt(111) markedly lower than on Ni(111), and also the dipole moment on Pt(111) is reversed (decrease of φ).

It was Grimley who as one of the first tried to model the H chemisorption by adopting the Anderson formalism [461]. In second quantization, for any orbital Φ_r there exist operators $c_{r\sigma}^+$ and $c_{r\sigma}$ which create and annihilate electrons in Φ_r with spin σ ($\uparrow$ or $\downarrow$). $c_{r\sigma}^+ c_{r\sigma} = n_{r\sigma}$ is the respective number operator. The chemisorption of H on W(100) was taken by Grimley as an example, and the Hamiltonian to enter the Schrödinger equation that describes the interaction with a H atom and a semi-infinite W surface was given as:

$$H = \sum_\sigma E_A n_{A\sigma} + J n_A \uparrow n_A \downarrow + \sum_{k\sigma} \varepsilon_k n_{k\sigma} + \sum_{k\sigma} \left(V_{Ak} c_{A\sigma}^+ c_{k\sigma} + V_{kA} c_{k\sigma}^+ c_{A\sigma} \right), \qquad (68)$$

where ε_k is the energy of the state Φ_k of the semi-infinite metal. All chemisorption effects depend strongly on a spectral density function

$$\Gamma(\varepsilon) = \pi \sum_k |V_{Ak}|^2 \, \delta(\varepsilon - \varepsilon_k), \qquad (69)$$

which in turn is dependent on the surface geometry through the matrix

element V_{Ak} that characterizes the interaction of the adsorbate state Φ_A and the metal states Φ_k. For a correct description of the coupling between metal and adatom it is essential to choose an appropriate group orbital (which depends on the position of the adatom) that provides the maximum overlap with the H 1s function. The calculation of the resulting spectral density in the d-band of the metal is a central problem in chemisorption theory.

For H on W Grimley has performed such calculations [462], and it was Newns [28] who transferred and modified the Anderson formalism to treat the H chemisorption on 3d transition metals (Ni, in particular). He considered only the coupling of the H 1s orbital with the metal d-band and neglected the influence of the s- and p-electrons completely. By fitting the adsorption energy of the hydrogen to the experiment he obtained a new metal state approximately 6 eV below the Fermi level, E_F. This is almost exactly the energy range in which UV-photoemission experiments [142,143] revealed H-induced extra emission features. Newns thus confirmed the concept of the formation of a surface molecule formed by adsorbed H and the underlying metal atom(s). However, the aforementioned agreement with the experiment must be taken with some care and could be rather accidentally, since later LCAO studies using tight-binding type calculations by Fassaert and van der Avoird [463] for H on Ni showed that also the sp-band was heavily involved in the chemisorptive binding. This was also confirmed by Doyen and Ertl in subsequent studies which are also based on the Anderson formalism [464]. They used an effective one-electron Hamiltonian to describe a surface molecule coupled to an indented metal, whereby their concept was based mainly on earlier considerations of Grimley and Pisani [465]. The results obtained with this model Hamiltonian provided some new aspects of the chemisorption bond. The model implies a reduced core hopping leading to a dominance of the hopping induced by the electron–electron repulsion. Therefore, the adsorption energy is not obtained by accumulating the charge in the space between the adatom and the metal surface, but rather by effectively keeping the electrons apart (by a correlated hopping) which in turn minimizes the electron–electron repulsion. If Doyen's model is applied to the mere interaction between the H 1s orbital and the metallic d-band (Newns' concept [28]) an adsorption energy of only 2 kcal/mol (as compared to the experimental value of 63 kcal/mol) is obtained, along with other inconsistencies. These results point to the fact that the 1s–3d coupling cannot be the dominating interaction. Rather, a coupling between the H 1s orbital and the valence s-band and the unoccupied H 2p orbitals to the metal d-band appear to be important. Also, non-orthogonality effects must be taken into account. Then the chemisorption-induced level should be centered around 7 eV below E_F, likewise in agreement with experimental findings.

Fassaert and van der Avoird [29b] have, among others, performed LCAO calculations for adsorption on periodic surfaces. The route they chose is based on a tight-binding extended Hückel method applied to finite (periodic) crystals

up to ~ 250 atoms, the non-orthogonal basis set comprising five 3d orbitals, one 4s orbital and three 4p orbitals per surface atom. In a first part the band structure of Ni is calculated, thereafter the effect of the (100), (110) and (111) surfaces on the local density of states as well as the charge distribution was examined. Their results are in good agreement with experimental photoemission spectra. The authors emphasize that both the 3d electrons and the conduction (sp) electrons take part in the chemisorptive bond. They find that the stability of the adsorption decreases in the order (110) > (100) > (111), the centered site should be less favorable than the bridge or atop site. These latter results though are at variance with the experimental findings and the general experience that H atoms prefer sites with high coordination. One may deduce that there are certain shortcomings of the LCAO–extended Hückel procedure which – although the adsorption energy is nicely verified – do not allow too detailed conclusions about bond lengths or binding sites.

4.3.2. Self-consistent layer calculations

During the last decade there has been a tendency to combine LCAO-type of calculations with self-consistent bulk band structure methods. According to Smith et al. [466], one usually departs in these self-consistent procedures from the Kohn–Sham equations

$$\left\{ -\frac{\hbar^2}{2m}\nabla^2 + V_{\mathrm{eff}}(n, \boldsymbol{r}) \right\} \Psi_i(\boldsymbol{r}) = E_i \, \Psi_i(\boldsymbol{r}), \tag{70}$$

in which the important quantity is the effective potential

$$V_{\mathrm{eff}}(\boldsymbol{r}) = \Phi(\boldsymbol{r}) + V_{\mathrm{xc}}(n(\boldsymbol{r})), \tag{71}$$

which is composed of the electrostatic potential

$$\Phi(\boldsymbol{r}) = \int \frac{n(\boldsymbol{r})}{|\boldsymbol{r}-\boldsymbol{r}'|}\mathrm{d}\boldsymbol{r}' - \sum_j \frac{Z_j}{|\boldsymbol{r}-m_j|} \tag{72}$$

and the exchange–correlation term, V_{xc}.

$$n(\boldsymbol{r}) = \sum_{i=1}^{N} |\Psi_i(\boldsymbol{r})|^2 \tag{73}$$

is the charge density. Z_j represents the nuclear charge of the atom at site m_j and $V_{\mathrm{xc}}(n(\boldsymbol{r}))$ is the exchange–correlation potential. This can be treated within a local density approximation. Within the LCAO approximation then the wave function Ψ_j is constructed from the atomic orbitals φ_i:

$$\Psi_i(\boldsymbol{r}) = \sum_j c_{ij} \, \varphi_i(\boldsymbol{r}). \tag{74}$$

Whilst Smith et al. have calculated among others the adsorption of nitrogen on Cu(100) [467] there are other authors who have treated the hydrogen chemisorption using related methods. As a first example we shall consider the work by Feibelman and Hamann [468] who have calculated the surface electronic structure of a H monolayer on top of a (0001) surface of titanium. The ability of H to readily diffuse into the interior of Ti was taken care of by considering a number of possible adsorption sites. The results showed that the surface electronic structure reacts very sensitively to the location of the H atom: Surface states close to E_F may be generated or removed, split-off states formed containing virtually all of the H 1s amplitude, a state split off from the bottom of the Ti with a large s wave function amplitude on the adsorbed H atoms, etc.

The aforementioned procedures included the core electrons explicitly in the calculations. There is a somewhat different approach being used by Louie [32,469,470] who solves eqs. (70)–(73) self-consistently by means of pseudopotentials. Furthermore, he does not use the atomic orbital basis set (eq. (74)) but rather a mixed basis consisting of plane waves *plus* d-like Gaussians.

A kind of a highlight in the chemisorption calculations represents Louie's work on the H/Pd(111) system [32]. Three different kinds of adsorption sites for H are calculated, among which are the two sites belonging to the stacking sequence of fcc Pd. According to Louie, the wave functions of the H adsorbate states are quite localized on the H atom and the topmost Pd atoms and are mostly d-like surrounding these Pd atoms. The essential ingredient in the chemisorptive bonding of H on Pd(111) seems the coupling of the H 1s to the Pd 4d wave function. Among others, a H–Pd bonding band ~ -6 eV below E_F results from Louie's calculation. A very recent investigation was concerned with coverage-dependent adsorption properties of H on a Pd(100) surface [471]. Therein the preferential adsorption sites (fourfold hollow below $\Theta_H = 1$, additional bridge-bonded sites for islands with $\Theta_{H,local} = 2$) are predicted, as well as bond lengths and vibration frequencies. All calculated properties seem to be in remarkable agreement with the experimental data.

4.3.3. Calculations using the effective medium theory

The effective medium theory was introduced by Stott and Zaremba [472] and by Nørskov and Lang [34] as a valuable concept to describe the perturbation imposed by an adsorbate atom (e.g., a H atom) on a host (the substrate) which may be a homogeneous free electron gas. As was pointed out by Nørskov [473,474], the binding energy ΔE of a H atom on a transition metal surface with a given electron density profile $n_0(\boldsymbol{r})$ is estimated from the binding energy ΔE_{hom} of the H atom in the homogeneous electron gas:

$$\Delta E = \Delta E_{hom}(\bar{n}_0) + \int_a \Phi_0(\boldsymbol{r})\, \Delta\rho(\boldsymbol{r})\, \mathrm{d}\boldsymbol{r} + \delta\left\{\int_{-\infty}^{E_F} \Delta n(\varepsilon)\, \varepsilon\, \mathrm{d}\varepsilon\right\}, \tag{75}$$

where $\bar{n}_0$ is the average of $n_0(\boldsymbol{r})$ using the atom-induced electrostatic potential $\Delta\Phi(\boldsymbol{r})$ of the atom in a homogeneous electron gas as the sampling function:

$$\bar{n}_0 = \frac{\int_a n_0(\boldsymbol{r})\, \Delta\Phi(\boldsymbol{r})\, \mathrm{d}r}{\int_a \Delta\Phi(\boldsymbol{r})\, \mathrm{d}r}, \tag{76}$$

$$\Delta\rho(\boldsymbol{r}) = \Delta n(\boldsymbol{r}) - \delta(\boldsymbol{r}) \tag{77}$$

is the charge density induced by the H atom in the homogeneous electron gas, $\Delta n(\varepsilon)$ is the atom-induced one-electron density of states, and $\Phi_0(\boldsymbol{r})$ is the electrostatic potential of the clean metal surface. "a" represents the region surrounding the atom. Inside "a", the one-electron potential is assumed independent of the host (= the homogeneous electron gas or the real surface), outside "a", the one-electron potential is considered independent of the presence of the H atom. This corresponds to a first-order perturbation treatment of the difference between the real surface and the homogeneous electron gas inside region "a" (where the properties of the H atom dominate) and of the region outside "a", where the properties of the host dominate. According to eq. (75) the charge density profile $n_0(\boldsymbol{r})$ of the surface is a very important quantity in the effective medium theory. The s- and p-electrons do not deviate very much from the free electron gas, but the d-electrons are somewhat problematic.

For most H–metal distances encountered in chemisorption the homogeneous electron gas approximation does not appear to cause problems. However, when the H atom has a very small distance to the metal atoms, localized charge densities, directed orbitals, etc. become important. This problem is circumvented in that the (extremely small) metal ion core is embedded in the H atom, and the respective energies are calculated again with the formalism of the EMT.

Despite some apparent shortcomings of this very simplistic procedure it is surprising how well the effective medium theory works in the case of H adsorption. There are numerous examples where the binding energies of H on various metal surfaces are predicted quite close to the experiment. Many of the respective publications have already been mentioned [45,51,98,99], we particular refer to the article by Nordlander et al. [99] which summarizes some of the results for H adsorption.

In order to demonstrate the remarkable agreement with experiment which can be reached by means of the effective medium theory applied to, e.g., H adsorption of Ni surfaces we present a table which was published by Nørskov [474]. It compares experimental results and theoretical predictions (obtained with EMT) for the adsorption site, H–metal layer distance, binding energy,

Table 12
Equilibrium properties of H on Ni(100) and (111) surface and between the first and second layer on Ni(111): site, Ni–H distance (d_{Ni-H}), binding energy (relative to the free atom) (ΔE), vibrational frequency (ω_0), and position of the hydrogen level with respect to the Fermi level (ϵ_b) are shown (after Nørskov [474])

Surface		Site	d_{Ni-H} (a_0)	ΔE (eV)	ω_0 (meV)	ϵ_b (eV)
Ni(100)	theory	Fourfold hollow	3.63	−2.7	79	−5.7
	experiment		–	−2.74	74	–
Ni(111)	theory	Threefold hollow	3.44	−2.7	134	−5.7
	experiment		3.47	−2.74	139	−5.9

ground state frequency ω_0 and position of the H1s resonance level for H on Ni(100) and Ni(111) (table 12). Besides the treatment of the H chemisorption problem, also (and this underlines the validity of the concept) the solution of hydrogen in the bulk of metals, its interaction with bulk defects and embrittlement phenomena can be tackled successfully. We only refer to the relevant literature [475,476].

4.3.4. The embedded atom method (EAM)

The effective medium theory dealt with in the foregoing section is not the only relatively simple theoretical concept that can successfully predict the binding properties of hydrogen. The somewhat related embedded atom method introduced by Daw and his group [35] has already been mentioned earlier, for example, in the context of the description of the H/Pd(111) system, its binding properties and collective behavior (cf. section 3.6.2). Like EMT, the EAM method is particularly capable of calculating bulk properties, e.g., heats of solution, embrittlement phenomena, etc. Again, the concept of embedding impurity atoms ($\triangleq$ the adsorbate) in a substrate of host atoms is used. In brief, the method models the total potential energy of a system of atoms by the expression:

$$E_{total} = \sum_i F_i(\rho_{h,i}) + \tfrac{1}{2}\sum_{ij} \varphi_{ij}(R_{ij}), \tag{78}$$

where $\rho_{h,i}$ is the total host electron density at site i, F_i is the energy to embed an atom in that electron density and φ_{ij} is the short-range core–core repulsion. This formula for the total energy comes out if density functional theory is applied to the problem of embedded each atom in a sea of electrons created by all other atoms. The embedding energy then contains the cohesive energy of the solid which manifests itself as binding between the atoms and the electron gas. The embedding function F_i for each element in the interaction system

depends only on the *electron density* of the host $\rho_{h,i}$ and not on the type of the host. $\rho_{h,i}$ is approximated by a superposition of the electron densities ρ_a of the atoms. The respective functions along with the embedding functions and the core–core repulsions can be determined semi-empirically for the bulk of the host and are assumed to be valid at the surface without further adjustment to surface properties. In the calculation procedure, the potential energy is then minimized. Taking the H/Pd(111) interaction system as an example [327], calculations were performed for a (111) slab of Pd atoms 4 layers thick and 10 layers thick; the results agreed the within 0.02%. A periodic cell consisting of six units in the [110] direction and in the [112] direction was imposed for the minimization calculations. The energy minimizations allowed for the motion of all atoms (Pd and H in this case) to find their site with lowest energy. Furthermore, the full harmonic zero-point energies of the H atoms were calculated including coupling effects among the atoms. In this way, e.g., clean surface layer relaxation effects (a slight contraction) could be predicted. Also, the H adsorption sites came out (the threefold surface sites and the octahedral subsurface site) as well as the respective H binding energies (2.67 and 2.65 eV, respectively), in excellent agreement with the experiment. Without going further into the details of the method it is simply stated here that not only the surface properties of the H on Pd(111) system have been correctly predicted including a determination of the phase diagram by joint Monte Carlo calculations, but also H-induced embrittlement phenomena can be calculated equally well [35a]. Furthermore, the EAM formalism allows a treatment of surface reconstruction of metals as recent developments show [477].

4.3.5. Embedded cluster model studies

Pursuing the "embedding" concept we also have to mention the theoretical developments introduced by Muscat (which we referred to occasionally before [222,316]. His method provides a calculational scheme devoted to the study of impurities at or near metal surfaces which allows an appropriate description of chemisorption and surface segregation phenomena. The method is presented in great detail in ref. [436]; here we only mention some key points in conjunction with H chemisorption.

Basically, a cluster of several metal atoms with one or more adsorbed H atoms is embedded in the surface of a semi-finite free electron gas. This allows a realistic and manageable treatment of both the sp- and the d-electrons of the transition metal host. Because of the tight-binding nature of the d-electrons of the transition metal it appears sufficient to model the interaction of these electrons and the valence electrons of the "impurity" atoms by a small cluster which surrounds the "impurity" site. The embedding of the cluster in a semi-finite free-electron medium then renders possible a sufficiently good handling of the interaction between the free-electron like sp electrons of the transition metal and the valence electrons of the "impurity" atom(s). Since it

is the intention to model the *chemisorption*, the process to be considered takes place at the *surface* of the metal and thus a reasonable choice of the surface potential is required. In principle, Muscat's method is capable of handling the H adsorption on both transition metals [222,478–480] and free-electron like metals [481] as well as H adsorption on, e.g., bimetallic (two-component) surfaces. Another advantage is that mutual adatom interactions can easily be introduced and calculated. Thus it is possible to also predict the long-range order properties of adsorbed hydrogen phases.

The theoretical approach starts off with the choice of an appropriate cluster potential. It is assumed that the potentials centered on the host or "impurity" sites are spherically symmetric within muffin-tins. Outside the muffin-tins the potential is regarded constant except in the surface region where a sudden change from the value of the bottom of the metal sp band to the vacuum potential is assumed:

$$V(\boldsymbol{r}) = \sum_{i=1}^{N} v_i(|\boldsymbol{r} - \boldsymbol{R}_i|)\, \Theta(a_i - |\boldsymbol{r} - \boldsymbol{R}_i|) + V_0 \Theta(z), \tag{79}$$

in which $\boldsymbol{R}_i$ is the position vector of the center of the muffin-tin i of radius a_i; z is the component of $\boldsymbol{r}$ which is perpendicular to the surface potential assumed to be at $z = 0$; Θ is the Heaviside unit step function; v_i denotes the (spherically symmetric) host or "impurity" potential, and V_0 stands for the surface barrier height (which is equal to the sum of the Fermi energy + the work function of the metal surface).

In order to represent the metal atoms self-consistent bulk band structure potentials are taken as they are available from literature data. The adsorbate atoms are modeled by parametrized screened potentials; for a hydrogen atom, in particular, the potential is parametrized so as to produce a bound state just below the bottom of the sp band for a proton in a bulk free electron gas as it is found in ab-initio calculations for H^+ in jellium. Next, the interaction of the muffin-tins with the electron gas must be considered. The Schrödinger equation can readily be solved for the presence of a single muffin-tin (single scattering). The introduction of a perturbation potential $v(\boldsymbol{r})$ induced by the muffin-tin causes a phase shift δ which, of course, depends strongly on $v(\boldsymbol{r})$ and plays a central role in describing the physical properties of the system. The next step then is to include the multiple scattering problem caused by the presence of all muffin-tins ($V(\boldsymbol{r})$). The change in the density of states by introducing the whole cluster is calculated as well as the change in the density of states caused by the "impurity" atom. Furthermore, the influence of the surface potential on the properties of the embedded cluster–metal system must be examined. Likewise, corrugation effects of the surface must explicitly be considered.

The result of all this effort is theoretical H chemisorption energies on transition metals and adsorption site geometries which are in good agreement

with most of the experimental data. This holds also as far as mutual H–H interactions are concerned which can be predicted as well. The capacity of Muscat's model may be demonstrated with the aid of some examples.

For the system Ni(111)/H the bonding UPS state is predicted to lie at $E_H = -5.8$ eV below E_F for a Ni–H distance fitted to the experimental value of 1.84 Å in a threefold coordinated site. A slight reduction of this value would cause a downshift of E_H to -7.1 eV below E_F. Owing to its very low binding energy value (-9.5 eV below E_F) which is not found experimentally, a H subsurface site can be ruled out. Muscat also estimates the influence of the Ni d-electrons on the binding energy ΔE: With jellium ($\hat{=}$ sp electrons) alone, $-1.7 < \Delta E^\circ < -2.6$ eV. Consideration of the d-electrons adds another -0.4 eV ($\Delta E'$) to this value (about 20%). He points out that it is just this additional amount of energy that enables the spontaneous dissociation of the H_2 molecule. Both types of threefold hollow sites are expected to be occupied, and the interaction between two H atoms in which the H–H bond extends all the way through an underlying Ni atom is found to be most attractive, in agreement with the honeycomb structure model derived from LEED [50] (cf. fig. 37).

Likewise, the system H/Ni(100) was calculated by Muscat. Calculations for an embedded 21 atom cluster + adsorbed H revealed the interesting result (although not experimentally verified) that the fourfold center site is not the energetically most favorable one but rather a location which is somewhat displaced from the fourfold hollow site towards the two adjacent Ni atoms so as to reach a shorter Ni–H bond length.

Another example concerns the H/Pd(111) system. Here, calculations for a Pd_{19} cluster plus an adsorbed H atom were made. Clearly, a H-induced resonance state -5.7 eV below E_Γ is produced in this model, whereby a Pd–H bond distance of 1.84 Å is chosen. Muscat underlines the good agreement between his calculations and Louie's self-consistent band structure [32]. As with Ni(111)/H either type of threefold hollow site comes out to be most favorable for adsorption. Looking at the H–H interactions, again the graphitic c(2 × 2) structure should yield the most attractive interactions. This is, however, at variance with the LEED data obtained by Felter et al. [327] who found $\sqrt{3}$ structures in which the H–H interaction does not run straight through an underlying Pd atom (cf. fig. 45). This disagreement reveals that so far this type of chemisorption calculations is probably not sophisticated enough to correctly predict subtle details of the chemisorption process, in particular the mutual interactions.

4.3.6. Self-consistent cluster calculations

We shall end our brief excursion into the area of H chemisorption theory by presenting some more recent results that have arisen from self-consistent cluster calculations using a formalism related to the SCF X_α approach mentioned in section 4.2.2. Several important developments were communi-

cated by Salahub and his group [153,483,484,488]. Their calculations are based on a SCF – local spin density – scattered wave (SCF–LSD–SW) model which is, among others, capable of predicting magnetic properties of clusters consisting of metal atoms such as nickel, and their alterations due to the presence of chemisorbed hydrogen [153]. Furthermore, adsorption sites, bond lengths, interaction energies for systems such as Pd + H can be calculated by modified LCAO X_α – SW techniques with a precision that is only governed by the size of the cluster and the choice of the basis set of the wave functions [483].

In their Ni cluster calculations Fournier and Salahub [153] use the above-mentioned SCF–LSD–SW method which differs from the well-known SCF X_α – SW procedure [448] in that the more accurate correlated potential of Janak et al. [445] was chosen instead of the X_α potential. The Ni spheres were assumed tangent to each other. A sphere radius for the H atom of 0.64 Å was taken except for the on-top geometry where 0.43 Å were considered in order to avoid excessive overlap. Partial waves up to $l = 1$ for H, $l = 2$ for Ni and $l = 4$ for the outher sphere were included.

The authors calculated three different clusters, viz., $Ni_{13} + 2H$ which provided a model of the fourfold hollow site on Ni(100), $Ni_{14} + 2H$ (modeling on-top adsorption on Ni(100)) and $Ni_{10} + H$ in order to mimic the threefold hollow site on a Ni(111) surface. Reasonable Ni–H bond distances were chosen as provided by structural analyses [50,482]. From the results the following picture emerges: In all three cases hydrogen decreases the magnetic moment of the Ni clusters. The change of total magnetic moment is surprisingly pronounced for the thirteen and fourteen Ni atom cluster, much larger than observed experimentally [486]. The apparent discrepancy can be explained by a too large H to Ni ratio, in other words, the size of the cluster is too small to correctly predict the magnetic properties. Three factors are made responsible for the decrease of the magnetization with hydrogen adsorption. As a main factor, hydrogen contributes one more electron into the band structure of Ni which most populate a down-spin orbital thus reducing the magnetization. The reduced magnetization causes a smaller exchange splitting which in turn makes some spin-up d-electrons (which do not directly participate in the chemisorptive bonding) move into the down-spin manifold. Furthermore, some d-character in the LDOS for Ni atoms near the hydrogen adsorbate is pushed above E_F due to the antibonding interactions thereby causing a further reduction of the magnetic moments.

Concerning cluster calculations performed for Pd and chemisorbed hydrogen we refer to the work by Baykara et al. [483] who apply the local-spin density (LSD) method to various Pd clusters. The usefulness of this formalism was proven for a variety of transition metal clusters [484], particularly, if the model potential (MP) method as proposed by Bonifac and Huzinaga [487] is combined with the LSD procedure. Also, all-electron (AE) calculations were made alternatively. In detail, a three atom cluster modeling the threefold site

on Pd(111) was considered, thereafter a larger, ten atom cluster containing a complete set of sites encountered along a possible H diffusion path in the fcc structure was subjected to calculation. For Pd, the authors used a sixteen electron (scalar) relativistic model potential [488]. A {6s, 5p, 4d} wave function basis set contracted to {4s, 4p, 3d} was employed for Pd. For hydrogen, a 5s orbital was contracted to the 2s level and augmented with a p-polarization function. In order to allow for angular polarization of the density and the exchange–correlation potential V_{xc}, 1d auxiliary functions were added, with exponents derived from the p-orbital functions using Dunlap's rule [489]. In addition, one p-type function was supplied for the density (exchange–correlation) fit.

The calculated chemisorption geometry (over a threefold site) shows a Pd–H distance of 1.72 Å, slightly longer that that proposed by Eberhardt et al. [145], but smaller than that derived from a LEED structure determination of the 2×1-2H phase on Pd(110) [490]. The calculated binding energies are likewise in quite good agreement with the available experimental data: The threefold chemisorption site is by 0.25 eV more stable than the most stable (octahedral) bulk site. The top of the activation barrier for diffusion, a triangular site, is predicted to be 0.34 eV (experimental value: 0.23 eV) above the octahedral site. A vibrational frequency of $\omega_0 = 892$ cm^{-1} is predicted for the H ground state on Pd_{10} cluster, in fair agreement with the experimental value of 998 cm^{-1} reported by Conrad et al. [162]. All of the sites lead to a split-off state of Pd d and H s character below the d-bands, located around -6.96 eV below E_F, in close proximity to the experimental value of -6.5 eV.

The consideration of structural relaxation (which was not addressed in the quoted calculations) could eventually lead to an even better agreement between theory and experiment.

In concluding this section, we simply want to point to some other cluster calculations, for example, by Andzelm [491] of H on various Ni and Pt clusters. Madhaven and Whitten [492] consider the H chemisorption on Cu be means of ab-initio valence (4s) CI calculations using a 38 atom Cu cluster. They predict atomic H adsorption on Cu(100) at several adsorption sites (fourfold, bridge and atop) with binding energies ranging from 13 to 22 kcal/mol with fourfold and twofold sites slightly preferred. The Cu–H bond distances range from 1.7 to 2.0 Å. Furthermore, high activation barriers for H_2 dissociation are calculated due to the repulsion of molecular H_2 by the surface and the energy required to stretch the molecule significantly above the surface. Walch [493] performed ab-initio/effective core potential cluster studies for the interaction of H with bcc iron. Twofold and fourfold sites on the (100) surface as well as octahedral, tetrahedral and trigonal interior sites were studied. The fourfold site (calculated binding energy 1.3–1.7 eV) was found to be the most favorable site for chemisorption, whereas the interior tetrahedral site was predicted to be occupied by bulk hydrogen. (Extensive cluster calculations

were also reported by Cox and Bauschlicher [494] for H on Ni(111) modeled by a Ni_{10} cluster.) The sophisticated first-principle calculations in the group of Koutecký [451–454] (H on Li, Be, Ge and Pd clusters) have already been mentioned. From these studies it appears that the consideration of excited electron states is of fundamental importance when aiming for good agreement between experiment and theory. Quite recently, an impressive ab-initio pseudo-potential total energy calculation for the system H on Ru(0001) was reported by Chou and Chelikowsky [495]. They find that at low H coverages the threefold hollow site is preferentially occupied whilst at higher coverages the additional population of subsurface sites is predicted, in agreement with experimental evidence reported by Peden et al. [383]. We must also mention layer band structure calculations by Umrigar and Wilkins [496] who consider H on Ni adsorption by modeling the system using a five Ni monolayer slab with a H layer on top of each side.

When we finish our short expedition into the theoretical chemistry here, we are, of course, aware that probably only a small part of all the many theoretical works has been considered in the context of this (mainly experimentally oriented) review article on H chemisorption. We may nevertheless state at this point that the calculations (in particular the cluster calculations) have meanwhile reached a level of sophistication that the correct prediction of even minor and subtle effects concerning the adsorption geometry, diffusion paths, split-off states, etc. appears to be manageable in the near future [497].

5. Conclusions and outlook

The interaction of hydrogen with solid surfaces resembles in many respects the surface interaction of other chemically active diatomic molecules such as CO, N_2 or O_2. However, there are also some peculiarities of H_2 or H which make its interaction pattern unique. This pattern may be summarized from the foregoing as follows:

A H_2 molecule that is supposed to interact with a solid usually has to surmount one or two activation energy barriers before it can enter the chemisorbed state. These may be located in the entrance or in the exit of the reaction coordinate. Accordingly, a direct or an indirect (molecular precursor) mechanism is effective. Depending on the height of the respective barrier dissociative (atomic) adsorption occurs at a slower (small sticking probability, s_0) or at a higher rate (high s_0). The existence of a high density of d-electron states at the Fermi level of the substrate appears to be the most crucial parameter that governs the dissociation process. Surfaces rich in d-electrons at E_F (transition metals) usually provide an easy dissociation path in that they exhibit only a small or even vanishing activation barrier. Nevertheless, they may in some cases (Ni(111)) build up small barriers in the entrance channel

thereby causing markedly reduced sticking probabilities. On the other hand, deep-lying d-bands (Ag, Au) or high concentrations of sp electrons (as with alkaline earth or alkali metals) do not provide an easy dissociation: Usually large activation barriers exist separating the molecular from the chemical (atomic) potential energy well. This holds as well for semiconductors and insulators with their localized or ionic bonds. Nevertheless, sp-metal and semiconductor surfaces may strongly interact with hydrogen if this element is offered in an atomic form. Then, metal–H or semiconductor–H bonds are formed instantaneously and the substrate becomes easily hydrogenated to the respective hydride compound with different electronic and geometrical structure. In this respect, hydrogen may well be compared with other diatomic molecules (O_2, N_2). The observed heats of adsorption for chemisorbed H range around 50 to 150 kJ/mol and, accordingly, the bond strengths of a H–substrate atom vary between 240 and 300 kJ/mol and are thus of the order of typical chemical bond energies.

However, the aforementioned *special* properties of the H atom may govern the interaction path outlined above quite markedly. The most significant property is the small physical size of a H atom of ~ 0.5 Å only. It is thus possible to bring a H atom very close to a surface – in some cases it does not even require much energy to move it *through* a surface layer of substrate atoms into the interior of a crystal. On a surface the small H atom then "feels" all sorts of irregularities in the electronic or geometrical structure very sensitively (foreign atoms, defects). From most of the experimental and theoretical data it appears that a highly coordinated site is preferred for chemical bonding to the surface. In this site the H atom can greatly perturb the local electronic structure and may polarize adjacent substrate bonds to a great extent. This then may lead to phenomena known as relaxation or reconstruction, whereby one can distinguish between a *local* perturbation and a long-range effect caused by the adsorbed H atom.

Another prominent feature of adsorbed H atoms is their ability to interact with each other via the substrate ("indirect" interactions). This has again to do with their small size which causes the indirect interactions to become effective before any direct orbital overlap can come into play. The through-bond interactions are responsible for the frequently observed phases with long-range order which are formed even at small surface concentrations. Also pertinent to H adsorption is the pronounced mobility of the H atoms even at fairly low temperatures which may, in some cases, lead to a "hydrogen fog" rather than to an ordered phase. Owing to its light mass the H atom, in particular, has a high tunneling probability as compared to other adsorbates, and consequently, greatly reduced activation energies for surface diffusion have been found in some cases. The high mobility of hydrogen at elevated temperatures is of importance also in the field of heterogeneous catalysis where H diffusion is seldom rate-limiting.

The consideration of various theoretical attempts to model the H–surface interaction has shown that an enormous progress has been made to understand the bonding mechanism. It appears as if the role of the sp-electrons on the one hand and that of the d-electrons on the other can be understood from the point of theory. Likewise, interaction potentials can be predicted quite well. However, there are still difficulties in handling mutual interactions quantitatively. Nevertheless, it is felt that – owing to the steady improvement of numerical computation capacity – larger clusters or larger wave function basis sets will become tractable in the near future. This will not only improve the precision of binding energy predictions but should also allow a reliable forecast of local bonding geometries, vibrational frequencies, dipole moments, etc. Furthermore, improved solid state calculations will be possible including effects of H-induced layer relaxations and surface reconstruction phenomena. By this time also the experimental data body will have reached a high level of reliability thanks to the more and more refined spectroscopic methods. All in all, the adiabatic potential energy be assumed manageable for the H-chemisorption system, even for the molecular hydrogen–surface interaction some properties of the potential energy hyperface can be qualitatively understood.

A still open problem is a quantitative description of the dynamics of the H_2–surface interaction, owing to the multi-dimensionality of the adiabatic potential. Here, a combined experimental and theoretical effort is needed to work out details of the motion of a H_2 molecule or a H atom when getting in contact with a solid surface.

It is sincerely hoped that the present article could help somewhat to direct the readers' attention to some still challenging questions encountered in H adsorption but also provide some confidence in the data body accumulated so far which may serve as a safe basis for future research in this field.

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